

H2-SHIFT: Convert. Don't Replace.

Round 2 Annexure: supporting evidence, calculations and plans

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This annexure supports the 10-slide main presentation and the workbook H2SHIFT_model.xlsx. All numbers here come from the recalculated workbook (base scenario) or from the source register. Evidence classes: A verified external evidence; B our calculated estimate; C explicit assumption or sensitivity range; D proposed target requiring validation. No customer interviews, letters of interest, test results, supplier quotations or partnerships are claimed; proposed partners are proposals.

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A1. Claim audit: Round 1 versus evidence

Being shortlisted is evidence of competition progress, not proof of technical performance or commercial feasibility. Each Round 1 claim was checked against the research; the table gives the verdict and its Round 2 consequence.

Original claim (Round 1)	Evidence provided in Round 1	Verification result	Keep / refine / correct	Round 2 implication
7.5 million CNG vehicles in India; 3x growth in 8 years	CRISIL CNG outlook (cited by name)	Latest official: 81.95 lakh on road (Mar 2025, PNGRB citing Vahan); 72.10 lakh (Jul 2024). '3x in 8 years' unverifiable (no official 2016-17 baseline retrieved)	Correct	Use 8.2 million (Mar 2025); quote +40 percent Mar 2023 to Mar 2025
About 65 percent of CNG vehicles are passenger cars	None	No official category split found; estimate 45 to 60 percent from factory CNG car sales and conversions	Correct	Use 55 percent as a labelled estimate (class B) with a range
Maruti Suzuki about 71 percent share of India's CNG passenger car market; 6.2 lakh CNG cars sold FY25	Maruti FY25 investor update	Verified: 71.6 percent (CY2024 industry data); 6.12 to 6.2 lakh FY25; FY26 above 7 lakh; June 2026 monthly share 42 percent	Keep and update	Donor pool argument stands
4,000 plus retrofit centres	None	Only a vendor blog; no official count (Vahan and IGL lists exist)	Correct	Say 'authorised retrofit centres exist in every CGD city; count to be confirmed from Vahan'
Spark-ignition engine, gaseous injection, existing cylinder packaging make CNG to hydrogen the 'shortest technical jump'	Engineering reasoning	Partly true: engine and cylinder bay help; but no CNG fuel-system part can be reused with pure hydrogen (ISO 11439 steel limited to 2 percent hydrogen; UN R134 qualification needed); the hydrogen cylinder is 84 mm larger in diameter	Refine	Retain the argument for the engine and bay; drop any implied component reuse
HCNG step: 18 to 20 percent hydrogen, no vehicle modification, zero customer investment, minus 76 percent CO, minus 25 percent HC, plus 4 to 5 percent economy	IOCL HCNG reformer patent	Legal since GSR 585(E) 25 Sep 2020 (IS 17314:2019). Delhi 50-bus BS-IV trial: about minus 70 percent CO, minus 25 percent HC, NOx unchanged, plus 4 to 5 percent economy (heavy-duty). Hydrogen supplies only about 6.2 percent of the blend's energy. HCNG dispensing exists only at demonstration scale	Correct	Reframe as an air-quality measure with about 6 percent fossil-carbon displacement, dependent on CGD blending; not a decarbonisation step
Bi-fuel H2 plus CNG with automatic switching, CNG cylinder retained, 1.2 to 1.5 kg Type IV at 350 bar, kit ₹1.8 to 2.2 lakh	Indicative BOM (cylinder 55 to 60 percent)	Two cylinders do not fit a Dzire or WagonR boot; the factory car already has petrol as backup. A 60 L composite cylinder holds 1.44 kg nominal, 1.12 kg usable. Installed cost ₹3.2 lakh at 10,000 per year (₹2.3 lakh ex-works), ₹7.7 lakh per pilot car	Correct	Architecture becomes H2 plus petrol (petrol retained); CNG hardware removed; cost ranges revised
Range 400 plus km combined, refuel 3 to 5 min	None	Hydrogen 61 km per fill plus 555 km on petrol; two depot fills give about 120 km per day on hydrogen; refuel 3 to 5 min plausible for a non-cooled 350 bar fill	Refine	Quote hydrogen range separately from petrol range
Step 3: dedicated H2-ICE, second cylinder, lean burn plus EGR, about 100 percent tailpipe CO2-free, upgrade ₹0.8 to 1.0 lakh	None	No space for a second cylinder in the sedan; a 700 bar cylinder of the same size gives 105 km. Zero tailpipe CO2 on hydrogen km is correct but NOx remains	Correct	Step 3 becomes a conditional 700 bar or hub-scale option after 2030
H2-ICE mileage 55 to 60 km/kg	Energy density plus retrofit studies	Energy balance at CNG parity gives about 60 km/kg on declared figures; fleet real-world with a lean PFI retrofit modelled at 54 km/kg (1.85 kg/100 km), range 43 to 63	Refine	Use 1.85 kg/100 km with a stated range; cross-check on the Energy_Range sheet
Green hydrogen price ₹397 (2026) to ₹260-310 (2028) to ₹160-200 (2030) per kg; 'at ₹200/kg delivered running cost equals CNG'	IEA and BNEF cost outlooks	These are production-cost trajectories, not pump prices. Delivered 2030 central estimate ₹360/kg (CEEW, USD 4.2); our conservative build-up ₹584/kg at a 500 kg/day hub at 70 percent utilisation; pilot point ₹1,433/kg. CNG fuel parity needs ₹188/kg delivered	Correct	Economics restated with delivered prices; parity condition made explicit
CNG ₹78/kg, 26 km/kg	Current market levels	Delhi CNG ₹86.98/kg (Sep 2026, after the 2026 gas price rises); fleet real-world 24 to 28 km/kg	Update	₹87/kg base, 25 km/kg base
Running-cost chart: BEV ₹1.2 to 1.8/km	None	Model gives ₹1.5/km energy cost with a 70/30 depot and public mix at ₹6 and ₹20/kWh	Keep	
Fleet taxi 150 km/day, 330 days: about ₹1.5 lakh annual saving versus petrol, 18-month payback	Taxi benchmark	The donor runs on CNG, not petrol; against CNG the retrofit never pays back in 5 years at any credible hydrogen price. NITI Aayog uses 70,000 km/yr for taxis	Correct	Comparator changed to CNG (and BEV); payback claim withdrawn
Safety table: hydrogen has 'different risks, not higher risks with the right design'; regulation AIS-157 plus PESO plus ISO/SAE	None	Properties verified (4 to 75 percent, 0.017 mJ, 3.8x diffusivity). AIS-157 is the fuel-cell standard and its scope excludes retrofitted vehicles; AIS-195 (2023, Amd 1 2025) is the hydrogen ICE standard for new M and N vehicles; no retrofit route exists	Correct	Regulatory pathway rebuilt around AIS-195, Rule 126/115 template, PESO GCR 2025; gate 1 added
Burst test at or above 2.25x	None	Consistent with GTR 13 / ISO 19881 minimum burst ratio for carbon-fibre composite containers	Keep	Cite standard

Flywheel: 7,400 to 17,500 CNG stations by 2030; 8.6 lakh t/yr capacity awarded, only about 8,000 t/yr commissioned	Government data	8,150 plus stations (Aug 2025); 17,700 (MWP commitment) or 18,000 (target) by 2030; 8.62 lakh tpa awarded (May 2025) versus about 8,000 tpa commissioned (Feb 2026): verified	Update	Flywheel logic kept for utilisation; price, not demand, is the binding constraint
Roadmap: 100 stations and 500 fleet cars (2026-27); 25,000 kits in 5 cities (2027-28); 1,00,000 plus cars (2029-30)	None	No approval route, no car-size cylinder approval, three captive dispensing points nationally; station-limited capacity about 1,400 cars by 2030	Correct	Roadmap replaced by four gated phases: 3 prototypes (2027-28), 30 pilot cars (2028-30), conditional 1,000 to 1,500 cars after 2030
Impact by 2030: 1,00,000 cars, 0.9 Mt CO ₂ avoided, 15,000 t/yr hydrogen demand, ₹2,000 crore retrofit industry, 10,000 jobs	None	Bounded model: about 1,400 cars, 3,600 t CO ₂ e/yr, 924 t/yr hydrogen; kit revenue about ₹45 crore; jobs not estimated	Correct	Impact restated with ranges; job claim withdrawn
Global proof points: Toyota GR Corolla H2 and Hilux prototypes	Toyota	GR Corolla H2 is a liquid-hydrogen race car; the Hilux prototype (UK) is a fuel-cell vehicle; the hydrogen combustion van is the HiAce (Australia, 2023-25, less than 200 km range, SCR for NO _x)	Correct	Cite HiAce, not Hilux, for hydrogen combustion
Suzuki hydrogen Burgman Street prototype, 70 MPa	Suzuki	Verified (JMS 2023, re-shown 2025, 'not launching anytime soon'); add Suzuki-AVL hydrogen Swift prototype (1.4 L DI, 100 kW, Vienna, May 2026)	Keep and add	Stronger small-car relevance, still pre-commercial
India: 70 pilot vehicles plus 16 stations, ₹410.77 crore	Government pilot projects	Verified (PIB 29 Jul 2026): 12 projects, 27 buses and 43 trucks, 16 stations, 21 routes, ₹41.58 crore released; no cars	Keep and qualify	Note absence of cars and the 4-wheeler eligibility of the scheme
Regulatory pathway: CMVR, AIS-157, ARAI/ICAT, PESO GCR, IS 16061, IS/ISO 17268, ISO 19880 / SAE J2601; 'mirror the CNG retrofit homologation pathway'	Standard names	Standards exist (IS 16061:2021 fuel quality; IS/ISO 17268:2020; IS 18538 series for stations) but none covers a hydrogen retrofit kit; AIS-157 excludes retrofits; the CNG template (AIS-024/028, Rule 115) is the right analogy but needs a new instrument	Refine	Compliance matrix with responsible parties and gate
NO _x control: lean burn lambda at or above 2, EGR, compact SCR if required, BS-VI compliance pathway	None	Consistent with literature: near-zero NO _x above lambda 2 at steady state; transients need aftertreatment; a DI engine met below 35 mg/km on WLTC only with SCR	Keep and qualify	Lean NO _x trap provision budgeted; compliance is a target
Carbon credits USD 10 to 15/t	Voluntary market range	India voluntary prices ₹1,350 to 2,800/t (2026); global average USD 3.5/t (2025); immaterial (at most ₹0.2/km)	Correct	Credits treated as upside only
Kit BOM: cylinder 55 to 60 percent, injectors and rail 15 percent, ECU 10 percent, sensors/valves/TPRD 15 percent	Indicative	Model BOM at volume: cylinder and valve 49 percent of parts; pilot cylinder cost dominates even more	Refine	BOM sheet with pilot and volume columns

A2. Assumptions, evidence classes and source register

A2.1 Model inputs (base, downside for the retrofit, upside)

Key	Parameter	Base	Downside	Upside	Unit	Class	Sources	Notes
General frame								
decision_year	Decision year for the owner's choice (first commercial kits after certification)	2,028	2,028	2,028	year	C		Pilot vehicles run from 2028; owner decision modelled at commercial availability
horizon	Analysis horizon (fixed at 5 years in the cash-flow table)	5.000	5.000	5.000	years	C		Remaining useful life of a 3-year-old fleet sedan
disc	Discount rate (nominal)	0.120	0.140	0.100	fraction	A	W4-S29	NBFC fleet loan rates 10-16 percent
days	Operating days per year	330.000	330.000	330.000	days	C	R1	Fleet taxi benchmark (Round 1)
km_day	Kilometres per operating day	167.000	212.000	150.000	km/day	A	W4-S30	55,000 km/yr base; NITI Aayog Feb 2026 plans 70,000 km/yr for 4W taxis (downside); Round 1 used 150
fuel_esc	Annual escalation of petrol and CNG prices	0.030	0.020	0.040	fraction/yr	C		Nominal
elec_esc	Annual escalation of electricity tariffs	0.020	0.010	0.030	fraction/yr	C		Nominal
vot	Value of in-shift time lost to refuelling or charging	150.000	150.000	150.000	Rs/hour	C		Approximate contribution margin per hour of a fleet taxi
Donor vehicle (Maruti Suzuki Dzire Tour S CNG, about 3 years old at decision)								
donor_value_now	Resale value of donor at decision date (taxi-registered)	350,000	400,000	300,000	Rs	C	W4-S28	New Tour S CNG ex-showroom about Rs 7.1 lakh; taxi retention 45-55 percent at 3-4 years; higher value favours selling for a BEV
donor_value_end	Residual value of donor at end of horizon (about 8 years old, high km)	100,000	70,000	130,000	Rs	C	W4-S28	
cng_price	CNG retail price, Delhi (IGL), September 2026	87.000	78.000	95.000	Rs/kg	A	W4-S16	Rs 86.98/kg on 14 Sep 2026 (aggregator); Ahmedabad Rs 94; lower CNG price is downside for the retrofit
cng_kmkg	CNG consumption in fleet use (real world, AC, load)	25.000	27.000	23.000	km/kg	C	W1-S3b;W1-S8a	Declared 32.1 km/kg (Tour S); owner reports 24-30; fleet use assumed lower
petrol_price	Petrol retail price, Delhi	102.000	95.000	111.000	Rs/L	A	W4-S19	Rs 102.12/L verified 25 May 2026; aggregators show Rs 111 in Sep 2026
petrol_kml	Petrol consumption in fleet use (real world, AC)	15.000	16.000	14.000	km/L	C	W1-S3a	Declared 26.1 km/L (Tour S); fleet use assumed lower
petrol_share_cng	Share of km run on petrol by the CNG car (CNG unavailable, queues)	0.050	0.050	0.050	fraction	C		
maint_cng	Maintenance and tyres, CNG car	0.900	0.700	1.200	Rs/km	C	W4-S22	CEEW 2025 Table 5 to be transcribed; Round 1 value retained
ins_cng	Insurance per year, CNG car	18,000	18,000	18,000	Rs/yr	C		
cng_retest	CNG cylinder hydrostatic test every 3 years	2,000	2,000	2,000	Rs per test	A	W2-S24	Gas Cylinder Rules: 3-yearly test
time_cng_h	In-shift time lost refuelling CNG per day (queues)	0.500	0.400	0.600	hours/day	C		
Hydrogen retrofit (H2-ICE, port injection, 350 bar composite cylinder, petrol retained as automatic backup)								
kit_cost	Installed conversion cost at commercial volume, incl. taxes (from BOM sheet)	322,123	402,654	273,805	Rs	B	BOM	Linked to BOM volume total
kit_cost_pilot	Installed conversion cost per pilot vehicle (from BOM sheet)	774,375	774,375	774,375	Rs	B	BOM	
salvage_cng	Credit for removed CNG cylinder and kit	10,000	0.000	20,000	Rs	C		Requalified cylinder resale
downtime_days	Conversion downtime	3.000	4.000	2.000	days	C		

lost_margin_day	Lost contribution per day of downtime	1,500	1,500	1,500	Rs/day	C		
h2_cons	Hydrogen consumption, H2-ICE mode, fleet use (real world, AC)	1.850	2.300	1.600	kg/100 km	B	W1-S12a;W1-S12c;W1-S9a	Energy balance versus CNG (Energy_Range); Ford Focus H2RV 1.4, Mazda RX-8 RE 2.4, HiAce van 3.1 kg/100 km
h2_share	Share of km driven on hydrogen (rest on petrol backup)	0.650	0.500	0.850	fraction	C		Two depot fills per day; check Energy_Range max share
h2_price_y1	Delivered hydrogen price at dispenser, year 1 (2028), unsubsidised, shared hub station	800.000	1,000	550.000	Rs/kg	B	W4-S22;W4-S46	CEEW: small-network refuelling today above USD 10/kg; H2 Supply build-up gives the pilot-station price
h2_change	Annual change in delivered hydrogen price	-0.100	-0.030	-0.150	fraction/yr	C	W4-S22	CEEW path to USD 4.2/kg (about Rs 360) delivered in 2030
maint_h2	Maintenance and tyres, retrofitted car	1.000	1.200	0.850	Rs/km	C	W4-S22	Adds injector, sensor, regulator service
ins_h2	Insurance per year, retrofitted car	21,000	24,000	19,000	Rs/yr	C	W2-S53	No IMT endorsement exists for hydrogen kits; assumes insurer accepts endorsed retrofit
h2_inspect_yr	Hydrogen system inspection and 3-yearly cylinder requalification, annualised	4,000	6,000	3,000	Rs/yr	C	W2-S24	
time_h2_h	In-shift time lost for hydrogen refuelling and detour per day	0.400	0.600	0.300	hours/day	C		Two short fills at a depot dispenser
mod_resid_factor	Residual value factor for a modified vehicle	0.900	0.800	1.000	fraction	C		
kit_resid	Residual value of the kit at end of horizon	0.000	0.000	20,000	Rs	C		
BEV alternative (sell donor, buy compact fleet BEV)								
bev_price	BEV ex-showroom price incl. GST (compact fleet BEV, 24-30 kWh)	1,050,000	900,000	1,300,000	Rs	A	W3-S27;W3-S25	Tata Tiago EV Rs 6.99-10.09 lakh, Punch EV Rs 9.69-12.59 lakh (2026), Xpres-T fleet EV from Rs 13.04 lakh
bev_onroad	Registration and first-year extras beyond ex-showroom	30,000	20,000	40,000	Rs	A	W4-S25	Delhi EV Policy 2026 waives road tax and registration fee to March 2030
bev_kwh100	BEV consumption measured at the meter (incl. charging losses)	15.000	13.000	17.000	kWh/100 km	A	W3-S30;W4-S23	Real-world tests 12.8-15.3 kWh/100 km at the battery (Tiago EV, Punch EV, Nexon EV) plus 5-15 percent charging losses
depot_tariff	Depot or home charging tariff, landed	6.000	5.000	8.000	Rs/kWh	A	W4-S20;W3-S32	Delhi DERC EV-charging tariff Rs 4.50/kWh (LT) plus surcharges and taxes
pub_tariff	Public DC fast-charging price	20.000	16.000	25.000	Rs/kWh	A	W4-S21;W3-S33	2026 network prices Rs 16-25/kWh incl. GST
pub_share	Share of energy from public fast charging	0.300	0.200	0.500	fraction	C		
maint_bev	Maintenance and tyres, BEV	0.500	0.400	0.800	Rs/km	C	W4-S22	
ins_bev	Insurance per year, BEV	30,000	28,000	34,000	Rs/yr	C		
bev_resid	BEV residual value at end of horizon	420,000	480,000	340,000	Rs	C	W4-S22	40 percent of price at 5 years (no Indian fleet-EV residual data found); higher residual is downside for the retrofit
time_bev_h	In-shift time lost for charging per day	0.500	0.300	1.000	hours/day	C		Overnight depot charging is idle time
Energy and storage physics								
cng_lhv	Lower heating value of CNG	48.000	48.000	48.000	MJ/kg	A	W4-S62	IPCC natural gas 48.0 TJ/Gg
petrol_lhv	Lower heating value of petrol	32.200	32.200	32.200	MJ/L	A	W4-S62	44.3 MJ/kg x 0.74 kg/L
h2_lhv	Lower heating value of hydrogen	120.000	120.000	120.000	MJ/kg	A	W1-S13a	
cyl_vol	Hydrogen cylinder internal (water) volume	60.000	60.000	60.000	L	A	W1-S2	Luxfer G-Stor Pro V060H35 (Type III, 350 bar): 60 L, OD 400 mm, length 767 mm, 36 kg, 1.45 kg H2
h2_density_350	Hydrogen density at 350 bar, 15 C	24.000	24.000	24.000	kg/m3	A	W1-S13a	Full-fill density per SAE J2601 definition
h2_density_700	Hydrogen density at 700 bar, 15 C	40.200	40.200	40.200	kg/m3	A	W1-S13a	
fill_soc	State of charge reached in a non-cooled 350 bar fill (hot ambient)	0.850	0.800	0.950	fraction	C	W1-S13d	SAE J2601: 95-100 percent needs pre-cooled dispensing; small tanks may fall outside light-duty tables
resid_density	Residual gas density at minimum usable pressure	1.700	2.000	1.500	kg/m3	C	W1-S13c	About 20 bar residual
fills_day	Depot fills per day available to the driver	2.000	1.500	2.500	fills/day	C		
cyl_mass	Composite cylinder mass (60 L, 350 bar)	36.000	42.000	32.000	kg	A	W1-S2	Luxfer Type III 36 kg; Type IV similar
kit_other_mass	Valve, frame, regulator, lines, ECU mass	14.000	16.000	12.000	kg	C		
cng_cyl_mass	Removed CNG cylinder (55 L Type I) mass	55.000	55.000	55.000	kg	A	W1-S14a;W1-S14b	50 L steel cylinder 48 kg; 60 L 61-69 kg

cng_kit_mass	Removed CNG regulator, rail, lines mass	8.000	8.000	8.000	kg	C		
petrol_ps	Donor engine power on petrol (Tour S 1.2 L)	81.600	81.600	81.600	PS	A	W1-S3a	60 kW at 5,700 rpm
cng_ps	Donor engine power on CNG	70.000	70.000	70.000	PS	C	W1-S6	Third-party 69 bhp; official CNG rating not captured
h2_power_factor	H2 lean-burn peak power relative to petrol	0.520	0.450	0.600	fraction	B	W1-S9a;W1-S9b	About 30 percent air displacement (20 percent loss at stoichiometric) plus 35-40 percent loss at lambda 2; strategy uses lambda 1.6-2.2
petrol_tank	Petrol tank capacity	37.000	37.000	37.000	L	A	W1-S3a	
Hydrogen supply and station								
pilot_cars	Pilot fleet size	30.000	30.000	30.000	cars	D		Statistical spread across three fleets, one dispensing point
h2_prod_cost	Green hydrogen production cost at plant gate (2026-28)	400.000	560.000	300.000	Rs/kg	A	W4-S33;W4-S34;W4-S38	CEEW USD 3.5-5/kg; industry Rs 397-560; SIGHT ammonia auctions imply up to Rs 302 for the hydrogen molecule
h2_prod_cost_2030	Green hydrogen production cost at plant gate (2030)	240.000	320.000	180.000	Rs/kg	A	W4-S22;W4-S57;W4-S36	CEEW USD 2-2.5; IEA about USD 3; NITI-RMI best case USD 1.6
transport_cost	Tube-trailer transport to station	45.000	60.000	35.000	Rs/kg	B	W4-S44	About 150 km, 300 kg payload
station_capex_pilot	Dispensing point capex, pilot (compressor, cascade, one dispenser, civil, safety, PESO)	100,000,000	150,000,000	80,000,000	Rs	B	W4-S56;W4-S43	Indian trade source Rs 14-20 crore for 200-500 kg/day; delivered hydrogen, no electrolyser
station_cap_pilot	Dispensing point design capacity, pilot	150.000	150.000	150.000	kg/day	C		
station_util_pilot	Station utilisation, pilot	0.500	0.350	0.650	fraction	C	W4-S22	Shared with bus/truck pilot users on the Sahibabad-Faridabad-Delhi route
station_capex_2030	Station capex, 2030 hub scale (500 kg/day)	120,000,000	200,000,000	90,000,000	Rs	B	W4-S56	
station_cap_2030	Station design capacity, 2030 hub	500.000	500.000	500.000	kg/day	C		
station_util_2030	Station utilisation, 2030 hub	0.700	0.500	0.850	fraction	C	W4-S22;W4-S46	CEEW assumes 30 percent in 2030 rising to 80 percent by 2050; ICCT 50-70 percent
station_life	Station capital recovery period	10.000	10.000	10.000	years	C		
station_om	Fixed operation and maintenance, share of capex per year	0.040	0.050	0.030	fraction/yr	A	W4-S22	CEEW 5 percent
comp_kwh	Compression and dispensing electricity	3.000	3.500	2.500	kWh/kg	C		Delivered gas at 200 bar to 450 bar cascade
ind_tariff	Industrial electricity tariff	9.000	10.000	8.000	Rs/kWh	C		
h2_loss	Venting, purge and metering losses	0.030	0.050	0.020	fraction	C		
h2_margin_tax	Retailer margin and taxes on delivered hydrogen	0.100	0.150	0.050	fraction	C		GST treatment of hydrogen as automotive fuel to be confirmed
trailer_kg	Tube-trailer payload	300.000	250.000	350.000	kg	C		Typical 200 bar tube trailer
Emissions factors								
ef_cng	Tailpipe CO2 per kg CNG	2.690	2.690	2.750	kg CO2/kg	A	W4-S62	IPCC 2006 default 56.1 t/TJ at 48 TJ/Gg
wt_cng	Well-to-tank uplift for CNG in India	0.250	0.150	0.350	fraction	C	W4-S63;W4-S17	Imported spot LNG share raises upstream emissions; India-specific factor not found
ef_petrol	Tailpipe CO2 per litre petrol	2.270	2.270	2.310	kg CO2/L	A	W4-S62	IPCC 2006 default
wt_petrol	Well-to-tank uplift for petrol	0.200	0.150	0.250	fraction	C	W4-S63	
ci_h2_green	Carbon intensity, green hydrogen incl. compression and delivery	2.000	3.500	1.200	kg CO2e/kg H2	A	W4-S63;W4-S64;W4-S65	India green hydrogen standard: 2 kg CO2e/kg well-to-gate
ci_h2_grid	Carbon intensity, grid-electrolysis hydrogen (India grid 2025)	39.000	43.000	35.000	kg CO2e/kg H2	B	W4-S59	0.71 t/MWh x 50-60 kWh/kg
ci_h2_grey	Carbon intensity, grey (SMR) hydrogen	12.000	14.000	10.000	kg CO2e/kg H2	A	W4-S64	
grid_now	Grid emission factor at the meter, 2025	0.860	0.860	0.860	kg CO2/kWh	A	W4-S59	CEA v21 weighted average 0.710 t/MWh (FY2024-25) adjusted for about 17 percent T&D losses
grid_2030	Grid emission factor at the meter, 2030	0.580	0.520	0.660	kg CO2/kWh	A	W4-S60	CEA NEP path 0.548 (2026-27) to 0.430 (2031-32) net, adjusted for T&D losses; lower grid factor is downside for the retrofit
bev_mfg	Manufacturing emissions, compact BEV incl. 27 kWh battery	7.000	6.000	8.500	t CO2e	B	W4-S63;W4-S66	Glider 4-6 t plus 27 kWh x 70 kg CO2e/kWh
bev_life_km	Lifetime km over which BEV manufacturing is amortised	150,000	180,000	120,000	km	C		
kit_mfg	Manufacturing emissions of the retrofit kit	0.500	0.700	0.400	t CO2e	B		Carbon-fibre cylinder dominant

nox_bs6	BS-VI NOx limit, M1 positive ignition	60.000	60.000	60.000	mg/km	A	W4-S71	
Market sizing								
cng_stock	CNG vehicles on road in India, all categories (March 2025)	8,195,000	7,500,000	8,800,000	vehicles	A	W3-S2	81.95 lakh (PNGRB citing Vahan, Aug 2025); 72.10 lakh in Jul 2024 (Lok Sabha reply); Round 1 said 7.5 million
cng_pv_share	Share of passenger cars in the CNG stock	0.550	0.450	0.600	fraction	B	W3-S10;W3-S12;W3-S15	No official split; estimated from cumulative factory-CNG car sales (Maruti about 2.6 million FY20-FY26 plus others) and aftermarket conversions; Round 1 said 65 percent
eligible_share	Share of CNG cars that are factory-fitted, boot-cylinder layout, under 6 years old at 2028	0.500	0.450	0.550	fraction	B	W3-S10;W3-S12	Factory CNG sales FY23-FY26 versus total CNG cars
fleet_share	Share of eligible cars in commercial or institutional fleets	0.200	0.150	0.250	fraction	C	W3-S18;W3-S25	Delhi alone registers about 60,000-85,000 taxis; no national fleet split published
hub_share	Share of fleet cars in cities with a hydrogen hub by 2030	0.400	0.300	0.500	fraction	C	W3-S36;W3-S37	MNRE corridors: Delhi-NCR (Sahibabad-Faridabad-Delhi), Ahmedabad-Vadodara-Surat, Pune-Mumbai, Kochi and others
adopter_share	Plausible early-adopter share (return-to-base fleets)	0.050	0.030	0.080	fraction	C		Hypothesis; needs customer validation
hub_stations_2030	Fleet-hub stations serving cars by 2030	8.000	4.000	15.000	stations	D	W3-S37	16 MNRE pilot stations sanctioned for buses and trucks; assumes half add light-duty dispensing, conditional on gates

A2.2 Source register (309 entries; access date 14 September 2026)

Prefix W1 engineering, W2 safety and regulation, W3 market, infrastructure and benchmarks, W4 economics and emissions. 'Excerpt' or 'summary' in Limitations means the original could not be opened through the build environment's network policy and the entry rests on indexed text of the page; re-verify before external use. Additional gap-closure sources read in full through the search index: PIB PRID 2006052 (14 Feb 2024), PIB PRID 2107795 (3 Mar 2025), Scribd copy of AIS-157 scope, Fuel Cells Works and RushLane on the Suzuki-AVL hydrogen Swift (May 2026), Luxfer hydrogen product page, Hexagon Purus Type 4 table (PDF read).

ID	Claim supported	Publisher	Title	URL	Date	Section	Status	Limitations
W1-S1	Hexagon Type IV cylinder dimensions/mass/H2 capacity (350 & 700 bar)	Hexagon Purus	Hydrogen Type 4 cylinder information	https://s3.eu-central-1.amazonaws.com/hexagonpurus-website/HPU_0222_12_HydrogenType4_Tabelle_2pages.pdf	REV. 08/22	p.2 table	spec	"frequently built units; minimum quantities may apply"
W1-S2	Luxfer Type III 350 bar 60/74/205 L cylinders	Luxfer Gas Cylinders	G-Stor Pro H2 Type 3 (EU) spec table	https://www.luxfercylinders.com/product/g-stor-h2-hydrogen-cylinders/	2023–25 (page)	EU table	spec	Firecrawl excerpt; Type III not Type IV
W1-S3a	Tour S petrol power/torque, FE, tanks, GC	Maruti Suzuki	Dzire Tour S page / Maruti Suzuki Commercial page	https://www.marutisuzuki.com/tour/dzire ; https://www.marutisuzukicommercial.com/dzire-tour-s	2025–26	spec block	spec	excerpt; CNG power not in excerpt
W1-S3b	Tour S S-CNG 32.12 km/kg	Maruti Suzuki	Media release: Maruti Suzuki launches the All-new Tour S	https://www.marutisuzuki.com/content/dam/arena-eds/corporate/press-release/assets/documents/2023/february/Media_Release_Maruti_Suzuki_launches_the_All-new_Tour_S.pdf	Feb 2023	body	spec (declared)	excerpt
W1-S3c	Dzire 4th gen kerb/GVW/FE/boot/tanks	Maruti Suzuki	Arena Dzire brochure; Dzire price page	https://www.marutisuzuki.com/content/dam/msil/arena/in/en/assets/cars/dzire/documents/Arena-Dzire-Brochure.pdf ; https://www.marutisuzuki.com/arena/dzire/price	2024–26	spec table	spec	excerpt
W1-S3d	WagonR kerb 850, GVW 1340, tank 32, boot 335; engines K10C/K12N	Maruti Suzuki	WagonR price page; WagonR page	https://www.marutisuzuki.com/arena/wagon-r/price ; https://www.marutisuzuki.com/arena/wagon-r	2025–26	spec table	spec	excerpt; variant for kerb/GVW not stated
W1-S3d2	WagonR Tour H3 25.22 km/l, 34.41 km/kg	Maruti Suzuki	WagonR Tour H3 page	https://www.marutisuzuki.com/tour/wagonr	2025–26	FAQ	spec	excerpt
W1-S3e	Swift 1197 cc 60 kW @5700	Maruti Suzuki	Swift page	https://www.marutisuzuki.com/arena/swift	2025–26	spec	spec	excerpt
W1-S3f	Ertiga 1462 cc, 75.8 kW @6000, 139 Nm @4300	Maruti Suzuki	Ertiga page	https://www.marutisuzuki.com/arena/ertiga	2025–26	spec	spec	excerpt
W1-S3g	1.5 L K-series S-CNG 64.6 kW @5500, 121.5 Nm	Maruti Suzuki	Press release: All New Brezza S-CNG	https://www.marutisuzuki.com/corporate/media/press-releases/2023/march/maruti-suzuki-introduces-all-new-brezza-s-cng	Mar 2023	body	spec	same engine as Ertiga S-CNG; excerpt
W1-S3h	Ertiga CNG 121.5 Nm @4200	Maruti Suzuki	Arena Ertiga brochure	https://www.marutisuzuki.com/content/dam/msil/arena/in/en/assets/cars/ertiga/document/Arena-Ertiga-Brochure.pdf	2024	spec	spec	excerpt
W1-S3i	Ertiga kerb 1150–1205/1250–1255, GVW 1760/1820, tanks 45/60	Maruti Suzuki	Ertiga price page; 6-page brochure	https://www.marutisuzuki.com/arena/ertiga/price ; https://www.marutisuzuki.com/content/dam/msil/arena/in/en/assets/cars/ertiga/document/ErtigaBrochure_6_Pgs_High_Res.pdf	2024–26	spec	spec	excerpt
W1-S3j	Ertiga 20.51 km/l, 26.11 km/kg	Maruti Suzuki	Press release Next-Gen Ertiga	https://www.marutisuzuki.com/corporate/media/press-releases/2022/april/inspire-togetherness-again-maruti-suzuki-next-gen-ertiga-arrives-at-arena-showrooms	Apr 2022	body	spec	excerpt
W1-S3k	Ertiga Tour M 21.10 km/l, 26.54 km/kg, 45 L	Maruti Suzuki	Ertiga Tour M page	https://www.marutisuzuki.com/tour/ertiga	2025–26	FAQ	spec	excerpt

W1-S3l	Eeco Tour V 60.5 kW @6000, 105.5 Nm @3000, 20.20 km/l	Maruti Suzuki	Eeco Tour V page	https://www.marutisuzuki.com/tour/eeco	2025–26	spec	spec	excerpt
W1-S3m	Eeco S-CNG 27.05 km/kg, 104.4 Nm @3000	Maruti Suzuki	Press release New Eeco	https://www.marutisuzuki.com/corporate/media/press-releases/2022/november/maruti-suzuki-new-eeco-more-power-more-fuel-efficient-and-more-style	Nov 2022	body	spec	excerpt
W1-S3n	Eeco kerb 970, 19.71 km/l, price from ₹5,20,900	Maruti Suzuki	Eeco price page	https://www.marutisuzuki.com/arena/eeco/price	2025–26	spec	spec	excerpt
W1-S3o	Eeco CNG kerb 1085, 26.78 km/kg	Maruti Suzuki	Arena Eeco brochure	https://www.marutisuzuki.com/content/dam/msil/arena/in/en/assets/cars/eeco/document/Arena-Eeco-Brochure.pdf	2024–25	spec	spec	excerpt
W1-S3p	Eeco S-CNG 65 L tank	Maruti Suzuki	Arena Eeco page	https://www.marutisuzuki.com/arena/eeco	2025–26	body	spec	excerpt
W1-S3q	Swift 24.8/25.75 km/l	Maruti Suzuki	Swift page	https://www.marutisuzuki.com/arena/swift	2025–26	spec	spec	excerpt
W1-S3r	Swift S-CNG 32.85 km/kg	Maruti Suzuki	Press release Epic New Swift S-CNG; Swift S-CNG brochure	https://www.marutisuzuki.com/corporate/media/press-releases/2024/september/epic-new-swift-s-cng ; https://www.marutisuzuki.com/content/dam/msil/arena/in/en/assets/cars/swift/document/Swift_SCNG_VerticalBrochure.pdf	Sep 2024	body	spec	excerpt; torque truncated
W1-S3s	Swift kerb 920/990, GVW 1355/1425, tanks 37/55	Maruti Suzuki	Swift price page	https://www.marutisuzuki.com/arena/swift/price	2025–26	spec	spec	excerpt
W1-S4	Aura CNG 50.5 kW, 95.2 Nm, tanks 37/65, dims, price	Hyundai Motor India	Aura specification/performance/price pages	https://www.hyundai.com/in/en/find-a-car/aura/specification ; https://www.hyundai.com/in/en/find-a-car/aura/performance ; https://www.hyundai.com/in/en/find-a-car/aura/price	2026	spec table	spec	excerpt; kerb weight absent
W1-S5	Tigor iCNG specs, ARAI mileage, tanks, price	CarDekho (third party)	Tata Tigor specifications	https://www.cardekho.com/tata/tigor/specs	2025–26	spec table	spec (third party)	not OEM page
W1-S5b	Twin-cylinder frees 200–205 L boot	carandbike (third party)	Tata Tigor page	https://www.carandbike.com/tata-cars/tigor	2025–26	body	claimed	third party
W1-S6	Tour S CNG 69 bhp/101.8 Nm, 34.3 km/kg, kerb 1045, boot 382, GC 163	CarDekho / ZigWheels (third party)	Dzire Tour S CNG overview; Tour S specs	https://www.cardekho.com/overview/Maruti_Dzire_Tour_S/ Maruti_Dzire_Tour_S_CNG.htm ; https://www.zigwheels.com/maruti-suzuki-cars/dzire-tour-s/specifications/	2026	spec	spec (third party)	search-summary
W1-S7	2025–26 ex-showroom prices (Tour S CNG ₹7.10 L; WagonR CNG ₹5.79–7.62 L; Ertiga CNG ~₹9.5 L; Eeco AC CNG ₹6.46 L; Swift VXi CNG ₹7.57 L)	CarDekho/CarWale/ ZigWheels (aggregators)	various price pages	https://www.cardekho.com/overview/Maruti_Dzire_Tour_S/ Maruti_Dzire_Tour_S_CNG.htm ; https://www.carwale.com/maruti-suzuki-cars/wagon-r/cng/ ; https://www.carwale.com/maruti-suzuki-cars/ertiga/cng/ ; https://www.cardekho.com/overview/Maruti_Eeco/Maruti_Eeco_5_Seater_AC_CNG.htm ; https://www.cardekho.com/overview/Maruti_Swift/Maruti_Swift_VXi_CNG.htm	Sep 2026 snapshot	price block	third-party	search-summary; unverified
W1-S7b	WagonR CNG 56 bhp/82.1 Nm, 60 L, 33.47 km/kg	CarWale (third party)	WagonR CNG page	https://www.carwale.com/maruti-suzuki-cars/wagon-r/cng/	2026	body	spec (third party)	search-summary
W1-S8a	Dzire CNG real-world 28–30 km/kg city	Cars24 (blog)	Maruti Suzuki Dzire Real World Mileage	https://www.cars24.com/article/maruti-suzuki-dzire-real-world-mileage/	2025–26	body	anecdotal	not a fleet study
W1-S8b	WagonR CNG real 24–28 km/kg; 3-yr hydrotest	VahanBazaar (blog)	CNG Mileage Real-World India	https://vahanbazaar.in/tips/cng-mileage-real-world-india-vahanbazaar	2026	body	anecdotal	low confidence
W1-S9a	H2 ≈30 % of mixture volume; ~20 % power loss at stoich PFI	Int. J. Hydrogen Energy (Elsevier)	Hydrogen-fuelled ICEs: Direct Injection versus Port-Fuel Injection	https://www.sciencedirect.com/science/article/pii/S0360319924028106	2024	intro/results	observed/ reviewed	excerpt + search-summary; full text blocked
W1-S9b	λ=2 torque ~34.9–40.7 %; BTE 40.4 % (DI e-turbo); NOx <10 ppm above λ 2.7; BTE ~38 % lean	multiple (ACS Omega 2024; IJHE 2025; Springer IJAT 2026)	see URLs	https://pubs.acs.org/doi/10.1021/acsomega.4c02222 ; https://www.sciencedirect.com/science/article/pii/S036031992501715X ; https://link.springer.com/article/10.1007/s12239-026-00469-z	2024–26	abstracts	observed	search-summary; attribution to individual papers not verified
W1-S9c	Supercharging + EGR power recovery	Int. J. Hydrogen Energy	Increasing the power output of hydrogen ICEs by means of supercharging and EGR (Verhelst et al.)	https://www.sciencedirect.com/science/article/abs/pii/S036031990900442X	2009	title/abstract	observed	content blocked
W1-S9d	">30 % of gasoline" power with supercharged stoich + EGR	LAPSE/PSE Community (review preprint)	Progress of Performance, Emission, and Technical Measures of H2 ICEs	https://psecommunity.org/wp-content/plugins/wporg/includes/file/2302/LAPSE-2023.9491-1v1.pdf	2023	excerpt	reviewed	wording ambiguous
W1-S9e	Backfire mitigation by injection control (review exists)	Int. J. Hydrogen Energy	Hydrogen port fuel injection: Review of fuel injection control strategies to mitigate backfire	https://www.sciencedirect.com/science/article/abs/pii/S036031992401348X	2024	title	reviewed	content blocked
W1-S9f	Timed manifold injection overcomes backfire (IIT Delhi)	Int. J. Hydrogen Energy	Performance characteristics of a hydrogen fuelled S.I. engine using timed manifold injection (Das) ; Hydrogen engine: R&D programmes in IIT Delhi	https://www.sciencedirect.com/science/article/abs/pii/S036031999190038K ; https://www.sciencedirect.com/science/article/abs/pii/S0360319991001781	1991; 2002	abstracts	observed	search-summary
W1-S9g	Knock at spark >26° CA; smaller advance than gasoline	OSTI ETDE record (Indian study)	Implications of combustion parameters on the ...	https://www.osti.gov/etdeweb/servlets/purl/21400931	n.d.	abstract	observed	search-summary; blocked
W1-S9h	NOx peak at λ=1.3, ~2.7× λ=1	KAUST repository (Onorati et al., Int. J. Engine Res.)	The potential of hydrogen ICEs for heavy-duty ... / role of H2 for future ICEs	https://repository.kaust.edu.sa/bitstreams/e16f5716-15c8-4c8f-9951-bab74c8db2c9/download	2022	excerpt	reviewed	excerpt only
W1-S10a	Bosch PFI for H2 engines product page	Bosch Mobility	Port fuel injection for hydrogen engines	https://www.bosch-mobility.com/en/solutions/powertrain/gas/port-fuel-injection-for-hydrogen-engines/	2024–25	page	announced	blocked; title via search
W1-S10b	PHINIA low-pressure H2 PFI injectors; DI-CHG10/15 ratings	PHINIA; MarkLines	Hydrogen Fuel Injectors; IAA 2024 note	https://www.phinia.com/technologies/technologies/hydrogen-solutions/fuel-injectors ; https://www.marklines.com/en/news/314084	2024	page	announced	search-summary
W1-S10c	Bosch DI injector 17 g/s @40 bar, 2026;	Power Progress;	Fuel injection must follow the hydrogen;	https://www.powerprogress.com/news/fuel-injection-must-	2024–25	body	announced	search-summary

	Liebherr H964/H966; ~60 bar limit	Mobility Engineering Tech	Liebherr Q&A	follow-the-hydrogen/8036089.article ; https://www.mobilityengineeringtech.com/component/content/article/tohe/pub/regulars/q-a/47237				
W1-S10d	OMB Saleri HPR 350 two-stage regulator; Bosch–OMB agreement; H2 valve catalogue	OMB Saleri; Green Car Congress	Hydrogen valves; catalogue 2023; Bosch partners with OMB Saleri	https://www.omb-saleri.it/products/hydrogen/ ; https://www.omb-saleri.it/uploads/categories/catalogo2023h2.pdf ; https://www.greencarcongress.com/2021/11/20211118-bosch.html	2021–23	page/ catalogue	production	search-summary
W1-S10e	Lexier 350 bar HPR and OTV	Lexier	Pressure Regulator 350 bar; On Tank Valve 350 bar	https://lexierweb.com/en/hydrogen/valves/pressure-regulators/hpr-350bar/ ; https://lexierweb.com/en/hydrogen/vehicles-components/on-tank-valve-350bar/	2025	page	production	search-summary
W1-S10f	Poppe+Potthoff TOPAQ OTV 2.0, R134/HGV 3.1 tests, TPRD variants	Poppe + Potthoff; Hydrogen Central	On Tank Valve; certification news	https://www.poppe-potthoff.com/hydrogen/on-tank-valve/ ; https://hydrogen-central.com/hydrogen-tank-valves-by-poppe-potthoff-successfully-complete-required-as-part-of-the-certification-according-to-r134/	Sep 2024	page	production-intent	search-summary
W1-S10g	Sensirion STC42A automotive H2 sensor specs	Sensirion	STC42A product page / press	https://sensirion.com/products/catalog/STC42A ; https://sensirion.com/company/news/press-releases-and-news/article/sensirion-announces-global-availability-of-the-stc42a-automotive-h2-sensor-for-thermal-runaway-detection	2024–25	datasheet summary	production	search-summary
W1-S10h	Bosch H2 leak-detection sensors	Bosch Mobility	Sensors for fuel cell-electric systems	https://www.bosch-mobility.com/en/solutions/powertrain/fuel-cell-electric/sensors-for-fuel-cell-electric-systems/	2025	page	production	search-summary
W1-S10i	NTM SenseH2	NTM Sensors / AzoSensors	SenseH2 Hydrogen Sensor	https://www.azosensors.com/equipment-details.aspx?EquipID=339	n.d.	page	production	search-summary
W1-S11a	ISO 11439: ≤2 % H2 for steel UTS >950 MPa; 0.1 % wet gas	iso11439.com FAQ (secondary)	CNG Cylinder standard ISO 11439 – FAQ	http://www.iso11439.com/faq.php	n.d.	FAQ	standard text (secondary)	standard itself not opened
W1-S11b	MoRTH allowed 18 % H-CNG in CNG engines; CMVR amendment	Autocar Professional	MoRTH allows use of H-CNG in CNG engines	https://www.autocarpro.in/news-national/morth-allows-use-of-hcng-in-cng-engines-67299	Sep 2020	body	policy	search-summary
W1-S11c	ARAI: 18 % HCNG –70 % CO, –25 % HC, +4–5 % fuel economy (HD BS-IV)	IAS Gyan (secondary, quoting PIB)	Notification allowing use of H-CNG issued	https://www.iasgyan.in/daily-current-affairs/notification-allowing-use-of-h-cng-issued	2020	body	reported	secondary; PIB blocked
W1-S11d	IOCL reformer + 50 DTC buses, Rajghat Depot-1, Oct 2020	IndianOil; pv magazine India	IndianOil press release 58744; Delhi HCNG trial	https://iocl.com/NewsDetails/58744 ; https://www.pv-magazine-india.com/2020/10/21/pollution-hit-delhi-starts-hydrogen-enhanced-cng-trial-for-buses/	Oct 2020	body	announced	IOCL page blocked
W1-S11e	SIAM-led trial: 18 % chosen; NMHC –58 %, THC –23 %, NOx –50 %	NPC / DOE (topic paper)	NPC Future Transportation Fuels Study – Topic paper 25 HCNG	https://www.npc.org/FTF_Topic_papers/25-HCNG.pdf ; https://www.energy.gov/sites/default/files/2022-10/25-HCNG.pdf	2012 (data ~2010)	body	reported	blocked; search-summary
W1-S12a	BMW Hydrogen 7: 191 kW, 8 kg LH2, ~201 km, 3.7 / 2.1 kg/100 km	Wikipedia summarising Wallner et al., IJHE 2008; BMW press	BMW Hydrogen 7; Fuel economy and emissions evaluation of BMW Hydrogen 7 Mono-Fuel demonstration vehicles	https://en.wikipedia.org/wiki/BMW_Hydrogen_7 ; https://www.sciencedirect.com/science/article/abs/pii/S0360319908011439 ; https://www.press.bmwgroup.com/global/article/detail/T0012215EN/	2006–08	body	observed (lab cycle)	primary blocked; summary
W1-S12b	Mazda RX-8 Hydrogen RE: 110 L @350 bar, 2.4 kg, ~107 hp, ~100 km	Wikipedia; Green Car Journal	Mazda RX-8 Hydrogen RE	https://en.wikipedia.org/wiki/Mazda_RX-8_Hydrogen_RE ; https://greencarjournal.com/dont-miss/mazdas-hydrogen-rx-8-re/	2006	body	claimed	secondary
W1-S12c	Ford Focus H2RV: 2.3 L SC H2 ICE, 110 hp + 33 hp, 45 mi/kg, 125 mi	EE Power; AutoIntell	Ford Unveils Hydrogen ICE; H2ICE Ford Focus	https://eepower.com/news/ford-unveils-hydrogen-internal-combustion-engine/ ; https://www.autointell.com/News-2003/September-2003/September-2003-4/September-24-03-p4.htm	2003	body	claimed	secondary
W1-S12c2	Ford E-450 H2ICE shuttle: 6.8 L V10 SC, 350 bar 26 GGE tank, ~150 mi	Green Car Congress; TRID	Ford H2ICE Shuttle Bus; Engineering the Ford H2 IC Engine Powered E-450	https://www.greencarcongress.com/2004/09/ford_hsub2subic.html ; https://trid.trb.org/view/1813766	2004–06	body	claimed	hp not confirmed
W1-S12d	Toyota Hydrogen HiAce: 3.4 L V6, 120 kW, 354 Nm, SCR/Euro VI, 7 vehicles, Nov 2023	Toyota Australia (via RACQ/CarExpert/WhichCar)	Toyota unveils Hydrogen HiAce; reviews	https://www.toyota.com.au/discover/news/toyota-unveils-hydrogen-powertrain-technology-for-pilot-program-in-australia ; https://www.racq.com.au/articles/drive/2023/11/toyota-tests-hydrogen-fuelled-v6-powered-hiace ; https://www.whichcar.com.au/news/hydrogen-powered-toyota-hiace-revealed-with-200km-range	Nov 2023	body	announced	Toyota pages blocked
W1-S12e	HiAce ~6.2 kg, 141 L, <200 km, >70 L/100 km; CPB month-long pilot	Drive.com.au; CarExpert; CPB Contractors	2024 Toyota HiAce hydrogen prototype review; CPB pilot news	https://www.drive.com.au/reviews/2024-toyota-hiace-hydrogen-prototype-quick-drive/ ; https://www.carexpert.com.au/car-reviews/2024-toyota-hydrogen-hiace-prototype-review-quick-drive ; https://www.cpbcon.com.au/news-and-media/2024/cpb-contractors-completes-australian-first-toyota-hydrogen-fuelled-vehicle-pilot	2024	body	claimed/ derived	consumption derived by me
W1-S12f	GR Corolla H2 liquid hydrogen; range doubled; tank >220 L; 2026 superconducting pump	Toyota Global Newsroom / Toyota Europe	Super Taikyu releases	https://global.toyota/en/newsroom/corporate/43550115.html ; https://newsroom.toyota.eu/evolved-liquid-hydrogen-powered-gr-corolla-to-participate-in-super-taikyu-fuji-24-hours-race/	2023–26	body	announced	no kg/100 km
W1-S12g	HySE formation and member roles	Honda Global	Ministerial approval granted to form HySE	https://global.honda/en/newsroom/news/2023/c230517eng.html	17 May 2023	body	announced	—
W1-S12h	Ashok Leyland/RIL H2-ICE truck; >20 trucks ~250,000 km	Business Standard; Hydrogen Central	Indian CV majors drive hydrogen transition; AL & RIL unveil	https://www.business-standard.com/industry/auto/indian-cv-majors-drive-hydrogen-transition-with-h2-vehicles-125061600725_1.html ; https://hydrogen-central.com/ashok-leyland-ril-unveil-indias-first-hydrogen-powered-heavy-duty-	2023–25	body	announced	press

				truck/ https://home.iitd.ac.in/show.php?id=16&in_sections=Press	n.d. (2023–24)	press	announced	—
W1-S12i	IIT Delhi H2 SI genset with KOEL and IOCL R&D	IIT Delhi	Technology for Hydrogen Utilization in SI Engine Generator	https://capacity4dev.europa.eu/discussions/worlds-first-hydrogen-powered-three-wheelers-launched-new-delhi_en	2012	body	announced	old
W1-S12j	Delhi hydrogen 3-wheeler fleet (15 vehicles)	capacity4dev (EU)	World's first hydrogen-powered three-wheelers launched in New Delhi	https://www.mdpi.com/1996-1073/12/4/645	2019	\$2	modelled (EOS)	excerpt
W1-S13a	24.0 g/L (35 MPa) and 40.2 g/L (70 MPa) at 15 °C	MDPI Energies	Effect of Hydrogen Refueling Parameters on Final State of Charge	https://demaco-cryogenics.com/blog/energy-density-of-hydrogen/	2010s–2020s	body	modelled	vendor blog
W1-S13b	39.75 kg/m³ at 700 barg, 20 °C	Demaco	The energy density of hydrogen	https://image-pubs.uspto.gov/dirsearch-public/print/downloadPdf/8927171	2010s–2020s	descriptions	design practice	patents, not standards
W1-S13c	Regulator to ≤20 bar; ~5 % unusable; 20 bar min	USPTO patents (Hyundai/Toyota-type)	e.g. US 8927171; US 11394041; US 10868315	https://webstore.ansi.org/standards/sae/sae26012020 ; https://efiling.energy.ca.gov/GetDocument.aspx?tn=66608	2020; 2014	abstract	standard	full tables not accessible
W1-S13d	J2601 scope ≥49.7 L; 85 °C; T40 3-min 95–100 % SOC; T20/T30 slower	SAE / ANSI store; CEC	SAE J2601-2020 abstract; SAE J2601 status presentation	https://www.osti.gov/servlets/purl/1389635 ; https://ataway.com/en/sae-j2601-understanding-the-international-standard-for-gaseous-hydrogen-refueling/	2017; 2025	body	reviewed	search-summary
W1-S13e	Non-communication fills need tight pre-cooling tolerance; 85 °C liner limit	OSTI (NREL); Ataway	Impact of hydrogen SAE J2601 fueling methods; Understanding SAE J2601	https://www.business-standard.com/markets/capital-market-news/time-technoplast-gets-final-approval-from-peso-for-manufacturing-composite-cylinders-124062100121_1.html ; https://themachinemaker.com/news/time-technoplast-receives-peso-approval-to-make-hydrogen-cylinders/	Mar–Jun 2024; May 2025	body	announced	press
W1-S13f	Time Technoplast PESO approvals (250 L Type IV H2; BV 150 L)	Business Standard; Machine Maker; Upstox	PESO approval news	https://ekcuae.com/3d-flip-book/type-4-composite-cylinder/	n.d.	page	announced	blocked
W1-S13g	Everest Kanto Type-4 composite cylinders	Everest Kanto (via search)	Type IV cylinders	https://encyclopedia.pub/entry/47333	2023	\$3	spec (secondary)	Toyota guide not opened
W1-S13h	Mirai 2021: 5.6 kg; 52/25.3/64.9 L = 142.2 L	Encyclopedia.pub (citing Toyota 2022 Mirai product guide)	Application of Hydrogen Storage Tanks	https://encyclopedia.pub/entry/47333 ; https://insideevs.com/news/755466/2026-hyundai-nexo-hydrogen-fuel-cell-official-specs/ ; https://www.hyundainews.com/assets/documents/original/52348-2023NEXOProductSpecifications20220720.pdf	2018–2025	spec	spec	secondary
W1-S13i	Nexo 156.6 L / 6.33 kg; new Nexo 162.2 L / 6.69 kg	Encyclopedia.pub; InsideEVs; Hyundai News	see URLs	https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/19008_onboard_storage_cost_performance_status.pdf	25 Nov 2019	p.1–2	modelled (DFMA)	blocked; search-summary
W1-S13j	DOE Record 19008 costs/capacities	US DOE HFTO (Strategic Analysis)	Onboard Type IV Compressed Hydrogen Storage System – Cost and Performance Status 2019	https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/24006-onboard-storage-cost-performance-status.pdf	28 Apr 2025	p.1–2	modelled (DFMA)	blocked; search-summary
W1-S13k	DOE Record 24006 costs	US DOE HFTO (Strategic Analysis)	Onboard Type IV Compressed Hydrogen Storage System – Cost and Performance Status	https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/13010_onboard_storage_performance_cost.pdf ; https://www.osti.gov/servlets/purl/1082754	2013	—	modelled	low confidence
W1-S13l	~\$33/kWh low-volume (2013 basis)	US DOE Record 13010 / SA 2013 final report	Onboard storage performance and cost	https://www.researchgate.net/publication/374204598 ; https://unece.org/sites/default/files/2023-07/ECE-TRANS-180-Add.13-Amend1e.pdf	2023	body	standard	search-summary
W1-S13m	15-yr life / 11,000 cycles (GTR 13, GB/T 42612)	ResearchGate (comparative analysis paper); UNECE GTR 13	A comparative analysis of the RCS for on-board high-pressure hydrogen storage cylinders; GTR 13 Amend.1	https://hmr.araiindia.com/api/AISFiles/AIS-195A_dc4d3241-c4b7-466f-8c37-6bd23851f0cc.pdf	2023–24	clause on containers	standard	blocked; search-summary
W1-S13n	AIS-195A: GCR 2016; PESO may use IS 16735, ISO 19881, R134, GTR 13	ARAI (HMR)	AIS-195A Safety and Procedural Requirements for hydrogen vehicles	https://peso.gov.in/web/en/gas-cylinders-amendment-rules-2024-hydrogen	2024	page	regulation	blocked
W1-S13o	PESO Gas Cylinders (Amendment) Rules 2024 (Hydrogen)	PESO	Gas Cylinders (Amendment) Rules, 2024 (Hydrogen)	https://gailcgd.gail.co.in/CGD/Entry/CNGfaq	n.d.	FAQ	spec (typical)	excerpt
W1-S14a	50 L CNG cylinder: 48 kg, 835 mm, 316 mm	GAIL Gas (CGD)	CNG FAQ	https://www.gascylindersupplier.com/diameter-325-mm-60l-cng-cylinder	n.d.	page	spec	non-Indian vendor
W1-S14b	60 L Type 1: OD 325 mm, 61.4 kg, ISO 11439, 200 bar	Anhui Clean Energy (vendor)	Diameter 325 mm 60L CNG cylinder	https://pwent.eu/en/products/products/gas-cylinder-products-high-pressure-low-pressure/cng-gas-cylinder/steel-60-liter-cng-gas-cylinder-200-bar/	n.d.	page	spec	vendor
W1-S14c	60 L Type 1: 69 kg	PWENT (vendor)	Steel 60 Liter CNG Gas Cylinder 200 Bar	https://www.unece.org/fileadmin/DAM/trans/main/wp29/wp29regs/2015/R110r3e.pdf	2015	container installation clause	regulation	excerpt (verbatim)
W1-S14d	R110: 20 g / 8 g for M1, N1 (10/5 M2,N2; 6.6/5 M3,N3)	UNECE	Regulation No. 110 Rev.3	https://hmr.araiindia.com/Control/AIS/825201593205AM4_AIS-024_and_AIS-028_CNG-Ver4_F.pdf	2015	—	standard	blocked
W1-S14e	AIS-024 / AIS-028 Ver.4 CNG code of practice	ARAI (HMR) / MoRTH	AIS-024 (Version 4) AIS-028 (Version 4)	https://morth.nic.in/sites/default/files/notifications_document/GSR%20585(E)%2025%20September%202020%20hydrogen%20CNG.pdf ; https://www.pib.gov.in/PressReleasePage.aspx?PRID=1659706	25-09-2020	Rule 115 insertion	In force	Blocked; wording from search summary
W2-S1	H-CNG added to CMVR Rule 115; IS 17314:2019	MoRTH (Gazette); PIB	G.S.R. 585(E) "hydrogen CNG"; PIB "Notification allowing use of H-CNG issued"	(BIS catalogue; not accessed)	2019	—	In force	18 % figure from press, not standard text
W2-S2	IS 17314:2019 H-CNG spec, 18 % H2	BIS (via PIB/press)	IS 17314:2019 Hydrogen enriched CNG for automotive purposes	https://indiankanoon.org/doc/10585089/ ; https://indiankanoon.org/doc/67316961/	—	Rule 115 (sub-rules not numbered here)	In force	Summary only
W2-S3	Rule 115 CNG kit provisions; s.52 14-day endorsement	Indian Kanoon (CMVR text)	Section 115 / 115B, CMVR 1989					

W2-S4	ARAI retrofitment certification (Rule 115, Rule 126 agencies)	ARAI	Vehicle Retrofitment Certification in India	https://www.araiindia.com/certification/certification/retrofitment	—	—	Current	Blocked
W2-S5	Retrofit framework (CNG/LPG/EV; AIS-123)	EVreporter (ARAI compiled)	Regulatory Framework for Retrofitting in India	https://evreporter.com/regulatory-framework-for-retrofitting-in-india/	—	—	—	Secondary
W2-S6	BS-VI CNG/LPG retrofit permitted	MoRTH; PIB; Business Standard	G.S.R. 625(E) "retrofitment in BS VI"	https://morth.gov.in/sites/default/files/notifications_document/7GSR%20625(E)%2011th%20August%202022%20retrofitment%20in%20BS%20VI.pdf ; https://pib.gov.in/PressReleasePage.aspx?PRID=1853835	11-08-2022	—	In force	Blocked
W2-S7	AIS-024/028 Version 4 exist	ARAI (HMR)	AIS-024 (Version 4) / AIS-028 (Version 4)	https://hmr.araiindia.com/Control/AIS/825201593205AM4_AIS-024_and_AIS-028_CNG-Ver4_F.pdf	—	—	Published	Blocked; title only
W2-S8	TA cost Rs 40–50 lakh, 3-yr validity	Autocar Professional	Delays in MoRTH guidelines for BS VI CNG kits hurting retrofit business	https://www.autocarpro.in/news-national/delays-in-morth-guidelines-for-bs-vi-cng-kits-hurting-retrofit-business-92660	2022 (approx.)	—	—	Industry claim
W2-S9	Rule 126 provisos (retrofit self-certification; imported CBUs for testing/research)	Commercial Law Publishers (CMVR e-book)	The Central Motor Vehicles Rules, 1989 — page 335 (Rule 126)	https://ebook.commerciallawpublishers.com/fa/cmvr/files/basic-html/page335.html	ed. incl. GSR 221(E) 30-03-2021	Rule 126 provisos	In force (pre-2026 text)	Verbatim via Firecrawl; may pre-date 2026 amendment
W2-S10	2026 amendment substituting Rule 126 provisos; self-certification	The Bar Bulletin; TeamLease RegTech	Centre Notifies Changes to Motor Vehicle Type Approval Norms...; Draft CMV (...Amendment) Rules 2025	https://thebarbulletin.com/centre-revises-motor-vehicle-type-approval-norms-self-certification-retrofitted-vehicles/ ; https://teamleaseregtech.com/updates/article/52441/draft-central-motor-vehicles-amendment-rules-2025/	2026	Rule 126	In force (per source)	GSR no./date NOT FOUND
W2-S11	AIS 157:2020 notified for M/N FCEVs; ISO 14687	MoRTH via India Code; GKToday; Mercom	9th GSR 579(E) dated 23 September 2020 hydrogen fuel cell	https://www.indiacode.nic.in/ViewFileUploaded?path=AC_CEN_30_42_00009_198859_1517807326286/notificationsindividualfile/&file=9th_gsr_579%28e%29_dated_23_september_2020_hydrogen_fuel_cell.pdf ; https://indiacode.ecourtsindia.com/rules/7ec3e5bb/ ; https://gktoday.in/current-affairs/standards-for-safety-evaluation-of-vehicles-propelled-by-hydrogen-fuel-cells ; https://www.mercomindia.com/inclusion-hydrogen-fuel-central-motor-vehicles-act	23-09-2020	new rule (see S12)	In force	Blocked; snippet
W2-S12	FCEV rule numbered 125MA; fuel IS 16061:2021; H2-ICE Rule 125M	Applus IDIADA	IRIS Highlights – July 2026	https://www.digitalsolutions.applusidiada.com/blogs/news/iris-highlights-july-2026	07-2026	—	—	Secondary digest
W2-S13	Rule 125M; AIS 195:2023; M & N	MoRTH	8th GSR 746(E) dated 16 October 2023 Hydrogen ICE	https://morth.nic.in/sites/default/files/notifications_document/8th%20GSR%20746(E)%20dated%2016th%20october%202023%20Hydrogen%20ICE.pdf	16-10-2023	Rule 125M	In force	Blocked; summary
W2-S14	GSR 720(E) 04-10-2023 also cites AIS 195:2023	(search index; MoRTH)	—	—	04-10-2023	—	Unresolved	Not verified
W2-S15	AIS-195 with Amd 1 exists; Annexure VIII LH2	ARAI (HMR); Scribd copy	AIS-195 with Amd 1	https://hmr.araiindia.com/api/AISFiles/AIS-195%20with%20Amd%201_eccd18bd-9eae-4e08-bbb2-7823b7964666.pdf ; https://www.scribd.com/document/856143651/AIS-195-cc6897f6-30be-4dc9-b188-b7c52017bebb	2023 / Amd 1 05-2025	1.0 Scope; Annex VIII	Published	Blocked; scope not captured
W2-S16	AIS-195A CEV; TPRD definitions/tests	ARAI (HMR); Scribd	AIS-195A Safety and Procedural Requirements for ... Hydrogen Powered CEV	https://hmr.araiindia.com/api/AISFiles/AIS-195A_dc4d3241-c4b7-466f-8c37-6bd23851f0cc.pdf ; https://www.scribd.com/document/1057373433/AIS-195-A	2024–25	1.0 Scope; TPRD clauses	Published	Fragments only
W2-S17	AIS-206 L-category scope; "AIS 157 and AIS 195 published for M & N"	Scribd (AIS-206 copy)	Hydrogen Vehicle Safety Standards AIS-206	https://www.scribd.com/document/902873977/AIS-206	2025	1.0 Scope	Published/ draft	Fragment
W2-S18	AIS-157A/DF (FC CEV) draft	ARAI (HMR)	Draft AIS-157A/DF June 2024	https://hmr.araiindia.com/api/AISFiles/AIS157A_CEV_DF_67e85335-2f04-40ee-ac66-ecb704d7bd46.pdf	06-2024	Scope	Draft	Blocked
W2-S19	ARAI AIS list updated 7 May 2026	ARAI (HMR)	List of published AIS (07-05-2026)	https://hmr.araiindia.com/storage/ais-details/1778147654_0_List_of_AIS_published-07_05_2026.pdf	07-05-2026	—	Current	Blocked
W2-S20	Hydrogen registration mark	MoRTH via Complinty/autopdp	CMV (Fourth Amendment) Rules 2025, G.S.R. 819(E)	https://complinty.com/legal-update/central-motor-vehicles-fourth-amendment-rules-2025-21166/	03-11-2025	—	In force	Secondary
W2-S21	7-year permit exemption incl. hydrogen	MoRTH via Autocar Pro/ChiniMandi	Electric, Ethanol, Hydrogen CVs Get 7 Year Permit Exemption	https://www.autocarpro.in/news/electric-ethanol-hydrogen-cvs-get-7-year-permit-exemption-133500	07-2026 (Gazette 6 July)	s.66(1) MV Act	In force	GSR no. not captured
W2-S22	Draft +5 yr age limit (Rule 88)	MoRTH via Business Standard/FPJ	Govt proposes 5-yr age extension for EV, CNG, hydrogen CVs	https://www.business-standard.com/industry/auto/govt-proposes-5-yr-age-extension-for-ev-cng-hydrogen-commercial-vehicles-126081800231_1.html	18-08-2026	Rule 88	Draft	Secondary
W2-S23	Draft GCR 2024 GSR 157(E) hydrogen provisions	DPIIT/PESO via TaxGuru, Lexplosion, RenewableWatch	Draft Gas Cylinders (Amendment) Rules, 2024	https://taxguru.in/corporate-law/draft-gas-cylinders-amendment-rules-2024.html ; https://peso.gov.in/web/en/gas-cylinders-amendment-rules-2024-hydrogen	05-03-2024	defs; rule 4; Sch. I	Draft (superseded by 2025 rules)	Blocked
W2-S24	GCR 2025 GSR 225(E): H2 cylinder on vehicle; 3-yr test; 15-yr life; codes	DPIIT via Lexplosion, LegalitySimplified, TeamLease, Sparrow RMS	Gas Cylinders (Amendment) Rules, 2025	https://lexplosion.in/ministry-of-commerce-and-industry-notifies-the-gas-cylinders-amendment-rules-2025/ ; https://www.legalitysimplified.com/gas-cylinders-amendment-rules-2025/ ; https://sparrowrms.in/sparrow-rms/wp-content/uploads/2025/04/GCR-amendment-2025-1.pdf	11-04-2025	rule 2, 4, 35(?), Sch. I/IV	In force	Blocked; clause numbers unverified
W2-S25	GSR 386(E), 501(E), 839(E) 2025 amendments	DPIIT via Worldtradescanner, LegalitySimplified, TeamLease	GCR Third/Fourth Amendment Rules 2025	https://worldtradescanner.com/DPIIT%20Notification%20G.S.R.%20501(E)%20dated%2028%20July,%202025.htm ; https://worldtradescanner.com/DPIIT%20Notification%20G.S.R.%20839(E)%20dated	16-06 / 28-07 / 12-11-2025	rule 4	In force	Secondary

				%2012%20November%202025.htm				
W2-S26	SMPV(U) 2025 amendment hydrogen	PESO; TaxGuru	SMPV(Unfired)(Amendment) Rules, 2025	https://peso.gov.in/web/en/smpvunfiredamendment-rules-2025 ; https://taxguru.in/corporate-law/static-mobile-pressure-vessels-unfired-amendment-rules-2025.html	eff. 05-06-2025	defs; Form AS-1	In force	Blocked
W2-S27	PESO Form-H online module for H2 dispensing stations	IndiaLaw; LegalitySimplified; VahanBazaar	PESO's New Online Module for Licensing Compressed Hydrogen Dispensing Stations	https://www.indialaw.in/blog/civil/pesos-module-hydrogen-stations-in-india/	14-05-2025	GCR Form H	Current	Secondary
W2-S28	PESO SOPs (SMPV 02-12-2024; GCR Jan 2025)	PESO	SOP for SMPV(U) Rules 2016; GCR SOP	https://peso.gov.in/web/sites/default/files/2024-12/SOP%20for%20SMPV(U)%20Rules,%202016-Final%20-%2002.12.2024.pdf ; https://www.peso.gov.in/web/sites/default/files/2025-01/GCR%20SOP.pdf	2024–25	—	Current	Blocked
W2-S29	PESO licence to IOCL Faridabad for H-CNG dispensing	Business Standard	PESO grants permission to Indian Oil to set up hydrogen plant	https://www.business-standard.com/content/b2b-chemicals/peso-grants-permission-to-indian-oil-to-set-up-hydrogen-plant-115051100485_1.html	05-2015	—	Historical	Secondary
W2-S30	Type IV H2 cylinder PESO approvals (Time Technoplast)	Business Standard; Autocar Pro; company PR	Time Technoplast gets final approval from PESO...; 250 L approval	https://www.business-standard.com/markets/capital-market-news/time-technoplast-gets-final-approval-from-peso-for-manufacturing-composite-cylinders-124062100121_1.html ; https://www.timetechnoplast.com/pdf/investors-calendar/press-release/approval-for-manufacturing-of-type-iv-composite-cylinders-for-hydrogen.pdf	03-2024 / 06-2024 / 01-2026	—	Current	Company-reported
W2-S31	Indian hydrogen standards landscape (AIS 195A, AIS 206, IS 19036, IS 16061, IS 18538-8)	MNRE with CEEW (PIB-hosted)	Green Hydrogen Standards and Approval Systems in India	https://static.pib.gov.in/WriteReadData/specificdocs/documents/2024/may/doc2024510336201.pdf ; https://ngdm.mnre.gov.in/admin/uploads/resources/171557269282523Green%20Hydrogen%20Standards%20and%20Approval%20Systems%20in%20India.pdf	05-2024	tables	Current	Blocked
W2-S32	BIS hydrogen standards list (23)	BIS	List of Indian Standards related to Hydrogen	https://www.bis.gov.in/wp-content/uploads/2023/10/List_Standards_Hydrogen.pdf	10-2023	—	Current	Blocked
W2-S33	MNRE transport pilot guidelines	MNRE; JSA; Business Standard	Scheme Guidelines for pilot projects ... green hydrogen in the transport sector	https://mnre.gov.in/en/notice/scheme-guidelines-for-implementation-of-pilot-projects-for-use-of-hydrogen-in-the-transport-sector-under-the-national-green-hydrogen-mission-nghm/	14-02-2024	objectives; steering committee	In force	Blocked
W2-S34	5 pilots, 37 vehicles, 9 HRS, 10 routes, Rs 208 cr	Lok/Rajya Sabha reply via DownToEarth, DD News, Elets	As told to Parliament (25 March 2025)	https://www.downtoearth.org.in/environment/as-told-to-parliament-march-25-2025-37-cross-country-vehicles-9-fuelling-stations-planned-under-green-hydrogen-mission ; https://ddnews.gov.in/en/india-rolls-out-pilot-projects-for-hydrogen-powered-buses-and-trucks/	25-03-2025	—	Current	Secondary
W2-S35	Mirai pilot with ICAT	Zee News; BusinessToday	Gadkari inaugurates pilot project ... Toyota Mirai	https://zeenews.india.com/electric-vehicles/nitin-gadkari-inaugurates-pilot-project-to-test-viability-of-hydrogen-vehicles-in-india-2445693.html	16-03-2022	—	Historical	Press
W2-S36	Tata H2 truck trials, 16 vehicles, 24 months	Tata Motors; News on AIR	Tata Motors Drives India's Green Future with Country's First Hydrogen Truck Trials	https://www.tatamotors.com/press-releases/tata-motors-drives-indias-green-future-with-countrys-first-hydrogen-truck-trials/ ; https://www.newsonair.gov.in/first-trials-of-hydrogen-powered-heavy-duty-trucks-flagged-off	04-03-2025	—	Current	Company/press
W2-S37	ISO 14687 grades; 2025 edition	ISO	ISO 14687:2019 / ISO 14687:2025 Hydrogen fuel quality — Product specification	https://www.iso.org/standard/69539.html ; https://www.iso.org/standard/82660.html	2019 / 2025	Table 1 grades	2025 current	Blocked; abstract
W2-S38	ISO 17268 pressure classes	ISO	ISO 17268:2020; ISO 17268-1:2025	https://www.iso.org/standard/68442.html	2020 / 2025	scope	Current	Abstract
W2-S39	SAE J2601 scope, edition	H2Tools (PNNL)	SAE J2601 Fueling Protocols for Light Duty Gaseous Hydrogen Surface Vehicles	https://h2tools.org/fuel-cell-codes-and-standards/sae-j2601-fueling-protocols-light-duty-gaseous-hydrogen-surface	4th ed. 05-2020	—	Current (revision open)	Verbatim page text captured
W2-S40	ISO 19881:2025; ISO 19880-1/-2	ISO; iTeh; H2Tools	ISO 19881:2025; ISO/AWI 19880-1; ISO 19880-2:2025	https://www.iso.org/standard/19881 ; https://www.iso.org/standard/93897.html ; https://www.iso.org/obp/ui/en/#iso:std:85839:en	2025	scope	Current	Abstracts
W2-S41	GTR 13 Amend. 1 (2023); R134 02 series	UNECE; InterRegs; GAR	ECE/TRANS/180/Add.13/Amend.1	https://unece.org/sites/default/files/2023-07/ECE-TRANS-180-Add.13-Amend1e.pdf ; https://www.interregs.com/articles/spotlight/297/updated-unece-regulations-relating-to-hydrogen-fuelled-vehicles-published	21-06-2023	—	In force	Blocked
W2-S42	GTR 13 numeric limits (4 %/8 %; leak; single failure)	Frontiers in Energy Research	Analysis of safety technical standards for hydrogen storage in fuel cell vehicles	https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2024.1376739/full	2024	—	—	Review; "118 NL/h" vs NL/min discrepancy
W2-S43	H2 properties	H2Tools (PNNL)	Basic Hydrogen Properties Chart; Flammability; Auto-Ignition	https://h2tools.org/basic-hydrogen-properties-chart ; https://h2tools.org/faq-keywords/flammability	—	—	Current	Blocked; summary
W2-S44	H2 vs other fuels (relative)	US DOE EERE	Hydrogen: Similar but Different (fact sheet)	https://www1.eere.energy.gov/hydrogenandfuelcells/pdfs/h2_safety_fsheet.pdf	—	—	Current	Blocked
W2-S45	NFPA 2 repair-garage ventilation	NFPA via UpCodes; NFPA	NFPA 2 (2023) Hydrogen Technologies Code — repair garages	https://up.codes/s/repair-garages-for-vehicles-fueled-by-lighter-than-air-motor-fuels ; https://www.nfpa.org/product/nfpa-2-hydrogen-technologies-code/p0002code	2023 ed.	Ch. 6; repair garages	Current	Secondary
W2-S46	ISO/TR 15916:2015 → ISO/TS 15916:2026	ISO	Basic considerations for the safety of hydrogen systems	https://www.iso.org/standard/56546.html ; https://www.iso.org/standard/89423.html	2015 / 2026	—	TS current	Abstract
W2-S47	UK guidance (HSG261; hydrogen hub;	HSE; GOV.UK	HSG261; Hydrogen Safety; Construction	https://www.hse.gov.uk/pubns/books/hsg261.htm ;	—	—	HSG261	Blocked

	withdrawn C&U note)		and use of hydrogen-powered vehicles [withdrawn]	https://solutions.hse.gov.uk/safe-net-zero/hydrogen-safety ; https://www.gov.uk/government/publications/construction-and-use-of-hydrogen-powered-vehicles/construction-and-use-of-hydrogen-powered-vehicles			current; GOV.UK note withdrawn	
W2-S48	Kjørbo root cause	Nel ASA; gasworld; Gexcon	Status and Q&A regarding the Kjørbo incident; Root cause ... identified	https://nelhydrogen.com/status-and-qa-regarding-the-kjorbo-incident/ ; https://www.gasworld.com/story/root-cause-of-norway-hydrogen-station-explosion-identified/	06/07-2019	—	—	Company statements
W2-S49	Santa Clara causes	Air Products (via H2Tools); Gexcon	Report on the June 2019 Hydrogen Explosion and Fire Incident in Santa Clara	https://h2tools.org/sites/default/files/2021-06/AP_Santa_Clara_Incident_Review_Report_Rev1.pdf	2021	—	—	Blocked; summary
W2-S50	Gangneung 2019; Busan 2024; HIAD	ScienceDirect (IJHE); CleanTechnica	Evaluating hydrogen risk management policy PR: Lessons learned from three hydrogen accidents in South Korea	https://www.sciencedirect.com/science/article/abs/pii/S0360319923012132 ; https://cleantechnica.com/2025/01/16/korean-hydrogen-explosions-leaks-auction-failure-impeachment-may-cause-pivot/	2023 / 2025	—	—	Abstract/press
W2-S51	Euro NCAP Mirai 2021; Nexo 2018	Euro NCAP; Hyundai	Toyota Mirai — Euro NCAP 2021 results — 5 stars; Hyundai Nexo 5-star	https://www.euroncap.com/en/results/toyota/mirai/43501 ; https://www.hyundai.com/worldwide/en/newsroom/detail/all-new-hyundai-nexo-achieves-maximum-five-star-rating-in-euro-ncap-safety-assessment-0000001089	2021 / 2018	—	—	Blocked; summary
W2-S52	DOE on NOx from hydrogen combustion	US DOE EERE	Does the use of hydrogen produce air pollutants such as nitrogen oxides?	https://www.energy.gov/eere/uelcells/does-use-hydrogen-produce-air-pollutants-such-nitrogen-oxides	—	—	Current	Requested /cmei/fuels/ URL not found; content from summary
W2-S53	IMT-25; Rs 60 TP loading	India Motor Tariff via GoDigit; IRDAI policy wordings; Acko; HDFC Ergo	India Motor Tariff (IMT) Endorsements — IMT.25	https://www.godigit.com/content/dam/godigit/directportal/en/downloads/car/Motor-IMT%20Endorsements.pdf ; https://www.hdfcergo.com/blogs/car-insurance/insurance-guide-for-cng-lpg-converted-cars-2025	—	IMT-25	Current	Blocked; summary
W2-S54	Form 22D (Rule 47B(1)); RTO documents	MoRTH (Parivahan); Policybazaar; RTOIndiaHelp	FORM 22D — Endorsement of alteration/retrofitment	https://parivahan.gov.in/sites/default/files/DownloadForm/cmvr/FORM-22D.pdf ; https://www.policybazaar.com/rto/form-22d/	—	Rule 47B(1)	Current	Blocked
W2-S55	Maruti warranty exclusion for retrofitted CNG	Maruti owner's manual (via GoMechanic/RideNRe pair)	Dzire Owner's Manual (warranty section)	https://marutistoragenew.blob.core.windows.net/mslintintvebpdf/Dzire_Owners_Manual_1.pdf	—	warranty exclusions	Current	Not verified
W2-S56	Rule 41 trade-certificate purposes	Indian Kanoon; transportinfo.in	The Central Motor Vehicles Rules, 1989 — Rule 41	https://indiankanoon.org/doc/28072254/ ; https://www.transportinfo.in/simba/cmvr/cmvr.htm	—	Rule 41	In force	Summary
W2-S57	Rule 126 test agencies	PIB; TÜV	Rule 126 CMVR	https://www.pib.gov.in/newsite/erelecontent.aspx?releid=279664 ; https://www.tuv.com/market-access-services/en/certification-filter/cmvr-(central-motor-vehicle-rules-notified-by-the-government-of-india).html	—	Rule 126	In force	Summary
W2-S58	AIS-137 (Part 9) draft Amd 1 for CNG/LPG retrofit	Scribd (regulatory update)	Regulatory Updates: CMVR Nov–Dec 2022	https://www.scribd.com/document/930900675/Regulatory-Update-Nov-Dec-2022	12-2022	—	Draft	Fragment
W2-S59	CMVR-TSC minutes (ARAI hydrogen projects)	ACMA	MoM 62nd CMVR-TSC	https://www.acma.in/uploads/docmanager/MoM_62_CMVR-TSC.pdf	—	—	—	Blocked
W2-S60	Analyst: H2-ICE retrofit not yet legalised	S&P Global Mobility	Hydrogen trucks in India: green hydrogen potential	https://www.spglobal.com/automotive-insights/en/blogs/2025/03/hydrogen-trucks-in-india-green-hydrogen-potential	03-2025	—	—	Opinion
W2-S61	CV OEM hydrogen shift	Business Standard	India's vehicles industry quietly begins shift to hydrogen-powered vehicles	https://www.business-standard.com/industry/auto/indian-cv-majors-drive-hydrogen-transition-with-h2-vehicles-125061600725_1.html	16-06-2025	—	—	Press
W2-S62	Gazette 27-03-2026 doc citing AIS 157:2020	Gazette of India via Scribd	CG-DL-E-27032026-...	https://www.scribd.com/document/1019065447/271279	27-03-2026	—	Unknown	Content unresolved
W2-S63	MNRE NGHM regulations directory	MNRE	National Green Hydrogen Mission Portal — Regulations	https://nghm.mnre.gov.in/regulations?language=en	—	—	Current	Blocked
W2-S64	State regulation of retrofit centres	Kerala MVD	Conversion of vehicles into CNG/LPG mode — Regulation and control of Retrofitment centres	https://mvd.kerala.gov.in/en/node/1386	—	—	—	Blocked
W2-S65	NCR age restrictions after CNG conversion	Delhi Govt (Switch Delhi) / press	Switch Delhi — retro-fitment	https://ev.delhi.gov.in/retro-fitment	—	—	—	Secondary
W2-S66	87 standards adopted / 59 under development	CEEW	How can India fix gaps for green hydrogen safety standards?	https://www.ceew.in/publications/how-can-india-adopt-international-standards-for-national-green-hydrogen-mission	2024	—	—	Think-tank
W2-S67	PESO hydrogen presentations at PNGRB	PESO via PNGRB	Statutory rules/regulations for CNG stations (PESO); Stakeholders interaction on hydrogen	https://www.pngrb.gov.in/pdf/press-note/PESO_15032024.pdf ; https://pngrb.gov.in/pdf/press-note/20250306_PESO.pdf	03-2024 / 03-2025	—	—	Blocked
W3-S1	72.10 lakh CNG vehicles registered	AIR News (newsonair.gov.in) reporting Lok Sabha written reply, MoS PNG Suresh Gopi	Total sales of CNG vehicles have increased over the past few years: Suresh Gopi	https://www.newsonair.gov.in/total-sales-of-cng-vehicles-have-increased-over-the-past-few-years-suresh-gopi			observed (Vahan)	page blocked; figure from search passage
W3-S2	58.61 lakh (Mar 2023) → 81.95 lakh (Mar 2025) CNG vehicles; +25% FY25; PNGRB VAT plea	ET Energyworld / UNI / The Hindu BusinessLine / Energywatch quoting PNGRB	PNGRB expands CGD network to 784 districts, sets 2034 target of 12.6 crore PNG connections; City Gas Distribution network now covers nearly 100 pc...; PNGRB in talks with states to rationalise	https://energy.economictimes.indiatimes.com/news/oil-and-gas/pngrb-expands-cgd-network-to-784-districts-sets-2034-target-of-12-6-crore-png-connections/123553997 ; https://www.thehindubusinessline.com/markets/commodities/pngrb-in-talks-with-states-to-rationalise-vat-on-natural-gas-			observed (PNGRB citing Vahan)	secondary reporting of PNGRB statement

			VAT	cng/article69991018.ece				
W3-S3	18,336 CNG stations & 12.63 crore PNG by 2034; ~1,206 stations added FY25	Business Standard; IBEF	PNGRB targets 12.6 crore PNG connections, 18,336 CNG stations by 2034	https://www.business-standard.com/economy/news/pngrb-targets-12-6-crore-png-connections-18336-cng-stations-by-2034-125082700329_1.html ; https://www.ibef.org/news/india-eyes-126-million-png-connections-18-000-cng-stations-by-2034			target / observed	secondary
W3-S4	738 (2014) → 8,150+ CNG stations; 18,000 by 2030	Hardeep Singh Puri (MoPNG) on X; The Hindu BusinessLine	One Nation, One Gas Grid (X post); India to double CNG stations to 18,000 by 2030 amid One Nation One Gas Grid plan: Puri	https://x.com/HardeepSPuri/status/1961324393344491893 ; https://www.thehindubusinessline.com/economy/logistics/india-to-double-cng-stations-to-18000-by-2030-amid-one-nation-one-gas-grid-plan-puri/article69988711.ece			observed (ministerial) / target	ministerial statement, not a PNGRB table
W3-S5	outdated "2,830 stations" figure	PRS Legislative Research	National Gas Grid Including PNG and CNG (Standing Committee report summary)	https://prsindia.org/policy/report-summaries/national-gas-grid-including-png-and-cng			outdated	used only to flag staleness
W3-S6	17,700 stations by 2030 (MWP, rounds to 11A); "17,500 by 2030"; "7,400 in 2025"	Autocar Professional; Construction World	India's CNG infra development in high gear; India Plans 17,500 CNG-PNG Stations... by 2030	https://www.autocarpro.in/feature/indias-auto-industry-set-for-a-cng-boom-as-gas-infra-inches-to-100-penetration-117791 ; https://www.constructionworld.in/policy-updates-and-economic-news/india-plans-17500-cng-png-stations-aims-for-120-million-connections-by-2030/52110			commitment/ target	industry press citing MoPNG/PNGRB
W3-S7	CNG ₹86.98/kg Delhi (IGL, +₹3.89); ₹88/kg Mumbai (MGL, +₹2); reasons	Goodreturns; Equentis; ETV Bharat; Infralog	CNG Price in New Delhi Today; CNG, PNG Price Hike: New Rates in Mumbai and Delhi-NCR; CNG Price Hiked By Rs 2 Per Kg In Mumbai; MGL hikes CNG... amid rising spot RLNG costs	https://www.goodreturns.in/cng-price-in-new-delhi.html ; https://www.equentis.com/blog/cng-png-price-hike-new-rates-in-mumbai-and-delhi-ncr/ ; https://www.etvbharat.com/en/business/cng-price-hiked-by-rs-2-per-kg-in-mumbai-enn26090104129			observed (retail)	aggregator pages; IGL/MGL notices not opened
W3-S8	Pune CNG ₹97.75/kg from 1 Sep 2026	MNGL (official rate card); Times of India	CNG Rate Card — Revision in Prices w.e.f. midnight of 1st Sep 2026: Pune GA Rs 97.75/kg; MNGL hikes CNG price by Rs2 per kg in Pune	https://mngl.in/cng/rate-card ; https://timesofindia.indiatimes.com/city/pune/mngl-hikes-cng-price-by-rs2-per-kg-in-pune/articleshow/133662320.cms			observed	primary text visible in passage
W3-S9	Ahmedabad CNG ₹94.02/kg	CarDekho / Goodreturns city price pages	CNG price in India Today	https://www.cardekho.com/cng-price ; https://www.goodreturns.in/cng-price.html			observed (aggregator)	supplier notice not verified
W3-S10	Maruti FY25 CNG 6.12–6.2 lakh, 33%; 71.6% CNG-segment share (2024)	Autocar India; Autocar Professional	One in every three Maruti sold in FY2025 was a CNG; One-in-Three Maruti Vehicles Sold in FY25 Ran on CNG	https://www.autocarindia.com/industry/one-in-every-three-maruti-sold-in-fy2025-was-a-cng-434994 ; https://www.autocarpro.in/news/one-in-three-maruti-vehicles-sold-in-fy25-ran-on-cng-125695			observed (company/in dustry data)	secondary
W3-S11	Maruti FY26 sales 24.22 lakh, exports 4.47 lakh; biogas plants ₹561 crore	Maruti Suzuki (press releases, not accessible); Automotive World; scanx	Maruti Suzuki announces Financial Results for FY2025-26; Annual Integrated Report 2025-26	https://www.marutisuzuki.com/corporate/media/press-releases/2026/april/maruti-suzuki-announces-financial-results-for-fy2025-26 ; https://www.marutisuzuki.com/corporate/media/press-releases/2026/august/maruti-suzuki-annual-integrated-report-2025-26 ; https://www.automotiveworld.com/news/annual-integrated-report-maps-maruti-suzukis-growth-push/			observed	primary blocked; via secondary
W3-S12	Maruti FY26 CNG >7 lakh; 42% share "last month"; FY27 target 9 lakh	ET Auto; Business Standard; Moneycontrol	Maruti Suzuki targets nearly 30% jump in CNG sales to 9 lakh units in FY27; Maruti Suzuki targets 900,000 units of CNG vehicle sales this fiscal	https://auto.economictimes.indiatimes.com/news/passenger-vehicle/maruti-suzuki-aims-for-30-increase-in-cng-vehicle-sales/132637759 ; https://www.business-standard.com/companies/news/maruti-suzuki-targets-900-000-units-of-cng-vehicle-sales-this-fiscal-126072600153_1.html			observed / target	42% is a monthly figure
W3-S13	PV fuel mix FY26: petrol 52.6% (24.61 of 46.83 lakh), CNG 21.7%; Jul 2026 CNG 24.67%	Autopunditiz (Vahan-based analyst)	Passenger Cars Fuel-wise Sales Analysis – India FY2026	https://www.autopunditiz.com/post/india-passenger-vehicle-fuelwise-sales-analysis-fy2026			observed (analyst compilation)	not SIAM; CNG % seen via social summary
W3-S14	SIAM quarterly highlights FY26 (not accessible)	SIAM	Key Highlights from Q1/Q2 FY2025-26	https://www.siam.in/pressrelease-details.aspx?mpgid=48&pgidtrail=50&pid=583;...pid=591			—	blocked; no CNG split retrieved
W3-S15	"4,000+ authorised installation centres"; aftermarket ₹2,500–3,500 crore/yr; brand shares	LetsGO CNG (vendor blog)	CNG Conversion Statistics India 2026: 50 Key Facts	https://letsgocng.in/cng-conversion-statistics-india-2026/			vendor estimate	unofficial; low confidence
W3-S16	RFC registries (Vahan CNG Maker portal; IGL list; Delhi RFC application)	MoRTH Vahan; IGL; Delhi Transport Dept	Available Fitment Center (CNG Maker); RTO Approved CNG Kit Retrofit Centres; Application for CNG Retro Fitment	https://vahan.parivahan.gov.in/cngmaker/vahan/user/retrofitment/availableFitmentCenter.xhtml ; https://www.iglonline.net/cng-retrofitment-center ; https://transport.delhi.gov.in/transport/application-cng-retrofitment			registry	no totals retrievable
W3-S17	Delhi aggregator scheme EV milestones; notified 21 Nov 2023	Legitquest (text reproduction); Hindustan Times; Delhi Transport Dept (X); Medianama	Delhi Motor Vehicle Aggregator and Delivery Service Provider Scheme, 2023	https://www.legitquest.com/act/delhi-motor-vehicle-aggregator-and-delivery-service-provider-scheme-2023/D483 ; https://www.hindustantimes.com/cities/delhi-news/delhi-govt-notifies-cab-aggregator-policy-fleets-to-switch-to-electric-101701281802617.html ; https://x.com/TransportDelhi/status/1730101116590059896			rule	6-month/1-year milestones only via HT
W3-S18	~60,000 taxis registered in Delhi; CNG/clean-fuel taxi permits 15 years; SC CNG requirement	Outlook India	Delhi: All Taxis Running On CNG, Other Clean Fuels To Have Permit Validity Of 15 Years	https://www.outlookindia.com/national/delhi-all-taxis-running-on-cng-other-clean-fuels-to-have-permit-validity-of-15-years-news-296571			observed (dept data, undated)	secondary
W3-S19	CAQM: only CNG/EV new aggregator/delivery vehicles in NCR from 1	Deccan Herald; Free Press Journal;	No new petrol or diesel vehicles for cab aggregators, delivery firms in NCR from	https://www.deccanherald.com/amp/story/india%2Fno-new-petrol-or-diesel-vehicles-for-cab-aggregators-delivery-firms-in			rule	secondary

	Jan 2026; Haryana rules	Swarajya	January 2026: CAQM; Haryana Cabinet Mandates CNG, EV Or Cleaner Fuel For All New Aggregator Vehicles In NCR From 2026	ncr-from-january-2026-caqm-3571433 ; https://www.freepressjournal.in/amp/india/haryana-cabinet-mandates-cng-ev-or-cleaner-fuel-for-all-new-aggregator-vehicles-in-ncr-from-2026				
W3-S20	Delhi 8.76 m vehicles (19 Mar 2026); taxis limited growth; e-buses 4,338	The Wire (reporting Delhi Economic Survey 2025-26); Delhi Planning Dept PDF (blocked)	Service Sector, Transport, Water: Highlights of Delhi's Economic Survey 2025-26	https://m.thewire.in/article/economy/service-sector-transport-water-highlights-of-delhis-economic-survey-2025-26/amp ; https://delhiplanning.delhi.gov.in/sites/default/files/2026-03/economic_survey_english_0.pdf			observed	PDF blocked
W3-S21	60-driver Delhi study; longer hours	Springer (book chapter)	Are Ride-Sharing Platforms Good for Indian Drivers? An Investigation of Taxi and Auto-Rickshaw Drivers in Delhi	https://link.springer.com/chapter/10.1007/978-3-030-64697-4_10			observed (survey)	small sample
W3-S22	ICCT ride-hailing TCO; BEV ₹8.8 – 5.8/km (2019 – 2020); 2019 not competitive	ICCT	Updated analysis of incentives for electrifying India's four-wheeler ride-hailing fleet; Are electric four-wheelers becoming more cost attractive as ride-hailing vehicles in India?; (vendor rule-of-thumb: Zevpoint EV fleet charging guide)	https://theicct.org/publication/updated-analysis-of-incentives-for-electrifying-indias-four-wheeler-ride-hailing-fleet/ ; https://theicct.org/are-electric-four-wheelers-becoming-more-cost-attractive-as-ride-hailing-vehicles-in-india/ ; https://www.zevpoint.com/blogs/guides/ev-fleet-charging-solutions-taxi-logistics-india			modelled	PDFs blocked; daily-km assumption not seen
W3-S23	BluSmart collapse; SEBI order 15 Apr 2025; ₹978 crore diversion; leasing model	Business Today; Down To Earth; Zentrum Law	SEBI turns heat on Gensol, BluSmart's house of cards wobbles; What BluSmart got wrong—and right	https://www.businesstoday.in/latest/corporate/story/sebi-turns-heat-on-gensol-blusmarts-house-of-cards-wobbles-472292-2025-04-16 ; https://www.downtoearth.org.in/air/what-blusmart-got-wrongand-right			observed	secondary
W3-S24	Everest (20,500 cars, ~2,000 EVs, 10,000 EV target); Refex 2,000+ EVs; Evera 4,500 by Jun 2026; fleet constraints	Business Standard; EVreporter; EV Drive	Next stop for BluSmart's EVs could be Everest Fleet; India's Leading Electric 4-Wheeler Passenger Fleet Operators; Evera Fleet Expansion 4500 EVs by June 2026	https://www.business-standard.com/companies/start-ups/everest-fleet-in-talks-to-acquire-blusmart-evs-for-uber-green-125051201066_1.html ; https://evreporter.com/indias-leading-electric-4-wheeler-passenger-fleet-operators/ ; https://evdrive.in/evera-fleet-expansion-4500-electric-vehicles/			observed / announced	company statements
W3-S25	Uber–Tata 25,000 Xpres-T EVs; Xpres-T price ₹13.04 lakh	Autocar India; Tata Motors Xpres-T site	Tata Motors to supply 25,000 XPRES-T EVs in largest ever fleet order; Tata XPRES-T EV Price	https://www.autocarindia.com/car-news/tata-motors-to-supply-25000-xpres-t-evs-in-largest-ever-fleet-order-427364 ; https://xprest.tatamotors.com/electric/price			announced / list price	delivery status unknown
W3-S26	used Dzire CNG from ₹3.71 lakh (Delhi); WagonR ₹1.6–6.0 lakh; new Dzire Tour S on-road ₹7.35–8.33 lakh	Cars24; Orange Book Value; CarDekho	Second Hand Maruti Dzire CNG Cars in New Delhi; Used Wagon R price and valuation; Dzire Tour S On Road Price in New Delhi	https://www.cars24.com/buy-used-maruti-dzire-cng-cars-new-delhi/ ; https://orangebookvalue.com/used-cars/maruti-suzuki-wagon-r-price ; https://www.cardekho.com/maruti/dzire-tour-s/price-in-new-delhi			listings (indicative)	not valuations
W3-S27	Tata EV prices/specs (Tiago, Punch, Nexon, Tigor)	Tata.ev; ZigWheels; Team-BHP; V3Cars	Tiago EV / Punch EV / Nexon EV price pages; Tigor EV 2026	https://ev.tata.cars/tiago/ev/price.html ; https://ev.tata.cars/punch/ev/price.html ; https://ev.tata.cars/nexon/ev/price.html ; https://www.v3cars.com/tata-cars/tigor-ev ; https://www.zigwheels.com/tata-cars/tiago-ev/			list price / ARAI	ex-showroom, variant-dependent
W3-S28	MG Comet, Windsor prices/specs	JSW MG Motor India; ZigWheels	MG Comet EV page; MG Windsor EV 2026	https://www.mgmotor.co.in/vehicles/comet-ev-electric-car-in-india ; https://www.zigwheels.com/mg-motor-cars/windsor-ev/			list price	—
W3-S29	e Vitara, Creta Electric, eC3X prices/specs	Autocar India; CarWale/Team-BHP	Price and specs of Maruti e Vitara, Hyundai Creta Electric, MG Windsor EV... compared; 2026 Citroen eC3X launched at Rs 11.99 lakh; Hyundai Creta Electric	https://www.autocarindia.com/auto-features/maruti-e-vitara-vs-rivals-prices-compared-440249 ; https://www.autocarindia.com/car-news/2026-citroen-ec3x-launched-at-rs-1025-lakh-439987 ; https://www.carwale.com/hyundai-cars/creta-electric/			list price	—
W3-S30	real-world range/efficiency tests	Autocar India; ZigWheels	Real world EV range tested of top EV cars in India; Tata EVs and their real-world range	https://www.autocarindia.com/auto-features/every-mass-market-ev-real-world-range-tested-434105 ; https://www.zigwheels.com/news-features/ev-news/heres-a-look-at-all-the-tata-evs-and-their-real-world-range/54737/			observed (media test)	single-test conditions
W3-S31	Tata battery warranty (8 yr/1.6 lakh km; lifetime for Curvv.ev/Nexon.ev 45)	Tata Motors; Tata.ev; Spinny	Tata Motors introduces Lifetime HV Battery Warranty...; Tata EV warranty page; Tata EV Warranty Explained	https://www.tatamotors.com/press-releases/reimagining-ev-ownership-tata-motors-introduces-lifetime-hv-battery-warranty-for-curvev-ev-and-nexon-ev-45-kwh/ ; https://ev.tata.cars/service/warranty.html ; https://www.spinny.com/blog/tata-ev-warranty/			company terms	fleet terms differ
W3-S32	Delhi EV tariff ₹4.50/kWh LT, ₹4.00/kVAh HT; DERC PM E-DRIVE order	Mercom India; thediscombill; EMobility+	Delhi Slashes Tariffs for EV Charging; Delhi Electricity Tariff 2025-26; DERC directs EV charging infra costs... borne by operators	https://www.mercomindia.com/delhi-tariffs-ev-charging-electric-mobility ; https://thediscombill.com/tariffs/delhi/ ; https://emobilityplus.com/2026/07/03/derc-directs-ev-charging-infrastructure-costs-under-pm-e-drive-scheme-to-be-borne-by-operators-not-consumers-in-delhi/			tariff order (secondary)	DERC order not opened
W3-S33	public DC ₹18–25/kWh; AC ₹8–18; Jio-bp ₹15.99	Pulse Energy; Team-BHP	EV Charging Rates in India 2026; Pricing of just Rs 15.99 per kWh — Jio BP	https://pulseenergy.io/blog/ev-charging-rates ; https://www.team-bhp.com/forum/electric-cars/273196-pricing-just-rs-15-99-per-kwh-will-jio-bp-disrupt-ev-charging-market.html			observed (operator prices)	vendor blog
W3-S34	GST: EV 5%; small cars 18%; large 40% (GST 2.0, Sep 2025)	Bajaj Finserv; The Munim	GST on Vehicles in 2026; GST on Electric Vehicles in India 2026	https://www.bajajfinserv.in/gst-on-vehicles ; https://themunim.com/blog/gst-on-electric-vehicles/			rule	secondary explainer
W3-S35	Delhi EV Policy 2026 (w.e.f. 1 Jul 2026) provisions	Business Standard; YourStory; EV Infrastructure News	Delhi govt notifies EV Policy 2026, rolls out subsidies, tax exemptions; Delhi government notifies Delhi EV Policy 2026	https://www.business-standard.com/india-news/delhi-govt-notifies-ev-policy-2026-rolls-out-subsidies-tax-exemptions-126070100488_1.html ; https://yourstory.com/2026/07/delhi-government-notifies-delhi-ev-policy-2026 ; https://www.evinfrastructurereads.com/ev-policies/delhi-notifies-ev-policy-2026-with-tax-exemptions-for-ev			rule (notified)	gazette not opened

W3-S36	5 pilot projects: 37 vehicles, 9 HRS, 10 routes, ₹208 crore; awardees; ARAI SIA	PIB (PRID 2107795, blocked); Down To Earth; DD News; Energetica India; Swarajya	As told to Parliament (March 25, 2025): 37 cross-country vehicles, 9 fuelling stations...; India rolls out pilot projects for hydrogen-powered buses and trucks; MNRE Initiates 5 Pilot Projects...	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2107795 ; https://www.downtoearth.org.in/environment/as-told-to-parliament-march-25-2025-37-cross-country-vehicles-9-fuelling-stations-planned-under-green-hydrogen-mission ; https://ddnews.gov.in/en/india-rolls-out-pilot-projects-for-hydrogen-powered-buses-and-trucks/ ; https://www.energetica-india.net/news/mnre-initiates-5-pilot-projects-for-using-hydrogen-in-buses-and-trucks			sanctioned	PIB blocked
W3-S37	12 pilot projects: 70 vehicles (27 buses, 43 trucks), 16 HRS, 21 routes, ₹410.77 crore, ₹41.582 crore released	PIB / MNRE (PRID 2291541, blocked); fiinews; EQ Mag; MNRE minister social posts	Centre Launches 12 Pilot Projects Under National Green Hydrogen Mission for Deployment of 70 Hydrogen-Powered Vehicles Across 21 Routes; Project: 12 NGHM are under implementation	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2291541&reg=48&lang=1 ; https://www.fiinews.com/2026/08/01/project-12-nghm-are-under-implementation/ ; https://www.eqmagpro.com/centre-launches-12-pilot-projects-under-national-green-hydrogen-mission-for-deployment-of-70-hydrogen-powered-vehicles-across-21-routes-eq/			sanctioned / under implementation	PIB blocked; 21-route list not retrieved
W3-S38	SIGHT 8.62 lakh tpa awarded (19 cos); 3,000 MW electrolyser awards; ~8,000 tpa commissioned (Feb 2026); ₹292 crore spent; 30 KTPA to 4 refineries; Tranche II 450,000 tpa; NGHM outlay/target	Business Today (column); Business Standard; ANI; Mercom India	India's green hydrogen ambitions face a reality check; Govt spends ₹292 crore under NGHM so far: MNRE; Centre awards 30 KTPA...; MNRE to Incentivize 450,000 MTPA...	https://www.businesstoday.in/opinion/columns/story/indias-green-hydrogen-ambitions-face-a-reality-check-536276-2026-06-11 ; https://www.business-standard.com/amp/industry/news/govt-spends-292-crore-under-national-green-hydrogen-mission-so-far-mnre-126081201853_1.html ; https://www.aninews.in/news/business/centre-awards-30-ktpa-green-hydrogen-capacity-to-four-oil-refineries-under-national-green-hydrogen-mission20260811180606/ ; https://www.mercomindia.com/mnre-450000-mtpa-green-hydrogen			awarded vs operating	secondary reporting of MNRE/SECI data
W3-S39	Numaligarh ₹279/kg; SECI green ammonia ₹55.75/kg; SIGHT auction Mar 2025; cost estimates	S&P Global; PIB (PRID 2153006, blocked); Renewable Watch; Reslink (vendor)	India's Numaligarh hydrogen tender ends at record low Rupees 279/kg; SECI conducts first-ever auction for procurement of Green Ammonia; SECI announces results of green hydrogen auction under SIGHT; India Green Hydrogen Price Per Kg 2026	https://www.spglobal.com/energy/en/news-research/latest-news/energy-transition/021226-indias-numaligarh-hydrogen-tender-ends-at-record-low-rupees-279kg ; https://www.pib.gov.in/PressReleasePage.aspx?PRID=2153006 ; https://renewablewatch.in/2025/03/20/seci-announces-results-of-green-hydrogen-auction-under-sight-programme/ ; https://www.reslink.org/blogs/india-s-green-hydrogen-mission-is-moving-from-paper-to-projects/			observed (tender)	₹397/kg and \$5.3–6.7/kg are vendor figures
W3-S40	California retail H2 \$32–36/kg; avg \$34.55 (Sep 2024); True Zero \$36	S&P Global; Stillwater Associates; hydrogenfuelfinder	California hydrogen pump prices for light-duty vehicles reach new highs; 2025 Cost Showdown for Drivers; Why Hydrogen Fuel Costs \$36/kg in 2025	https://www.spglobal.com/commodity-insights/en/news-research/latest-news/energy-transition/100124-california-hydrogen-pump-prices-for-light-duty-vehicles-reach-new-highs ; https://stillwaterassociates.com/2025-cost-showdown-for-drivers-is-hydrogen-fuel-cheaper-than-gasoline/ ; https://hydrogenfuelfinder.com/blog/hydrogen-fuel-cost-2025-reality-check.html			observed (retail)	—
W3-S41	Germany €12.95/kg avg; €12.85; Hylane–H2 Mobility €8/kg (2026, trucks)	Atawey; Energy News; Driving Hydrogen; glpautogas	Understanding the Cost and Price of Hydrogen (2026); Hydrogen price goes to €12.85/kg; €8/kg hydrogen enables cost parity with diesel: Hylane & H2 Mobility	https://atawey.com/en/hydrogen-prices-in-2026-cost-per-kg-refuelling-costs-comparisons-and-2030-outlook/ ; https://energynews.biz/hydrogen-price-goes-up-to-e12-85-kg/ ; https://drivinghydrogen.com/2025/11/20/e8-kg-hydrogen-enables-cost-parity-with-diesel-hylane-h2-mobility-milestone/			observed / contracted	dates of averages not exact
W3-S42	Japan ¥700/kg subsidy; Tokyo auction ~\$23/kg; METI priority regions	Driving Hydrogen; METI; Hydrogen Insight	Japan to introduce ¥700/kg hydrogen fuel subsidy...; First Selection of Priority Regions for FC Commercial Vehicles; Hydrogen prices of \$23/kg — Tokyo... auction	https://drivinghydrogen.com/2025/05/30/japan-to-introduce-%C2%A5700-kg-hydrogen-fuel-subsidy-to-kickstart-commercial-uptake/ ; https://www.meti.go.jp/english/press/2025/0519_002.html ; https://www.hydrogeninsight.com/production/hydrogen-prices-of-23-kg-tokyo-city-government-announces-results-for-first-double-sided-green-hydrogen-auction/2-1-1830862			policy / auction	pump price NOT retrieved
W3-S43	Standards: 34 standards in 14 months; PESO/OISD/PNGRB/MoRTH/BIS; Gas Cylinder Rules 2016; ISO 19880 harmonisation; CMVR 4th Amdt 2025 H2 plates; MoRTH 18 Aug 2026 draft	PIB (static background); PNGRB/PESO; autopdp; Business Standard	Green Hydrogen Standards and Approval Systems in India; Stakeholders interaction on pathways for transmission of hydrogen; Automotive Regulation Updates India: Govt proposes 5-yr age extension for EV, CNG, hydrogen commercial vehicles	https://static.pib.gov.in/WriteReadData/specificdocs/documents/2024/may/doc2024510336201.pdf ; https://www.pngrb.gov.in/pdf/press-note/PESO_15032024.pdf ; https://autopdp.com/regulatory-updates ; https://www.business-standard.com/industry/auto/govt-proposes-5-yr-age-extension-for-ev-cng-hydrogen-commercial-vehicles-126081800231_1.html			rule / draft	PDFs blocked; passages only
W3-S44	NTPC Leh: 5 FC buses; 80 kg/day; 350 bar; 1.7 MW solar; Nov 2024 flag-off; Jun 2025 handover	NTPC; UT Ladakh; pv magazine India; AIR News	NTPC Hands Over Five Hydrogen Fuel Cell Buses to Leh Administration; Union Minister Manohar Lal flags off the fleet of green hydrogen buses of NTPC in Ladakh	https://ntpc.co.in/media/press-releases/ntpc-hands-over-five-hydrogen-fuel-cell-buses-leh-administration ; https://ladakh.gov.in/ntpc-hands-over-five-hydrogen-fuel-cell-buses-to-sidco-at-green-hydrogen-mobility-station-leh/ ; https://www.pv-magazine-india.com/2025/06/19/the-hydrogen-stream-ntpc-hands-over-five-hydrogen-fuel-cell-buses-to-leh-administration/ ; https://www.newsonair.gov.in/union-minister-manohar-lal-flags-off-the-fleet-of-green-hydrogen-buses-of-ntpc-in-ladakh			operating	costs/H2 price not disclosed
W3-S45	Tata 16-truck trials (4 Mar 2025); 40 H2-ICE trucks (Feb 2026); IOCL HRS at 4 sites	Tata Motors; AIR News; electrive	Tata Motors drives India's green future with country's first hydrogen truck trials; First trials of hydrogen powered heavy	https://www.tatamotors.com/press-releases/tata-motors-drives-indias-green-future-with-countrys-first-hydrogen-truck-trials/ ; https://www.newsonair.gov.in/first-trials-of-hydrogen-			trial / announced	primary blocked

			duty trucks flagged off; Tata Motors to deploy 40 H2 ICE trucks in India	powered-heavy-duty-trucks-flagged-off ; https://www.electrive.com/2026/02/27/tata-motors-to-deploy-40-h2-ice-trucks-in-india/				
W3-S46	IOCL Faridabad & Vadodara stations; 15-bus trials; Panipat 10 kTPA by Dec 2027; Air Products "first station"	IOCL; Indian Chemical News; Air Products; VahanBazaar (unofficial)	Green Drive: IOC plans hydrogen production units; Panipat green hydrogen unit on track for December 2027 completion; India's First Hydrogen Fuelling Station Inaugurated	https://iocl.com/MediaDetails/59096 ; https://www.indianchemicalnews.com/interviews/panipat-green-hydrogen-unit-on-track-for-december-2027-completion-dr-alok-sharma-director-rd-iocl-551 ; https://investors.airproducts.com/news-releases/news-release-details/indias-first-hydrogen-fuelling-station-inaugurated			operating (captive) / under construction	dates & pressures not verified
W3-S47	HCNG 50-bus Delhi trial (20 Oct 2020); 70% CO / 25% HC reduction; NOx same	fuelcellsworks; Busworld; IOCL media; pv magazine India	Shri Pradhan inaugurates H-CNG plant and launches trials in Delhi of 50 hydrogen-powered buses; Trial run of H-CNG powered city buses to begin in Delhi Oct 20; Hydrogen CNG, biogas: the many faces of new auto fuel	https://fuelcellsworks.com/news/india-shri-pradhan-inaugurates-h-cng-plant-and-launches-trials-in-delhi-of-50-hydrogen-powered-buses ; https://news.busworld.org/article/11296/trial-run-of-hcng-powered-city-buses-to-begin-in-delhi-oct-20 ; https://www.iocl.com/MediaDetails/56688 ; https://www.pv-magazine-india.com/2020/10/21/pollution-hit-delhi-starts-hydrogen-enhanced-cng-trial-for-buses/			pilot (IOCL-reported)	formal report not found
W3-S48	Ashok Leyland-Reliance H2-ICE truck; launch in 18–24 months (Oct 2024)	Business Standard; Rushlane	Ashok Leyland to launch first H2-powered truck in the next 2 years; Ashok Leyland, Reliance Unveil India's First Hydrogen-ICE Powered Heavy Duty Truck	https://www.business-standard.com/companies/news/ashok-leyland-to-launch-first-h2-powered-truck-in-the-next-2-years-124100700992_1.html ; https://www.rushlane.com/first-hydrogen-ice-powered-heavy-duty-truck-12460051.html			prototype / announced	launch not verified
W3-S49	Toyota Mirai-ICAT pilot, Mar 2022	Autocar Professional; India TV	Toyota Kirloskar, iCAT begin pilot project with Toyota Mirai; Nitin Gadkari drives India's first green hydrogen-powered car to Parliament	https://www.autocarpro.in/news-national/toyota-kirloskar-icat-begin-pilot-project-with-toyota-mirai-81384 ; https://www.indiatvnews.com/news/india/nitin-gadkari-drives-green-hydrogen-powered-car-to-parliament-toyota-mirai-pics-2022-03-30-766664			pilot	—
W3-S50	TVS hydrogen iQube; Bajaj exploring H2 2W/3W	Autocar India	TVS working on hydrogen powered scooter; Bajaj exploring hydrogen-powered two-wheelers under Chetak brand	https://www.autocarindia.com/bike-news/hydrogen-powered-tvs-scooter-in-the-works-425268 ; https://www.autocarindia.com/bike-news/bajaj-exploring-hydrogen-powered-two-wheelers-under-chetak-brand-431355			concept	—
W3-S51	Mirai sales US 499 (2024) / 210 (2025); global 1,702 (Jan–Nov 2024); cumulative 21,475 (Nov 2022); 2026 fuel credit	GoodCarBadCar; CarFigures; CarBuzz; Wikipedia; Toyota.com	Toyota Mirai Sales Figures; Toyota's Biggest Flop Of 2025...; Toyota Mirai (Wikipedia, "Markets and sales"); 2026 Toyota Mirai	https://www.goodcarbadcar.net/toyota-mirai-sales-figures-usa-canada-monthly-yearly/ ; https://carfigures.com/us-market-brand/toyota/mirai ; https://carbuzz.com/toyotas-biggest-flop-2025-sold-fewer-than-1000/ ; https://en.wikipedia.org/wiki/Toyota_Mirai ; https://www.toyota.com/mirai/			observed	2025 global total not found
W3-S52	Nexo gen-2 (Seoul 2025; Korea sales Jun 2025; 720 km); CR-V e:FCEV (~300/yr; 2,000 plan)	electrive; TopElectricSUV; Honda News; Green Car Reports	Next-gen Hyundai Nexo to debut in Q2 2025; First Look Review: Next-gen Hyundai Nexo; Honda Reveals 2025 Honda CR-V e:FCEV; First Drive: 2025 Honda CR-V e:FCEV	https://www.electrive.com/2024/05/10/next-gen-hyundai-nexo-2025/ ; https://topelectricsuv.com/first-look-review/hyundai-nexo-next-gen/ ; https://hondanews.com/en-US/honda-automobiles/releases/honda-reveals-2025-honda-cr-v-efcev-americas-first-production-plug-in-hydrogen-fuel-cell-electric-vehicle ; https://www.greencarreports.com/news/1142702_2025-honda-cr-v-efcev-test-drive-review			production	sales figures partial
W3-S53	Global FCEV sales +24.4% (2025); H1 2026 4,643 units (+13%); Europe 14	electrive; SNE Research	Global H ₂ vehicle sales rise by nearly 25%; Global FCEV Sales Reach 4,643 Units in H1 2026, Up 13.0% YoY	https://www.electrive.com/2026/02/23/global-h2-e2%82%82-vehicle-sales-rise-by-nearly-25/ ; https://www.sneresearch.com/en/insight/release_view/707/page/0			observed (SNE)	absolute 2025 total not seen
W3-S54	IEA GHR 2025: heavy trucks only fast-growing FCEV market; China ~95% FC CV stock; OMV closure; low-emissions H2 <1%; ~1,160 HRS end-2024; country counts	IEA; TÜV SÜD; Interact Analysis; IEA AFC TCP	Global Hydrogen Review 2025 (Executive summary; PDF); Over 1,000 hydrogen refuelling stations in operation worldwide in 2024; 80% of global HRS in five countries	https://www.iea.org/reports/global-hydrogen-review-2025/executive-summary ; https://iea.blob.core.windows.net/assets/12d92ecc-e960-40f3-aff5-b2de6690ab6b/GlobalHydrogenReview2025.pdf ; https://www.tuvsud.com/en/newsroom/press-releases/2025/february/over-1000-hydrogen-refuelling-stations-in-operation-worldwide-in-2024 ; https://interactanalysis.com/insight/80-of-global-hydrogen-refueling-stations-are-located-in-just-five-countries/			observed	IEA site blocked; FCEV stock number not retrieved; "72,193 FCEVs / 1,022 HRS" seen without year
W3-S55	Hydrogen HiAce pilot (CPB, 2024) results; 2025 hybrid trial	Toyota Australia; CPB Contractors; Roads Australia; just-auto	Toyota Unveils Hydrogen HiAce; CPB Contractors completes Australian-first Toyota hydrogen-fuelled vehicle pilot; Toyota to test hydrogen-electric hybrid vans in 2025	https://www.toyota.com.au/discover/news/toyota-unveils-hydrogen-powertrain-technology-for-pilot-program-in-australia ; https://www.cpbcon.com.au/news-and-media/2024/cpb-contractors-completes-australian-first-toyota-hydrogen-fuelled-vehicle-pilot ; https://roads.org.au/cpb-contractors-completes-historic-hydrogen-vehicle-trial/ ; https://www.just-auto.com/news/toyota-to-begin-testing-hydrogen-electric-hybrid-vans-in-2025/			pilot	2025 results not found
W3-S56	GR Corolla H2 (LH2) 2025 racing status	Toyota Global Newsroom; Driving Hydrogen	Pushing Hydrogen Engine Technology to New Heights in the Super Taikyu Series Final Race; Hydrogen-Powered GR Corolla... Fuji 24 Hours; Toyota's superconducting pump and liquid H2 take to the grid	https://global.toyota/en/newsroom/corporate/43550115.html ; https://global.toyota/en/newsroom/corporate/44458612.html ; https://drivinghydrogen.com/2025/11/21/toyotas-superconducting-pump-and-liquid-h2-take-to-the-grid/			development	—

W3-S57	HySE members; Dakar 2024 4th, 2025 2nd in class	Global Suzuki; Toyota Times	HySE to participate in the Dakar 2025 with the HySE-X2; HySE Buggy 2nd in Class at Dakar 2025	https://www.globalsuzuki.com/globalnews/2024/1112.html ; https://toyotatimes.jp/en/spotlights/1082.html			research	—
W3-S58	Burgman Hydrogen at JMS 2025; "not launching anytime soon"; Access CNG concept	Global Suzuki; BikeWale; Team-BHP	Hydrogen Engine Burgman — Japan Mobility Show 2025; Suzuki Burgman Hydrogen Showcased at JMS 2025; Suzuki Access CNG & hydrogen Burgman showcased	https://www.globalsuzuki.com/mobilityshow2025/burgman/ ; https://www.bikewale.com/news/suzuki-burgman-hydrogen-showcased-at-japan-mobility-show-2025/ ; https://www.team-bhp.com/forum/motorbikes/301240-suzuki-access-cng-hydrogen-powered-burgman-400-showcased-japan-mobility-show-2025-a.html			concept	70 MPa spec not restated
W3-S59	Cummins B6.7H production Mar 2025 (India, Tata); X15H 2027	The Buzz EV News; Cummins; Truckinginfo; autonocion	Cummins starts H2-ICE production at new India facility; Cummins X15 Off-Highway... Agritechnica; Cummins Debuts 15L Hydrogen Engine	https://www.thebuzzevnews.com/cummins-h2-ice/ ; https://www.cummins.com/news/releases/2025/11/10/cummins-x15-highway-receives-powertrain-magazines-2026-alternative-engine ; https://www.truckinginfo.com/10170361/cummins-debuts-15l-hydrogen-engine			production (limited)	X15H date secondary
W3-S60	JCB H2 engine EU type-approval (May 2025); 11 authorities (Jan 2025); 130+ engines	JCB; Power Progress	JCB secures full EU type-approval for hydrogen engine; Landmark start to 2025...; JCB receives type approval from 11 licensing authorities	https://www.jcb.com/en-GB/explore/insight/news/2025/05/jcb-secures-full-eu-type-approval-for-pioneering-hydrogen-engine/ ; https://www.jcb.com/en-GB/explore/insight/news/2025/01/landmark-start-to-2025-as-jcbs-hydrogen-engine-approved-for-use/ ; https://www.powerprogress.com/news/jcb-receives-type-approval-from-11-licensing-authorities-for-h2-ice/8049767.article			approved / evaluation	off-highway only
W3-S61	KEYOU conversions; Daimler H2-ICE 2027	Daimler Truck; gasworld; Allison Transmission	Daimler Truck Partners with KEYOU on Hydrogen Combustion Technology; Daimler hydrogen ICE truck to launch in 2027; KEYOU hands over first hydrogen 18-ton truck... to EP-Trans	https://www.daimlertruck.com/en/newsroom/pressrelease/daimler-truck-partners-with-keyou-on-hydrogen-combustion-technology-53511488 ; https://www.gasworld.com/story/daimler-hydrogen-ice-truck-to-launch-in-2027-under-keyou-partnership/2252665.article/ ; https://www.allisontransmission.com/newsroom/article/2024/12/12/zero-emission-and-efficient---keyou-hands-over-first-hydrogen-18-ton-truck-with-allison-fully-automatic-transmission-to-ep-trans			pioneer fleet / announced	—
W3-S62	MAN hTGX 200 units; Deutz TCG 7.8 H2; Westport H2 HPDI	Hydrogen Insight; MAN; Deutz; IAA/trans.info	MAN to produce limited series of 200 vehicles with H2 combustion engines; DEUTZ TCG 7.8 H2; Westport unveils H2 HPDI; Westport successfully demonstrates H2 HPDI	https://www.hydrogeninsight.com/transport/hydrogen-sceptical-truck-maker-man-to-produce-limited-series-of-200-vehicles-with-h2-combustion-engines/2-1-1622908 ; https://press.mantruckandbus.com/corporate/man-expands-its-zero-emission-portfolio/ ; https://www.deutz.com/en/products/product-highlights/deutz-tcg-78-h2 ; https://trans.info/en/westport-successfully-demonstrates-h2-hpdi-fuel-system-374302			limited series / prototype	—
W3-S63	ULEMCo conversions (Aberdeen 35 vehicles; ambulance; H2/HVO)	Van Fleet World; ULEMCo; Autocar UK	ULEMCo wins deal for largest fleet-wide hydrogen conversion to date; ULEMCo completes first hydrogen/HVO dual-fuel conversion; Ulemco reveals first hydrogen-fuelled ambulance	https://vanfleetworld.co.uk/ulemco-wins-deal-for-largest-fleet-wide-hydrogen-conversion-to-date/ ; https://ulemco.com/ulemco-completes-first-hydrogen-hvo-dual-fuel-conversion/ ; https://www.autocar.co.uk/car-news/new-cars/ulemco-reveals-first-hydrogen-fuelled-ambulance			commercial (small)	prices NOT FOUND
W3-S64	Stellantis ends FC programme 16 Jul 2025; Symbio fallout	Stellantis (blocked); Driving Hydrogen; US News/Reuters; Global Banking & Finance	Stellantis Discontinues Hydrogen Fuel Cell Technology Development Program; Stellantis halts hydrogen van plans...; Symbio cuts 70% of staff...; Stellantis to exit Symbio JV, faces €235 million cost	https://www.stellantis.com/en/news/press-releases/2025/july/stellantis-discontinues-hydrogen-fuel-cell-technology-development-program ; https://drivinghydrogen.com/2025/07/16/stellantis-halts-hydrogen-van-plans-ends-fuel-cell-development-programme/ ; https://money.usnews.com/investing/news/articles/2025-07-16/stellantis-discontinues-hydrogen-fuel-cell-programme-and-van-production ; https://drivinghydrogen.com/2025/12/08/symbio-cuts-70-of-staff-as-stellantis-walks-away/			decision (observed)	primary blocked
W3-S65	Shell closes 7 California LD stations (6 Feb 2024)	S&P Global; InsideEVs; Automotive News	Shell permanently closes light-duty hydrogen fueling stations; Shell is immediately Closing All Of Its California Hydrogen Stations	https://www.spglobal.com/energy/en/news-research/latest-news/electric-power/021224-shell-permanently-closes-light-duty-hydrogen-fueling-stations ; https://insideevs.com/news/708156/shell-closes-california-hydrogen-stations/ ; https://www.autonews.com/mobility-report/shell-closes-california-light-duty-hydrogen-fueling-stations/			observed	—
W3-S66	H2 Mobility: 22 closures by mid-2025; 14 more 31 Dec 2025	electrive; MobilityEnergy; H2.LIVE; CleanTechnica	H2 Mobility to shut down 22 hydrogen fuel stations in Germany; Fourteen hydrogen filling stations in Germany to close this year; Network consolidation 700 bar	https://www.electrive.com/2025/03/03/h2-mobility-to-shut-down-22-hydrogen-fuel-stations-in-germany/ ; https://www.mobilityenergy.com/en/industry/2025/11/17/fourteen-hydrogen-filling-stations-in-germany-to-close-this-year-focusing-on-forward-looking-locations/ ; https://h2.live/en/news/3815/			observed	—
W3-S67	Honda Clarity ended Aug 2021; GM–Honda FCSM JV to end before end-2026	InsideEVs; Automotive Manufacturing Solutions; Headlight.news	Official: Honda Discontinues Clarity PHEV And Clarity Fuel Cell; Honda-GM fuel cell venture ends as EV priorities reshape	https://insideevs.com/news/514412/honda-discontinues-clarity-phev-fcv/ ; https://www.automotivemanufacturingsolutions.com/electrification/hondagm-fuel-cell-venture-ends-as-ev-priorities-			observed	—

				reshape/2591276 ; https://headlight.news/2025/02/24/honda-general-motors-continue-to-drift-apart-ending-fuel-cell-partnership/				
W3-S68	Suzuki Growth Strategy FY2030 (CNG/biogas/ethanol; ~70% CNG share); multi-pathway statements 2025	Suzuki Motor Corp (globalsuzuki.com blocked; mirror suzuki.hu; Automotive World); Autocar Professional; S&P Global Mobility	Suzuki Announces Growth Strategy for FY2030; Suzuki backs multi-pathway route to carbon neutrality, says only BEVs not practical; India's multifuel path to automotive sustainability	https://www.globalsuzuki.com/globalnews/2023/0126.html ; https://www.suzuki.hu/corporate/en/company/growth-strategy-2030/growth-strategy ; https://www.autocarpro.in/news/suzuki-backs-multi-pathway-route-to-carbon-neutrality-says-only-bevs-not-practical-129454 ; https://www.spglobal.com/automotive-insights/en/blogs/2025/11/indias-multifuel-path-to-automotive-sustainability			company strategy	context only
W3-S69	SRDI established 28 Aug 2022; role; Banas Suzuki biogas plant 6 Dec 2025; 11 biogas plants	Global Suzuki; SRDI site; MarketScreener; BioEnergy Times	Suzuki establishes R&D center in India; About Suzuki R&D India; Suzuki holds the opening ceremony of Banas Suzuki Biogas Plant in India; Suzuki, Maruti Suzuki to add 11 biogas plants	https://www.globalsuzuki.com/globalnews/2022/0828b.html ; https://www.suzukirindia.com/about-us ; https://uk.marketscreener.com/news/suzuki-motor-december-9suzuki-holds-the-opening-ceremony-of-banas-suzuki-biogas-plant-in-india-ce7d51d2db8ff22d ; https://bioenergytimes.com/suzuki-maruti-suzuki-to-add-11-biogas-plants-using-cow-dung-farm-waste/			company	context only
W3-S70	Delhi draft EV Policy 2.0 proposed phasing out CNG autos; previous policy extended to 31 Mar 2025	Deccan Herald	Draft EV policy 2.0 recommends phasing out of CNG-driven autorickshaws in Delhi; Delhi electric vehicle policy extended till March 31, 2025	https://www.deccanherald.com/amp/story/india%2Fdelhi%2Fdraft-ev-policy-20-recommends-phasing-out-of-cng-driven-autorickshaws-in-delhi-3482789 ; https://www.deccanherald.com/amp/story/india%2Fdelhi%2Fdelhi-electric-vehicle-policy-extended-till-march-31-2025-3295696			draft / rule	superseded by 2026 policy
W3-S71	Context: India's alternative-fuel shift "becoming a CNG story"	Down To Earth	India's CNG Dominance: A Step Towards Cleaner Transport or a Roadblock to Zero Emissions?	https://www.downtoearth.org.in/energy/indias-transport-sectors-alternative-fuel-shift-is-becoming-a-cng-story-but-what-comes-next			commentary	—
W3-S72	CNG kit prices ₹40–70k (sequential)	Cars24; Spinny	CNG Kit Price in India; CNG Kit in India: Price, Function, Conversion	https://www.cars24.com/article/cng-kit-price-in-india/ ; https://www.spinny.com/blog/cng-kit-in-india-price-function-conversion/			market guide	vendor content
W3-S73	Creta Electric BaaS ₹10.99 lakh + ₹3.90/km	Team-BHP / CarDekho	Hyundai Creta Electric 2026	https://www.team-bhp.com/new-cars/hyundai/creta-electric/			list price	—
W3-S74	~85,000 taxis in Delhi (2022)	Statista (summary only)	Number of taxis in Delhi	(URL not captured — search summary only)			observed (secondary)	unverified; low confidence
W3-S76	expert scepticism on 2030 gas targets	IEEFA	Is India's 2030 gas consumption target feasible?	https://ieefa.org/resources/indias-2030-gas-consumption-target-feasible			analysis	title only seen
W3-S77	PNGRB 2030/2040 projections (CGD 87 mmscmd 2030 base case) and critique	PNGRB press note; PSU Watch	Natural Gas Projections 2030–40 (press note 7 Apr 2025); PNGRB projects a natural gas boom, but experts urge a sobering reality check	https://pngrb.gov.in/pdf/press-note/20250407_NG_Projections2030_40.pdf ; https://psuwatch.com/newsupdates/pngrb-projects-a-natural-gas-boom-but-experts-urge-a-sobering-reality-check			modelled	PDF blocked
W4-S1	700-bar Type IV system \$12.1–13.3/kWh (median 12.7, 2020\$) at 100k/yr; carbon fibre dominant	US DOE Hydrogen Program	Record #24006 "Onboard Type IV Compressed Hydrogen Storage System —Cost and Performance Status"	https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/24006-onboard-storage-cost-performance-status.pdf	28 Apr 2025 / 2024 analysis	Record summary (page not verified)	modelled	Host blocked; values from search summary; 10k & 500k rows unread
W4-S2	\$14.2 / 15.7 / 21.3 per kWh at 500k / 100k / 10k systems/yr (2016\$)	US DOE	Record #19008 onboard storage cost & performance status	https://www.hydrogen.energy.gov/pdfs/19008_onboard_storage_cost_performance_status.pdf	25 Nov 2019	Record p.1 summary	modelled	search summary
W4-S3	2013: 350-bar \$15/kWh, 700-bar \$19/kWh single tank (500k/yr); Record 13010 \$16–20/kWh	US DOE / Strategic Analysis	Record #13010: "Hydrogen Storage Cost Analysis Final Public Report" (OSTI 1082754)	https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/13010_onboard_storage_performance_cost.pdf ; https://digital.library.unt.edu/ark:/67531/metadc827980/m2/1/high_res_d/1082754.pdf	11 Jun 2013 / 2013	abstract/summary	modelled	old dollar-year
W4-S4	DOE targets \$10/9/8 per kWh = \$333/300/266 per kg	US DOE	Technical Targets for Onboard Hydrogen Storage for LDVs	https://www.energy.gov/cmei/fuels/doe-technical-targets-onboard-hydrogen-storage-light-duty-vehicles	current page	Storage system cost row	target	targets not costs
W4-S5	12% cost reduction 2013 – 2015; volumes 10k–500k/yr	US DOE FCTO	Webinar: Onboard Type IV compressed H2 storage system cost	https://www.energy.gov/sites/prod/files/2016/03/f30/fcto_webinarslides_compressed_h2_storage_system_cost_022516.pdf	25 Feb 2016	slide 8	modelled	snippet
W4-S6	Compressed-gas storage USD 400–700 per kg H2 (Cerri et al.)	MDPI Energies 16(13):5233	A Review on the Cost Analysis of Hydrogen Gas Storage Tanks for Fuel Cell Vehicles	https://www.mdpi.com/1996-1073/16/13/5233	2023	Table citing Cerri et al. [33]	literature	secondary citation
W4-S7	Type IV 700-bar tanks ~\$700–1,200 per kg capacity	Dataintel (market report)	Cryo-Compressed Hydrogen Storage Market	https://dataintel.com/report/cryo-compressed-hydrogen-storage-market	2025	Opportunities & threats	estimate	low-grade commercial source
W4-S8	Time Technoplast PESO approvals (Mar/Jun 2024), 250 L, BV cert, ₹2.30 crore trial order, ₹555 crore 9MFY26	Business Standard; Autocar Professional; Time Technoplast press release; Scanx	multiple	https://www.business-standard.com/markets/capital-market-news/time-technoplast-gets-final-approval-from-peso-for-manufacturing-composite-cylinders-124062100121_1.html ; https://www.autocarpro.in/news/time-technoplast-secures-international-certification-for-hydrogen-composite-cylinder-126444 ; https://www.timetechnoplast.com/pdf/investors-calendar/press-release/approval-for-manufacturing-of-type-iv-composite-cylinders-for-hydrogen.pdf	2024–2026	news	observed	no unit prices
W4-S10	Bosch H2 PFI "suitable for retrofitting existing fleets"	Bosch Mobility	Port fuel injection for hydrogen engines	https://www.bosch-mobility.com/en/solutions/powertrain/gas/port-fuel-injection-for-hydrogen-engines/	current	product page	observed	no price
W4-S11	CNG kit prices ₹45–65k (BS6 OBD-II incl.	mycarhelpline; Cars24;	Best CNG kit brands 2026; CNG kit	https://www.mycarhelpline.com/index.php?	2025–2026	retail/trade	observed	listings vary by city

	install), Lovato ₹31,999–58,000, Landi Renzo ₹32,000, install ₹3–7k, RTO ₹2–3k	IndiaMART; Autonexa; IOAGPL	price in India; listings	option=com_easyblog&view=entry&id=557&Itemid=91 ; https://www.cars24.com/article/cng-kit-price-in-india/ ; https://www.indiamart.com/proddetail/landirenzo-cng-kit-23070488673.html ; https://autonexa.com/cng-fitting-cost-in-india-complete-guide-price-break-down-kits-and-savings ; https://ioagpl.com/cng-kit-price/		listings	(retail)	
W4-S12	ARAI retrofit approval under Rule 115; "full approval in 45* days"; fee schedule not public; Scribd cost-breakdown doc exists; ₹15–25 lakh homologation anecdote	ARAI; Scribd; Quora	Retrofitment certification; Vehicle type approval; "Retrofitment Kit Approval"	https://www.araiindia.com/certification/certification/retrofitment ; https://www.araiindia.com/certification/certification/vehicle ; https://www.scribd.com/document/975757543/Retrofitment-Kit-Approval-1	current	pages	observed / anecdote	fees NOT FOUND
W4-S13	PFI gasoline conversion needs injectors + plugs; DI needs ignition/injection; custom pistons	Ultimate Cell UK; OSTI 1008335 (NETL small H2 engine)	trade guides	https://ultimatecelldistributors.co.uk/insights/hydrogen-retrofit-for-diesel-petrol-engines/ ; https://www.osti.gov/servlets/purl/1008335	2020s / 2011	—	qualitative	low grade
W4-S14	HORIBA H2-ICE test bed Pune, ₹28–30 crore, ≤380 kW, Dec 2024	HORIBA; Autocar Professional	HORIBA inaugurates first H2-ICE test bed facility in India	https://www.horiba.com/int/mobility/news/detail/news/12/2024/20241210/ ; https://www.autocarpro.in/news/horiba-opens-hydrogen-internal-combustion-engine-test-bed-facility-in-pune-123903	10 Dec 2024	press release	observed	—
W4-S15	ARAI ₹500 crore H2-ICE test infra (from 2025); Takwe cylinder test facility; ₹100 crore labs	Autocar Professional; Pune.news; Telematics Wire	ARAI to invest Rs 500 crore...; ARAI to establish hydrogen cylinder testing facility	https://www.autocarpro.in/news/arai-to-invest-rs-500-crore-to-enhance-infra-h2ice-testing-likely-from-2025-120865 ; https://www.autocarpro.in/news/arai-to-establish-a-hydrogen-cylinder-testing-facility-at-greenfield-takwe-near-pune-115086	2023–2024	news	announced	plans, not completions
W4-S16	CNG Delhi ₹86.98/kg	Goodreturns; CarDekho	CNG price in New Delhi today	https://www.goodreturns.in/cng-price-in-new-delhi.html ; https://www.cardekho.com/cng-price-in-delhi-state	14 Sep 2026	price page	observed (aggregator)	IGL site blocked
W4-S17	MGL CNG ₹88/kg from 1 Sep 2026 (+₹2), PNG ₹53; spot RLNG cited	BusinessToday; IndianPSU; Goodreturns	MGL hikes CNG, PNG prices in Mumbai from September 1	https://www.businesstoday.in/latest/story/mgl-hikes-cng-png-prices-in-mumbai-from-september-1-check-revised-rates-552419-2026-08-31 ; https://indianpsu.com/mgl-cng-price-hike-mumbai-88-september-2026/ ; https://www.goodreturns.in/cng-price-in-mumbai.html	31 Aug–14 Sep 2026	news	observed	—
W4-S18	Ahmedabad CNG ₹94.02 (7 Sep 2026); ₹86.02 (16 May 2026); ₹83.77 (3 Apr 2026)	Goodreturns; DeshGujarat; Vibes of India	CNG price in Ahmedabad; Adani CNG hikes	https://www.goodreturns.in/cng-price-in-ahmedabad.html ; https://deshgujarat.com/2026/05/16/adani-gas-imposes-another-price-hike-cng-rate-in-ahmedabad-now-at-rs-86-02-per-kg/ ; https://deshgujarat.com/2026/04/03/adani-cng-price-hiked-by-%E2%82%B91-50-rate-reaches-%E2%82%B983-77-amid-global-gas-crisis/	2026	news/price page	observed	—
W4-S19	Petrol Delhi ₹102.12/L (25 May 2026), crude basket \$113–114; aggregator ₹111.21 (14 Sep 2026); Ahmedabad ₹102.15 vs ₹94.59 conflict	Univest; PingTV; CarDekho; Goodreturns; DriveSpark; Business Standard	Petrol price hike to Rs 100 in Delhi...; petrol price pages	https://univest.in/blogs/petrol-price-hike-crosses-100-delhi-fourth-hike-10-days-may-2026 ; https://www.pingtvindia.com/fuel-price-hike-india-petrol-price-delhi-crossed-100/ ; https://www.cardekho.com/petrol-price ; https://www.goodreturns.in/petrol-price-in-ahmedabad.html ; https://www.drivespark.com/petrol-price-in-ahmedabad/	May–Sep 2026	news/price pages	observed	Sep figures unverified vs IOCL
W4-S20	DERC EV tariff ₹4.50/kWh LT, ₹4.00/kVAh HT, zero fixed; domestic ₹3–8; DERC 1 Jul 2026 upstream-cost order	thediscombill; bridgewaypower; easyEVplug; Energetica India; MeraEV	Delhi electricity tariff 2025-26; DERC directs...	https://thediscombill.com/tariffs/delhi/ ; https://www.easyevplug.com/ev-charger-installation-delhi ; https://energetica-india.net/news/derc-directs-full-upstream-ev-charging-infra-costs-to-be-included-in-demand-notes-under-pm-e-drive	2025–2026	tariff summaries	observed (secondary)	DERC order not opened
W4-S21	Public DC fast charging ₹16–24/kWh; Tata Power ₹17; ChargeZone ₹20–25; Statiq ₹15–22	Pulse Energy; searchev.co; MeraEV; GreenTax	EV charging rates in India 2026	https://pulseenergy.io/blog/ev-charging-rates ; https://searchev.co/cost-for-charging-electric-cars/ ; https://www.meraev.com/blog/ev-charging-stations-india-types-costs-guide	2026	guides	observed (secondary)	operator tariffs vary by state
W4-S22	CEEW TCO: charging tariffs; H2 production/dispensing/delivered cost 2030–2050; taxi/4W findings; H2 bus competitiveness; maintenance Table 5; 3W/2W ₹/km	CEEW	What Drives Vehicle Ownership Costs in India? / Cost of Ownership for India's Road Transport Sector	https://www.ceew.in/publications/cost-of-ownership-for-road-transport-sector-for-different-vehicle-segments-fuels-and-powertrains ; https://www.ceew.in/sites/default/files/cost-of-ownership-for-different-vehicle-segments-fuels-and-powertrains.pdf	Jun–Jul 2025 / 2024 base	Exec summary; Fuel costs; Green hydrogen; Maintenance; Results \$4 (verbatim extracts)	modelled	Tables 3 & 5 values not readable
W4-S23	ADAC charging losses AC 5–10% (wallbox), 24–30% (socket), DC 5–15%	Zecar; InsideEVs; TheNextWeb; vision-mobility (reporting ADAC)	EV charging losses up to 24%...; How efficient is each type of EV charger	https://zecar.com/reviews/ev-charging-losses-adac-study ; https://insideevs.com/features/711659/ev-charger-efficiency-losses/ ; https://vision-mobility.de/en/news/adac-analysis-less-energy-is-lost-at-fast-chargers-384109.html	2022–2024	test summaries	measured	ADAC report not opened
W4-S24	GST: CNG small cars & parts 28% → 18% (Sep 2025); CNG-tank classification dispute 18% vs 28%	EY; Taxo; SAG Infotech	GST Council announces major rate rationalization (4 Sep 2025); GST notices to CNG kit providers	https://www.ey.com/content/dam/ey-unified-site/ey-com/en-in/alerts-hub/2025/09/gst-council-announces-major-rate-rationalization-and-trade-facilitation-measures.pdf ; https://taxo.online/latest-news/28-12-2024-gst-notices-sent-to-over-a-dozen-cng-kit-providers/	Sep 2025; Dec 2024	alert; news	observed	kit rate unresolved
W4-S25	Delhi EV Policy 2026: notified 30 Jun 2026, in force 1 Jul 2026–31 Mar 2030; 100% road tax & registration waiver BEV ≤₹30 lakh; hybrids excluded	EV Infrastructure News; Akashvani News; Coverfox; MeraEV	Delhi notifies EV Policy 2026...	https://www.evinfrastructurereports.com/ev-policies/delhi-notifies-ev-policy-2026-with-tax-exemptions-for-ev ; https://newsnair.gov.in/electric-cars-priced-upto-30-lakh-rupees-and-registered-in-delhi-to-be-granted-100-exemption-on-road-tax/	Jun–Jul 2026	news	observed	gazette not opened
W4-S26	Tata 8 yr/160,000 km (Tiago.ev, Punch.ev, Curvv.ev, Nexon.ev 45), lifetime first owner;	Tata.ev; Cars24 news; ev.care	Tata EV warranty; MG ZS EV warranty explained	https://ev.tata.cars/service/warranty.html ; https://www.cars24.com/news/auto/tata-nexon-ev-lifetime-	2025–2026	warranty pages	observed	—

	MG ZS EV 8 yr/150,000 km; Windsor/Comet lifetime			battery-warranty-details-revealed/ ; https://ev.care/blog/mg-zs-ev-warranty-guide				
W4-S27	Geotab degradation 1.8%/yr (2024) → 2.3%/yr (2025 update); >90% after 160,000 km	Geotab	EV battery health study; 2024 battery degradation update	https://www.geotab.com/blog/ev-battery-health/ ; https://www.geotab.com/uk/press-release/2024-battery-degradation/	2024–2025	blog/press	measured	fleet mix not Indian
W4-S28	Used Dzire CNG 2022 ₹6.09 lakh (56k km); True Value 2022 ₹6.40 lakh / 2024 ₹7.40 lakh; new Tour S CNG ₹7.09 lakh; on-road ₹6.24–8.51 lakh	Cars24; Maruti True Value; Cars24 new cars; CarDekho	listings	https://www.cars24.com/buy-used-maruti-dzire-cars-new-delhi/ ; https://www.marutisuzukitruevalue.com/used-dzire ; https://www.cars24.com/new-cars/maruti-suzuki/dzire-tour/ ; https://www.cardekho.com/maruti/dzire-tour-s/price-in-new-delhi	Sep 2026	listings	observed (asking prices)	not taxi-registered cars
W4-S29	NBFC rates: Shriram from 10%; taxi 9–15%; Chola 9.35–17.35% used car, 10–21% CV; NBFC 13–18%	Shriram Finance; upniva (Cholamandalam); Asymmetrica/calcwise	rate pages	https://www.shriramfinance.in/articles/used-car-loan/2025/understanding-used-car-loan-interest-rates-for-taxi-drivers ; https://upniva.com/loans/cholamandalam-used-car-loan ; https://asymmetrica.in/calculators/car-loan-rates/	2025–2026	rate pages	observed	indicative
W4-S30	Taxi 70,000 km/yr; car 11,500; bus 75,000 (Table A.3)	NITI Aayog	Scenarios Towards Viksit Bharat and Net Zero — Sectoral Insights: Transport	https://www.niti.gov.in/sites/default/files/2026-02/Scenarios-Towards-Viksit-Bharat-and-Net-Zero-Sectoral-Insights-Transport.pdf	Feb 2026	Annex Table A.3 (verbatim)	modelling assumption	—
W4-S31	ICCT ride-hailing TCO ₹7.6/km; ~34% Delhi / ~20% Hyderabad with incentives; ₹60,000 bulk discount	ICCT	Updated analysis of incentives for electrifying India's four-wheeler ride-hailing fleet	https://theicct.org/wp-content/uploads/2021/06/update-electrifying-india-ride-hailing-fleet-apr2021.pdf	Apr 2021	working paper (snippets)	modelled	daily-km assumption not extracted
W4-S32	Driver anecdotes ~300 km/day incl. empty running	Quora	—	https://www.quora.com/How-many-kilometers-on-average-do-Ola-or-Uber-cab-drivers-drive-in-a-single-day	n.d.	—	anecdotal	low grade
W4-S33	Green H2 USD 3.5–5/kg (industry); breakeven <USD 2; USD 1.8/kg policy levers	CEEW	How can India Boost Investment for Domestic Green Hydrogen?	https://www.ceew.in/publications/how-can-indian-policy-makers-boost-investments-for-domestic-green-hydrogen-financing	2024–2025	Challenge 1 (verbatim)	estimate	—
W4-S34	₹397–560/kg (\$4.6–6.7/kg). 2–3× grey	GH2 India	India's Green Molecules Moment: ACME's new deals...	https://www.gh2.org.in/indias-green-molecules-moment-acmes-new-deals-show-the-market-is-finally-moving-from-announcements-to-offtake	2025–2026	article (verbatim)	estimate	advocacy body
W4-S36	Green H2 \$4–7/kg today; \$1.60 (2030), \$0.70 (2050) best case; \$1.4/kg most aggressive; \$1/kg goal	NITI Aayog & RMI	Harnessing Green Hydrogen	https://www.niti.gov.in/sites/default/files/2022-06/Harnessing_Green_Hydrogen_V21_DIGITAL_29062022.pdf ; https://www.pib.gov.in/PressReleasePage.aspx?PRID=1838104	29 Jun 2022	cost chapter (summary)	modelled/target	search summary
W4-S37	At-pump green H2 ~\$5/kg (₹350) mid, \$3.5 (₹245) optimistic (2050)	ICCT; EurekAlert	Hydrogen fuel for transport in India	https://theicct.org/publication/india-fuel-hydrogen-cost-jan22/ ; https://www.eurekalert.org/news-releases/941197	Jan 2022	summary	modelled	year attribution uncertain
W4-S38	Green ammonia auctions ₹55.75 (first), ₹49.75 lowest, ₹53.27 weighted avg, 724,000 t/yr; ₹54.73 8th; ₹52.50 Onix	PIB; pv magazine India; S&P Global; ACME; Fuel Cells Works	SECI first green ammonia auction...; Green ammonia only 10% costlier...; Low-priced Indian renewable ammonia auctions	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2153006 ; https://www.pv-magazine-india.com/2025/08/28/green-ammonia-is-only-10-costlier-than-grey-in-the-latest-sight-mode-2a-auctions/ ; https://www.spglobal.com/commodity-insights/en/news-research/latest-news/energy-transition/090425-low-priced-indian-renewable-ammonia-auctions-a-meaningful-signal-h2global	Jul–Sep 2025	news	observed (auction)	conversion to H2 is a calculation
W4-S39	SIGHT Mode-1 Tranche-II 450,000 t/yr, ₹2,239 crore; ₹50/40/30 per kg	Renewable Watch; JMK; SolarQuarter	SECI announces results of green hydrogen auction under SIGHT	https://renewablewatch.in/2025/03/20/seci-announces-results-of-green-hydrogen-auction-under-sight-programme/ ; https://jmkresearch.com/seci-awards-450000-mt-annual-capacity-under-sight-tranche-ii-for-green-hydrogen-production/	Mar 2025	news	observed	—
W4-S40	Electrolyser Tranche-II 1.5 GW, 11 winners, ₹4,440 crore FY26–30	Mercom; pv magazine India; MNRE	MNRE allocates ₹44.4 bn...; SECI awards LoAs...	https://www.mercomindia.com/mnre-electrolyzer-manufacturing-sight-ii-program ; https://www.pv-magazine-india.com/2025/01/21/the-hydrogen-stream-seci-awards-loas-for-electrolyser-production-linked-incentives/	Jan 2025	news	observed	—
W4-S41	Electrolyser ₹30,000/kW alkaline, ₹42,000/kW PEM (1 MW, 300 MW/yr)	JMK Research / World Bank technical report	India's \$2.1bn leap towards its green hydrogen vision; Technical and Economic Characteristics...	https://jmkresearch.com/wp-content/uploads/2025/02/Indias-2.1bn-leap-towards-its-Green-Hydrogen-vision.pdf ; https://documents1.worldbank.org/curated/en/09022326123011297/pdf/P506220-b45b0865-8706-4862-8792-d8cc71f84569.pdf	Feb 2025 / 2026	summary	modelled	attribution between the two docs unclear
W4-S42	Grey H2 \$1.5–1.6/kg stable, \$3–7.5 under LNG shock; gap \$1.6–3.2; SMR ₹181–229/kg (LinkedIn)	RMI; LinkedIn post	Green Hydrogen Production Pathways for India	https://rmi.org/green-hydrogen-production-pathways-for-india/ ; https://www.linkedin.com/posts/sanjeevsharmagh2_refineryhydrogen-greenhydrogen-iocl-activity-7454393902108889088-B4dk	2024–2026	exec summary (verbatim)	estimate	LinkedIn low grade
W4-S43	Station cost \$1,200–3,000 per kg/day (770–1,620 kg/day, 700 bar)	US DOE	Record 21002 Hydrogen Fueling Station Cost	https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/21002-hydrogen-fueling-station-cost.pdf	2 Nov 2020	summary	observed (grants)	blocked; snippet
W4-S44	Delivery+dispensing ~\$11/kg @450 kg/d, ~\$8/kg @1,000 kg/d (liquid; HDSAM 3.1)	US DOE	Record 20007 Hydrogen Delivery and Dispensing Cost	https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/20007-hydrogen-delivery-dispensing-cost.pdf	25 Aug 2020	summary	modelled	blocked; snippet
W4-S45	350-bar delivery+dispensing \$2.70–3.70/gge (HDSAM 2.3, mature)	US DOE	Record 16011 H2 delivery cost 2015	https://www.hydrogen.energy.gov/pdfs/16011_h2_delivery_cost_2015.pdf	2015	summary	modelled	mature-market case
W4-S46	200 kg/day station \$6–8/kg (gaseous), \$8–9 (liquid); utilisation 50% (2030), 70% (2050)	ICCT	Cost of renewable hydrogen produced onsite at HRS in Europe	https://theicct.org/publication/fuels-eu-onsite-hydro-cost-feb22/	Feb 2022	white paper summary	modelled	snippet
W4-S47	NREL 88818 scope (700 bar, 2022\$); ATB	NREL	Levelized Cost of Dispensed Hydrogen	https://docs.nrel.gov/docs/fy24osti/88818.pdf ;	Mar 2024	ATB table	modelled	88818 tables unread

	2024 H2 price ranges 6.2–12.0 / 8.2–18.0; future 6.2–7.0; 4.0 target		for Heavy-Duty Vehicles (TP-5400-88818); ATB 2024 Hydrogen	https://atb.nlr.gov/transportation/2024/Hydrogen		(verbatim)		
W4-S48	LCOH \$11.51 (2025) → \$4.26/kg (2050); UR 0.3/0.5/1.0; breakeven \$4.90–5.85	Elsevier (journal article, S2590116826000421)	Breakeven levelized cost of hydrogen for the zero-emission freight transition	https://www.sciencedirect.com/science/article/pii/S2590116826000421	2026	Highlights; §4.3 (verbatim)	modelled	freight context
W4-S49	350-bar compressor/storage cost inputs used by CEEW	Wiley (Chem. Ing. Tech. 96)	Detailed Cost Analysis of Hydrogen Refueling Costs for Fleets (Eißler et al.)	https://onlinelibrary.wiley.com/doi/full/10.1002/cite.202300123	2024	—	modelled	not opened
W4-S50	California \$32–36/kg; True Zero \$36	S&P Global; hydrogenfuelfinder; Stillwater	California hydrogen pump prices reach new highs; Why hydrogen fuel costs \$36/kg in 2025	https://www.spglobal.com/energy/en/news-research/latest-news/energy-transition/100124-california-hydrogen-pump-prices-for-light-duty-vehicles-reach-new-highs ; https://hydrogenfuelfinder.com/blog/hydrogen-fuel-cost-2025-reality-check.html	2024–2025	news	observed	—
W4-S51	Germany €10–15/kg retail; €8/kg hylane deal from 1 Jan 2026	H2.LIVE / H2 MOBILITY; Driving Hydrogen; glpautogas	Hydrogen for 8 euros per kilogram now possible	https://h2.live/en/press/hydrogen-for-8-euros-per-kilogram-now-possible-hylane-and-h2-mobility-enable-cost-effective-operation-of-hydrogen-trucks/ ; https://www.glpautogas.info/en/hydrogen-sale-price-germany.html	Nov 2025	press release	observed	fleet deal
W4-S52	Japan Iwatani ¥1,210 → ¥1,650/kg (Jun 2025); ¥700/kg subsidy	Hydrogen Insight; Driving Hydrogen	Price hike...; Japan to introduce ¥700/kg subsidy	https://www.hydrogeninsight.com/transport/price-hike-hydrogen-vehicle-drivers-to-be-charged-33-more-for-fuel-at-majority-of-h2-filling-stations-in-japan/2-1-1628765 ; https://drivinghydrogen.com/2025/05/30/japan-to-introduce-%C2%A5700-kg-hydrogen-fuel-subsidy-to-kickstart-commercial-uptake/	2025	news	observed	—
W4-S53	NTPC Leh: 1.7 MW solar, 80 kg/day 350-bar station, 5 FC buses (25 kg, 300 km)	NTPC	NTPC hands over five hydrogen fuel cell buses to Leh Administration	https://ntpc.co.in/media/press-releases/ntpc-hands-over-five-hydrogen-fuel-cell-buses-leh-administration	Jun 2025	press release	observed	no cost data
W4-S54	MNRE Phase-1 pilots: 5 projects, 37 vehicles (15 FCEV + 22 H2-ICE), 9 HRS, ₹208 crore, 10 routes, 18–24 months	PIB; Down To Earth	Pilot Projects on Hydrogen Fuelled Buses and Trucks Launched under NGHM	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2107795 ; https://www.downtoearth.org.in/environment/as-told-to-parliament-march-25-2025-37-cross-country-vehicles-9-fuelling-stations-planned-under-green-hydrogen-mission	Mar 2025	press release	observed (sanction)	—
W4-S55	Phase-2: 12 projects, 70 vehicles (27 buses/43 trucks), 16 HRS, 21 routes, ₹410.77 crore (₹41.58 released); ₹292 crore NGHM spend to date	pv magazine India; Fuel Cells Works; EQ; Business Standard; Renewable Watch	India awards 12 green hydrogen mobility pilots...; Govt spends ₹292 crore under NGHM so far	https://www.pv-magazine-india.com/2026/08/03/the-hydrogen-stream-india-awards-12-green-hydrogen-mobility-pilots-with-inr-410-77-crore-support/ ; https://fuelcellworks.com/2026/08/02/fuel-cells/india-s-hydrogen-vehicle-pilot-nearly-doubles-70-vehicles-16-refuelling-stations-410-77-crore ; https://www.business-standard.com/amp/industry/news/govt-spends-292-crore-under-national-green-hydrogen-mission-so-far-mnre-126081201853_1.html	Jul–Aug 2026	news	observed (sanction)	MNRE order not opened
W4-S56	HRS ₹14–20 crore; 200–500 kg/day break-even; "₹4/km" hydrogen	VahanBazaar; Tata AIG blog	Hydrogen refuelling stations India 2030	https://vahanbazaar.in/tips/hydrogen-refuelling-stations-india-2030-vahanbazaar ; https://www.tataaig.com/knowledge-center/car-insurance/hydrogen-cars-in-india	2026	blog	estimate	low grade
W4-S57	IEA GHR 2026: India/Middle East/Chile "just above USD 3/kg" by 2030; China <USD 2	Green Hydrogen Organisation (reporting IEA GHR 2026)	IEA Global Hydrogen Review 2026...	https://gh2.org/iea-global-hydrogen-review-2026-green-hydrogen-can-help-prevent-next-food-and-energy-crisis	18–19 Jun 2026	article (verbatim)	modelled	IEA site blocked
W4-S58	IEA GHR 2025: project pipeline shrank; electrolysis >80% of drop; assumptions annex	IEA	Global Hydrogen Review 2025 (+ Assumptions Annex)	https://www.iea.org/reports/global-hydrogen-review-2025 ; https://iea.blob.core.windows.net/assets/15673ab3-a86a-4434-bff4-490bb42d3563/GlobalHydrogenReview2025AssumptionsAnnex.pdf	Sep 2025	exec summary	modelled	blocked; India LCOH NOT FOUND
W4-S59	CEA WAEF FY2024-25 0.710 (prov.), FY2023-24 0.727; BM 0.512; CM definition	CEA (user guide v21.0); batchwise.ai; reclimate; LinkedIn (R. Goenka)	CO2 Baseline Database for the Indian Power Sector, User Guide v21.0	https://cea.nic.in/wp-content/uploads/baseline/2025/12/User_Guide_V_21.0.pdf ; https://batchwise.ai/methodology/cea-grid-emission-factors/all-india/	Dec 2025 / FY2024-25	CM definition (verbatim); WAEF (secondary)	observed (provisional)	primary blocked; T&D note unread
W4-S60	NEP 2022-32: EF 0.548 (2026-27), 0.430 (2031-32) kg/kWh net; non-fossil 57.4% / 68.4%	CEA; PIB	National Electricity Plan 2022-32 (Vol. I Generation)	https://cea.nic.in/wp-content/uploads/irp/2023/05/NEP_2022_32_FINAL_GAZETTE-1.pdf ; https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=1928750&reg=48&lang=2	May 2023	summary	projected	snippet
W4-S61	India default fuel EFs exist (GLEC-aligned)	Smart Freight Centre India	India Default GHG Emission Values V1.0	https://smart-freight-centre-media.s3.amazonaws.com/documents/India_Default_GHG_Emission_Values_V1_SFC_INDIA_latest.pdf	2023–2024	—	published	not opened
W4-S62	IPCC default EFs 69,300 (gasoline), 56,100 (natural gas) kg CO2/TJ; NCV 44.3 / 48.0 TJ/Gg	IPCC	2006 IPCC Guidelines Vol. 2 Energy, Ch. 1 Tables 1.2 & 1.4	https://ghgprotocol.org/Third-Party-Databases/IPCC-Emissions-Factor-Database	2006	Tables 1.2, 1.4	default	from memory — verify before use
W4-S63	ICCT 2025 LCA values (235/234/203/188/163/63/52/50/175 g/km; H2 13.6 & 1.2 kg CO2e/kg; battery 72.8 kg/kWh; +40% production; 17,000 km; TTW non-CO2)	ICCT (Negri & Bieker)	Life-cycle GHG emissions from passenger cars in the EU: a 2025 update	https://theicct.org/wp-content/uploads/2025/07/ID-392-%E2%80%93Life-cycle-GHG_report_final.pdf ; https://theicct.org/publication/electric-cars-life-cycle-analysis-emissions-europe-jul25/	8 Jul 2025	Key findings; Fuel-cycle §; Figure 4 (verbatim extracts)	modelled	battery chemistry attribution unverified
W4-S64	Unabated NG 10–14 kg CO2e/kg (upstream 1–5); solar PV electrolysis 0.9–2.5; downstream 0.4–2.7	IEA; CRS R48196	Towards hydrogen definitions based on their emissions intensity; Hydrogen Production: Overview and Issues for	https://www.iea.org/reports/towards-hydrogen-definitions-based-on-their-emissions-intensity ; https://www.congress.gov/crs-product/R48196	Apr 2023; Oct 2024	exec summary	modelled	snippet

			Congress					
W4-S65	India green-H2 standard ≤2 kg CO2e/kg well-to-gate	MNRE (via CSEP/RMI summaries)	Green Hydrogen Standard for India	https://csep.org/key-insights/green-hydrogen-pathways-in-india-from-a-grid-perspective/	Aug 2023	—	regulation	primary not opened
W4-S66	Medium petrol car 54.1 t, PHEV 36.9 t, BEV 25.0 t (15 yr, 42 km/day, STEPS); BEV lower after 2 yr	IEA	EV Life Cycle Assessment Calculator	https://www.iea.org/data-and-statistics/data-tools/ev-life-cycle-assessment-calculator	2024–2025	tool defaults (verbatim)	modelled	global assumptions
W4-S67	b24X 26 t (25–31); Corolla Cross hybrid 53 t (44–70)	Green NCAP	European LCA results 2024	https://www.greenncap.com/european-lca-results/	2024	results table (verbatim)	modelled	EU mix
W4-S68	DI H2 engine λ 2.4–3.0, BTE 43%, WLTC tailpipe NOx <35 mg/km with urea-SCR	ACS Omega 11(7):12788 (PMC12947045)	NOx/N2O Formation and Coordinated Urea-SCR Control in an Ultra-Lean Hydrogen-Fueled Automotive SI Engine	https://pmc.ncbi.nlm.nih.gov/articles/PMC12947045/	2026	abstract	measured/ modelled	full text not indexed
W4-S69	55 kW 4-cyl PFI H2ICE NRMM tailpipe NOx variants	Environ. Sci.: Processes & Impacts (doi 10.1039/d4em00448e)	Evaluating low NOx hydrogen engines designed for off-road and construction applications	https://doi.org/10.1039/d4em00448e	Feb 2025	abstract	measured	values inaccessible
W4-S70	Base H2 car cannot stay ultra-lean at high load; hybrid EMS ~90% engine-out NOx, ~16% H2; Euro 7 60 mg/km fuel-neutral	arXiv (ETH Zürich / Bosch)	Energy Management of Hydrogen Hybrid Electric Vehicles — A Potential Study	https://arxiv.org/abs/2309.09804	Sep 2023	\$1, \$3.1–3.2, Table 4 (verbatim)	simulated	simulation only
W4-S71	BS-VI M1 NOx 60 mg/km petrol; 80 diesel	ICCT; DieselNet; ARAI	India Bharat Stage VI emission standards (policy update); Indian Emission Regulation Booklet	https://theicct.org/wp-content/uploads/2021/06/India-BS-VI-Policy-Update-vF.pdf ; https://dieselnet.com/standards/in/d.php ; https://www.araiindia.com/pdf/Indian_Emission_Regulation_Booklet.pdf	2016/2021	limit tables	regulation	CNG row not verbatim
W4-S72	H2 indirect GWP100 11 ± 4; GWP20 37 ± 18; OH-bias 20% overestimate; leakage 1–10%	Nature 2025 (PMC12711571); Ocko & Hamburg 2022 (ACP); arXiv 2408.05127; PMC10976891	The global hydrogen budget; Climate consequences of hydrogen leakage; Model underestimates of OH reactivity...	https://pmc.ncbi.nlm.nih.gov/articles/PMC12711571/ ; https://acp.copernicus.org/preprints/acp-2022-91/acp-2022-91.pdf ; https://arxiv.org/abs/2408.05127	2022–2025	abstracts	modelled	O&H exact values not verified
W4-S73	India VCM ₹1,350–2,800/t (2026); MSCI index \$3.50/t (2025), \$4.30 (2024); CCTS offset mechanism	Grow Billion Trees; Senken; Regreener; CEEW	Carbon credit rate in India 2026; Carbon credit prices 2026; India's voluntary offset scheme	https://growbilliontrees.com/pages/carbon-credit-rate-in-india-updated-monthly-prices ; https://www.senken.io/academy/understanding-carbon-markets/carbon-credit-price ; https://www.ceew.in/publications/voluntary-carbon-offset-mechanism-and-challenges-in-carbon-credit-trading-scheme-market-for-india	2025–2026	price trackers	observed	Indian tracker low grade
W4-S74	Transport pilot guidelines 14 Feb 2024: ₹496 crore to FY26; buses/trucks/cars <3.5 t; steering committee	JSA; Lexology; Business Standard; Deloitte	MNRE issues Scheme Guidelines for pilot projects on use of green hydrogen in the transport sector	https://www.jsalaw.com/newsletters-and-updates/mnre-issues-scheme-guidelines-for-pilot-projects-on-use-of-green-hydrogen-in-the-transport-sector/ ; https://www.business-standard.com/industry/news/govt-issues-guidelines-for-transport-pilot-projects-on-green-hydrogen-s-use-124021401729_1.html	Feb 2024	summaries	regulation	guideline PDF blocked; funding share per vehicle not extracted
W4-S75	R&D scheme ₹400 crore; guidelines 15 Mar 2024; 1st call 23 projects; 2nd call 14 Jul–21 Aug 2025; TRL ≥4; 100%/80% funding; ₹200 crore testing-infra call	MNRE; Mercom; SolarQuarter; CSIR; pv magazine India	Second Call for Proposals under R&D Scheme of NGHM	https://mnre.gov.in/en/notice/second-call-for-proposals-under-research-and-development-scheme-of-national-green-hydrogen-mission-nghm/ ; https://www.mercomindia.com/mnre-invites-second-round-of-rd-proposals-under-green-hydrogen-mission ; https://www.csir.res.in/en/news-calls-and-events/second-call-proposals-under-research-and-development-scheme-national-green ; https://www.pv-magazine-india.com/2025/08/12/mnre-launches-second-call-for-proposals-to-build-and-upgrade-green-hydrogen-testing-infrastructure/	Mar 2024–Aug 2025	notices/news	regulation	remaining budget NOT FOUND
W4-S76	Gujarat GH2 Policy 2025 incentives (electrolyser, HRS, buses, MSME)	Mercom India; iamrenew; JMK (policy PDF)	Gujarat's new Green Hydrogen Policy targets 3 MMTPA by 2035; Gujarat Green Hydrogen Policy 2025	https://www.mercomindia.com/gujarats-new-green-hydrogen-policy-targets-3-mmtpa-by-2035 ; https://jmkresearch.com/wp-content/uploads/2025/12/Gujarat-Green-Hydrogen-Project-Policy-2025-.pdf	late 2025	summaries	regulation	gazette not opened
W4-S77	UP GH2 Policy 2024: 30% (40% first 5) capex, cap ₹225 crore; land ₹15,000/acre/yr	Invest UP; Business Standard	Uttar Pradesh Green Hydrogen Policy 2024	https://invest.up.gov.in/wp-content/themes/investup/sector-assets/policies_schemes/Uttar_Pradesh_Green_Hydrogen_Policy_2024.pdf ; https://www.business-standard.com/economy/uttar-pradesh/uttar-pradesh-green-hydrogen-policy-2024-key-features-incentives-eligibility-and-implementation-126090700729_1.html	2024 / Sep 2026 explainer	summaries	regulation	no mobility incentive found
W4-S78	Maharashtra GH2 Policy 2023: ₹50/kg blending; 500 FCEV cars 30% ≤₹60 lakh; 20 HRS 30% ≤₹4.5 crore; ₹8,500 crore	MAHAURJA (FAQ); JMK; Swaraja; Global Trade Alert	Maharashtra Green Hydrogen Policy 2023 FAQs	https://www.mahaurja.maharashtra.gov.in/Site/Upload/Pdf/GHfaq.pdf ; https://jmkresearch.com/wp-content/uploads/2023/11/Maharashtra-Green-Hydrogen-Policy-2023.pdf	Oct 2023	FAQ	regulation	fuel-cell wording
W4-S79	AIS 195:2023 for hydrogen ICE M/N vehicles (GSR 746(E)); AIS 195A:2024 CEV	MoRTH	GSR 746(E) dated 16 Oct 2023 — Hydrogen ICE	https://morth.nic.in/sites/default/files/notifications_document/8th%20GSR%20746(E)%20dated%2016th%20october%202023%20Hydrogen%20ICE.pdf	16 Oct 2023	notification	regulation	not opened
W4-S80	AIS-137 durability × DF; hydrogen reference fuel; CMVR 115/116/126	ARAI (AIS-137 Parts 3/4/6)	AIS-137 Automotive Industry Standard	https://hmr.araiindia.com/Control/AIS/45201991403AM AIS_137_Part_3_F.pdf ;	2019 + amendment	procedure sections	regulation	snippet

				https://hmr.araiindia.com/Control/AIS/53201963840PMAIS_137_Part_4_F.pdf	s			
W4-S81	IOCL Faridabad: H2 dispensing since 2005; solar-electrolysis refuelling; PESO permission; Fluitron dispenser Jan 2023	IndianOil; BusinessWire	R&D Centre pages; Fluitron installs first domestically built hydrogen dispenser	https://iocl.com/pages/r-and-d-centre ; https://www.businesswire.com/news/home/20230116005549/en/Fluitron-Installs-First-Domestically-Built-Hydrogen-Dispenser-in-Faridabad-India	2023–2025	pages	observed	—
W4-S82	Tata Prima H.55S H2 specs; trials 4 Mar 2025; 40 trucks VOC Port by 2028; Cummins Jamshedpur production Mar 2024; \$425 m JV	Tata Motors; electrive; Cummins; Hydrogen Insight; Truck Junction	Tata Motors drives India's green future...; Tata Motors to deploy 40 H2 ICE trucks; Cummins produces H2-ICE for Tata Motors	https://www.tatamotors.com/press-releases/tata-motors-drives-indias-green-future-with-countrys-first-hydrogen-truck-trials/ ; https://www.electrive.com/2026/02/27/tata-motors-to-deploy-40-h2-ice-trucks-in-india/ ; https://www.cummins.com/en-na/news/releases/2024/05/01/cummins-produces-hydrogen-internal-combustion-engines-tata-motors-new	2024–2026	press	observed	—
W4-S83	Ashok Leyland–Reliance H2-ICE truck; AL fleet 2026	Team-BHP; Business Standard	Ashok Leyland & Reliance unveil India's first hydrogen-ICE truck; India's vehicles industry quietly begins shift...	https://www.team-bhp.com/news/ashok-leyland-reliance-unveil-india%E2%80%99s-first-hydrogen-ice-truck ; https://www.business-standard.com/industry/auto/indian-cv-majors-drive-hydrogen-transition-with-h2-vehicles-125061600725_1.html	Feb 2023; Jun 2025	news	observed	—
W4-S84	Suzuki hydrogen Burgman (JMS 2025); hydrogen Swift 1.4 L DI 100 kW/220 Nm; Toshihiro Suzuki BMGE 2025 quote; Maruti H-CNG	Suzuki (globalsuzuki); DriveSpark; Automotive World; evo India; V3Cars	Hydrogen Engine Burgman; Hydrogen-powered Suzuki Swift revealed; Suzuki unveils three new motorcycle models...	https://www.globalsuzuki.com/mobilityshow2025/burgman/ ; https://www.drivespark.com/four-wheelers/2026/suzuki-hydrogen-powered-swift-revealed-084763.html ; https://www.automotiveworld.com/news-releases/suzuki-unveils-three-new-motorcycle-models-including-an-ev-scooter-at-the-bharat-mobility-global-expo-2025/ ; https://www.v3cars.com/news/maruti-suzuki-hydrogen-cng-hcng-engine-for-future-vehicles	2025–2026	press/news	observed	no retrofit statement
W4-S86	IIT Madras–Hyundai HTWO centre operational by 2026	Careers360	IIT Madras unveils Hyundai HTWO Centre	https://news.careers360.com/iit-madras-unveils-hyundai-htwo-centre-on-campus-operational-by-2026-iitm-hydrogen-india-research/amp	2025	news	announced	—

A3. Cost model: bill of materials, ownership cost, break-even and sensitivities

A3.1 Retrofit bill of materials (pilot batch versus commercial volume)

#	Component or activity	Qty	Unit cost pilot (₹)	Unit cost volume (₹)	Pilot total (₹)	Volume total (₹)	Class	Basis
	Storage and mounting							
1	Composite hydrogen cylinder, 60 L water volume, 350 bar (Type III or IV), 15-year life	1.00	250,000	95,000	250,000	95,000	B	Scaled from DOE/Strategic Analysis Type IV system cost records (S-CYLCOST): about \$33/kWh at low volume and \$21/kWh at 10k/yr for a 5.6 kg 700 bar system, applied to a 1.45 kg (48 kWh) cylinder with a small-size penalty; example product Luxfer G-Stor Pro V060H35 (S-CYL)
2	On-tank valve with solenoid, excess-flow, manual shut-off and TPRD (UN R134 class)	1.00	55,000	18,000	55,000	18,000	C	Production parts exist (S-OTV); no public price; estimate
3	Mounting frame, straps, boot bulkhead shield, TPRD vent duct, boot ventilation	1.00	30,000	11,000	30,000	11,000	C	Fabricated steel frame per UN R110-type 20 g / 8 g loads; estimate
	Fuel delivery							
4	H35 receptacle, check valve, high-pressure filter	1.00	25,000	8,000	25,000	8,000	C	ISO 17268 H35 receptacle; estimate
5	Stainless (316L) high-pressure lines and fittings	1.00	18,000	7,000	18,000	7,000	C	Estimate
6	Two-stage pressure regulator 350 bar to injection pressure, with pressure and temperature sensors	1.00	55,000	15,000	55,000	15,000	C	Production FCEV regulators exist (S-REG); estimate for ICE outlet pressure
7	Hydrogen port-fuel injectors (4) and rail	1.00	40,000	12,000	40,000	12,000	C	Bosch / PHINIA H2 PFI injectors announced 2024-25 (S-INJ); estimate
8	Low-pressure hose, fuel shut-off, pressure relief	1.00	8,000	3,500	8,000	3,500	C	Estimate
	Controls and safety							
9	Hydrogen ECU (dual-map, CAN, automatic petrol fallback, diagnostics) with harness	1.00	45,000	15,000	45,000	15,000	C	Analogous to sequential CNG ECU at volume; pilot uses low-volume controller
10	Hydrogen leak sensors (boot and engine bay), controller, cabin warning, vent fan	1.00	22,000	7,000	22,000	7,000	C	Automotive H2 sensors exist (S-SENS); estimate
	Engine and emissions							
11	Cold-grade spark plugs, PCV/oil-separator upgrade, coolant and knock sensor check	1.00	8,000	4,000	8,000	4,000	C	Backfire and pre-ignition mitigation items (S-H2ICE)
12	Lean-NOx aftertreatment provision (lean NOx trap) and lambda sensor	1.00	35,000	15,000	35,000	15,000	C	Needed if lean calibration alone does not meet BS-VI NOx over the cycle (S-NOX); estimate
	Installation and validation							
13	Removal of CNG kit and cylinder, installation labour (hours)	40.00	600	600	24,000	8,400	C	Pilot 40 h; volume 14 h at Rs 600/h (see qty note)
14	Per-vehicle leak test, commissioning, road test and documentation	1.00	10,000	2,500	10,000	2,500	C	Estimate
15	Certification and engineering amortisation per kit	1.00	0	8,000	0	8,000	B	Volume: one-time development and type-approval cost (Pilot_Budget) spread over 50,000 kits; pilot carries it in the programme budget
	Subtotal parts, labour and validation				625,000	229,400		
	Warranty provision (share of subtotal)		0	0	31,250	9,176		
	Installer and kit-maker margin (share of subtotal)		0	0	0	34,410		
	GST on kit and installation (rate assumed; CNG kits attract 28 percent, hydrogen kit classification to be confirmed)		0	0	118,125	49,137		
	Installed conversion cost per vehicle (incl. taxes)			Pilot (G) / Volume (H):	774,375	774,375	3221 23.48	
	Share of cylinder and valve in volume kit cost				0			
	Volume kit cost in Rs lakh				3			
	Pilot kit cost in Rs lakh				8			
	Reference: Round 1 assumed Rs 1.8 to 2.2 lakh installed. Reference: sequential CNG retrofit kit incl. cylinder and fitting in India about Rs 0.6 to 0.9 lakh (S-CNGKIT).							

Installed conversion cost per vehicle including taxes: pilot ₹774,375; commercial volume ₹322,123. Cylinder and valve are 49 percent of parts cost at volume. One-time development and type-approval costs sit in the programme budget (A9) and are amortised at ₹8,000 per kit over 50,000 kits in the volume case.

A3.2 Ownership cost comparison (base scenario)

Owner decision in 2028 with a 3-year-old Dzire Tour S CNG; horizon 5 years; 55,000 km per year (167 km x 330 days); nominal discount rate 12 percent; fuel escalation 3 percent, electricity 2 percent, delivered hydrogen minus 10 percent per year from ₹800/kg. Option A keeps the car (residual value at year 5), option B converts it (kit at t0, residual at year 5 reduced by 10 percent), option C sells it at t0 (₹3.5 lakh) and buys a compact BEV (₹10.5 lakh, residual ₹4.2 lakh at year 5). In-shift refuelling or charging time is costed in all three options at ₹150 per hour; overnight charging is free idle time. No depreciation is charged in addition to capital; no battery replacement is assumed (8-year, 160,000 km warranties).

Total cash cost by year (₹)	t0 (2028)	2029	2030	2031	2032	2033
A: continue on CNG	0	293,947	299,975	306,183	312,578	219,165
B: hydrogen retrofit	316,623	761,230	712,149	668,488	629,719	505,371
C: sell and buy BEV	730,000	166,623	168,310	170,030	171,784	-246,426
Delivered hydrogen price path (₹/kg)		800	720	648	583	525

Result (base scenario)	Value
NPV of costs: A / B / C (₹ lakh)	10.4 / 27.3 / 11.0
Levelised cost per km: A / B / C (₹/km)	5.25 / 13.73 / 5.55
Year-1 fuel or energy cost per km: CNG mode / hydrogen mode / petrol mode / BEV (₹/km)	3.48 / 14.80 / 6.80 / 1.53
Year-1 blended fuel cost per km: A / B (₹/km)	3.65 / 12.00
Year-1 non-fuel operating cost per km: A / B / C (₹/km)	1.69 / 1.81 / 1.49
Extra cost of B versus A; B versus C; C versus A over 5 years (₹ lakh)	16.8; 16.2; 0.6
Fuel-parity hydrogen price versus CNG; versus petrol (₹/kg)	188; 368
Full-TCO break-even year-1 hydrogen price versus A; versus C (₹/kg)	-41 (negative: no positive price breaks even); -11
Maximum conversion cost for parity with A; with C at the base hydrogen price (₹)	0; 0
Simple payback of the conversion versus A	none
Minimum annual km for payback within the horizon	n/a
Decomposition check (should be zero)	0.00 and 0.00

A3.3 Sensitivity grids (₹ lakh extra cost over 5 years; positive means the retrofit costs more)

Grid 1: retrofit minus continue-on-CNG by year-1 hydrogen price (rows) and annual km (columns)

Rs/kg \ km/yr	30,000 km	40,000 km	50,000 km	60,000 km	70,000 km
₹100/kg	+3.0	+2.9	+2.9	+2.8	+2.7
₹150/kg	+3.6	+3.7	+3.8	+3.9	+4.0
₹200/kg	+4.1	+4.4	+4.7	+5.0	+5.2
₹250/kg	+4.7	+5.1	+5.6	+6.1	+6.5
₹300/kg	+5.2	+5.9	+6.5	+7.1	+7.8
₹400/kg	+6.3	+7.3	+8.3	+9.3	+10.3
₹500/kg	+7.4	+8.8	+10.1	+11.5	+12.9
₹600/kg	+8.5	+10.2	+11.9	+13.7	+15.4
₹800/kg	+10.7	+13.1	+15.6	+18.0	+20.5

Grid 2: retrofit minus sell-and-buy-BEV by hydrogen price (rows) and BEV price in ₹ lakh (columns)

Rs/kg \ BEV price	BEV ₹9 lakh	BEV ₹10 lakh	BEV ₹11 lakh	BEV ₹12 lakh	BEV ₹13 lakh	BEV ₹14 lakh
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₹100/kg	+3.7	+2.7	+1.7	+0.7	-0.3	-1.3
₹150/kg	+4.7	+3.7	+2.7	+1.7	+0.7	-0.3
₹200/kg	+5.7	+4.7	+3.7	+2.7	+1.7	+0.7
₹250/kg	+6.7	+5.7	+4.7	+3.7	+2.7	+1.7
₹300/kg	+7.7	+6.7	+5.7	+4.7	+3.7	+2.7
₹400/kg	+9.7	+8.7	+7.7	+6.7	+5.7	+4.7
₹500/kg	+11.7	+10.7	+9.7	+8.7	+7.7	+6.7
₹600/kg	+13.7	+12.7	+11.7	+10.7	+9.7	+8.7
₹800/kg	+17.7	+16.7	+15.7	+14.7	+13.7	+12.7

Grid 3: retrofit minus continue-on-CNG by hydrogen price (rows) and installed kit cost in ₹ lakh (columns)

Rs/kg \ kit cost	kit ₹1.5 lakh	kit ₹2.0 lakh	kit ₹2.5 lakh	kit ₹3.0 lakh	kit ₹4.0 lakh	kit ₹6.0 lakh
₹100/kg	+1.1	+1.6	+2.1	+2.6	+3.6	+5.6
₹150/kg	+2.1	+2.6	+3.1	+3.6	+4.6	+6.6
₹200/kg	+3.1	+3.6	+4.1	+4.6	+5.6	+7.6
₹250/kg	+4.1	+4.6	+5.1	+5.6	+6.6	+8.6
₹300/kg	+5.1	+5.6	+6.1	+6.6	+7.6	+9.6
₹400/kg	+7.1	+7.6	+8.1	+8.6	+9.6	+11.6
₹500/kg	+9.1	+9.6	+10.1	+10.6	+11.6	+13.6
₹600/kg	+11.1	+11.6	+12.1	+12.6	+13.6	+15.6
₹800/kg	+15.1	+15.6	+16.1	+16.6	+17.6	+19.6

One-at-a-time driver	Low value	High value	Extra cost at low (₹ lakh)	Extra cost at high (₹ lakh)
Year-1 delivered hydrogen price (Rs/kg)	550.00	1,000	+11.8	+20.8
Installed kit cost (Rs)	273,805	402,654	+16.4	+17.6
Annual km	30,000	70,000	+10.7	+20.5
Hydrogen consumption (kg/100 km)	1.60	2.30	+14.7	+20.7
CNG price (Rs/kg)	95.00	78.00	+16.2	+17.6
Share of km on hydrogen	0.85	0.50	+18.9	+15.3

Reading: the extra cost of the retrofit stays positive across every grid cell; the smallest values occur at low hydrogen prices, low mileage and low kit cost. The BEV comparison is closest at high BEV prices and low hydrogen prices.

A4. Packaging, energy balance and performance calculations

Quantity	Value	Unit	Method or note
Energy use per km on CNG (fleet real-world)	1.920	MJ/km	LHV / km per kg
Energy use per km on petrol (fleet real-world)	2.147	MJ/km	
Hydrogen consumption assumed (input)	1.850	kg/100 km	Input h2_cons
Energy use per km on hydrogen	2.220	MJ/km	
Implied H2-ICE efficiency relative to CNG operation (>1 = better)	0.865	ratio	Cross-check: lean PFI hydrogen at part load can match or exceed CNG efficiency; retrofit calibration usually loses some of that
Hydrogen economy	54.054	km/kg	
Cylinder internal volume	60.000	L	
Nominal hydrogen at 350 bar, 15 C	1.440	kg	Density 24.0 kg/m3 (S-DENS)
Nominal hydrogen if 700 bar cylinder of same volume	2.412	kg	Design option; needs 700 bar dispensing
Hydrogen after a non-cooled 350 bar fill	1.224	kg	Fill SOC input; hot ambient, no pre-cooling
Residual hydrogen at minimum usable pressure	0.102	kg	About 20 bar residual
Usable hydrogen per fill	1.122	kg	Key output
Range on hydrogen per fill	60.649	km	Key output
Range on hydrogen per fill, 700 bar option	105.308	km	
Petrol backup range	555.000	km	Existing tank retained
Daily hydrogen need at full hydrogen operation	3.090	kg/day	
Fills per day needed for 100 percent hydrogen operation	2.754	fills/day	
Maximum share of km on hydrogen with the available depot fills	0.726	fraction	Compare with input h2_share
Share of km on hydrogen used in the model (input)	0.650	fraction	
Hydrogen used per operating day in the model	2.008	kg/day	
Mass added: cylinder + valve/frame/regulator/lines/ECU	50.000	kg	
Mass removed: CNG cylinder + CNG kit	63.000	kg	
Net kerb mass change	-13.000	kg	Negative = lighter than CNG configuration
Peak power on petrol	81.600	PS	Donor spec
Peak power on CNG (factory)	70.000	PS	Third-party figure; official CNG rating not captured
Estimated peak power on hydrogen (lean PFI)	42.432	PS	About 30 percent air displacement plus lean operation (S-H2PWR)
Power loss versus petrol	0.480	fraction	
Power loss versus CNG	0.394	fraction	
Power-to-mass, hydrogen mode, 4 occupants (kerb 1045 kg + 300 kg)	31.856	PS per tonne	Urban duty reference: a loaded 800 cc hatchback is about 40 PS per tonne
HCNG 18 vol% blend: hydrogen share of energy	0.062	fraction	Volumetric LHVs 10.8 and 35.8 MJ/Nm3 (S-HCNG); no vehicle modification

Packaging basis (concept, to be verified by mock-up): the Dzire Tour S carries a 55 L Type I steel CNG cylinder (OD 316 to 325 mm, about 0.85 to 1.0 m long, 48 to 69 kg) transversely in the boot. The replacement is a 60 L composite cylinder of the Luxfer G-Stor Pro V060H35 class (Type III, 350 bar, OD 400 mm, 767 mm long, 36 kg, 1.45 kg hydrogen; Type IV equivalents from Quantum or NPROXX exist in the 26 to 80 L range but no accessible datasheet was found). The larger diameter consumes an estimated further 40 to 60 L of luggage space; seating is unchanged; ground clearance of 163 mm rules out under-floor mounting. Mounting frame designed to the UN R110 M1 loads of 20 g longitudinal and 8 g lateral (hydrogen-specific installation clauses of AIS-195 to be applied once accessible). Non-cooled 350 bar fills are assumed to reach 85 percent state of charge in hot ambient conditions; SAE J2601 light-duty tables start at 49.7 L storage volume, so the fill protocol must be agreed with the station.

Consumption cross-check: the fleet CNG car at 25 km/kg uses 1.92 MJ/km; the hydrogen car at 1.85 kg/100 km uses 2.22 MJ/km, an efficiency ratio of 0.87 (lean port-injection retrofit calibration below CNG stoichiometric efficiency). Comparable vehicles: Ford Focus H2RV 1.4 kg/100 km (supercharged hybrid, claimed), Mazda RX-8 Hydrogen RE 2.4 kg/100 km (rotary), BMW Hydrogen 7 3.7 kg/100 km (V12, FTP-75), Toyota hydrogen HiAce about 3.1 kg/100 km (3.4 L V6 van, derived). Power: port-injected hydrogen displaces about 30 percent of intake air (about 20 percent power loss at stoichiometric) and lean operation at lambda 2 costs a further 35 to 40 percent; the model uses 52 percent of petrol power (42 PS) with lambda 1.6 to 2.2.

A5. Regulatory compliance matrix and hazard analysis

A5.1 Compliance matrix

Requirement	Applicable technology / category	Current source	Required evidence or test	Responsible party	Status / uncertainty
Whole-vehicle safety type approval for a hydrogen ICE vehicle	H2-ICE, categories M (incl. M1) and N	CMVR Rule 125M (GSR 746(E), 16 Oct 2023) referencing AIS 195:2023 with Amendment 1 (May 2025)	AIS-195 tests: storage qualification, TPRD, leak and concentration limits, post-crash, labelling	Kit maker with a Rule 126 test agency (ARAI or ICAT)	In force for new vehicles; applicability to retrofitted in-use vehicles not found in the document (not accessible in this study)
Legal gateway for alteration or retrofit	Any retrofit	MV Act s.52; CMVR Rule 126 proviso (retrofitted vehicles tested and type-approved by test agencies, or self-certified by OEM or State-authorized workshops; 2026 amendment reported); Rule 115 CNG/LPG kit provisions	Kit type approval; installer authorisation; installation certificate	Kit maker, test agency, State transport department, RTO	Gateway exists in the abstract; no hydrogen kit standard or test procedure exists
Emissions of the converted vehicle	H2-ICE conversion	AIS-137 (retrofit part used for CNG/LPG kits); BS-VI M1 NOx 60 mg/km	NOx (and lubricant-derived HC, PM) on the certification cycle; OBD	Test agency	No H2-ICE retrofit emission procedure found; hydrogen combustion produces thermal NOx (US DOE)
On-board hydrogen cylinder design approval	Type III or IV composite, 350 bar (700 bar option)	Gas Cylinders Rules 2016 as amended by GSR 225(E), 11 Apr 2025 (definition covers hydrogen cylinders fitted to motor vehicles); Schedule I codes incl. UN R134, CGA S-1.x, IS 5903 for relief devices	PESO design and prototype approval; manufacturer approval; QR or RFID marking	Cylinder manufacturer; PESO Chief Controller	Type IV approvals exist for 150 to 250 L truck cylinders (Time Technoplast); no car-size approved cylinder found
Periodic cylinder inspection	All on-board hydrogen cylinders	GCR 2025: hydrostatic or hydrostatic-stretch test every 3 years; 15-year service life	Test certificate from a PESO-approved test station with composite-trained staff	Owner; test station; PESO	Test-station network for composite hydrogen cylinders is immature
Installation, venting, TPRD discharge direction	H2-ICE conversion	AIS-195 (assumed) plus UN GTR 13 / R134 discharge rules; GCR nitrogen purge for venting	Installation layout vetted by the test agency (CNG precedent); TPRD routing away from occupants, wheel wells and electrics	Kit maker; installer; test agency	AIS-195 clauses not captured
Hydrogen fuel quality	H2-ICE (ISO 14687 grade A/B technically sufficient); FCEV grade D	GCR 2025: fuel conforming to IS 16061 or ISO 14687; FCEV rule (GSR 579(E)) ISO 14687, later IS 16061:2021	Fuel-quality certificate at the dispenser; IS 18538 (Part 8) station quality control	Fuel supplier; station operator	Indian rules silent on an ICE grade
Refuelling station licence	H35 dispensing for cars	GCR 2016 Form-H licence (online module since 14 May 2025); SMPV(U) 2025 amendment for bulk storage; NFPA 2 / ISO 19880-1 (IS 18538 series); receptacle IS/ISO 17268:2020	PESO licence; station safety case	Station operator; PESO	Very few stations; no public retail station as of 2026
Fuelling protocol	H35 light duty	SAE J2601 (2020): storage systems of 49.7 L or more; 35 MPa tables	Dispenser compliance; agreed fill protocol for a 60 L system	Station operator	No Indian adoption found; small systems near the scope boundary
Registration mark	Hydrogen-fuelled vehicle	Central Motor Vehicles (Fourth Amendment) Rules 2025, GSR 819(E), 3 Nov 2025	Plate or mark per rule	Owner; RTO	Application to converted vehicles is an inference
RC endorsement of the retrofit	Any retrofit	Rule 47B(1), Form 22D; s.52 (14 days)	Form 22D with kit type approval, installer certificate, PESO cylinder certificate, layout drawings	Owner; RTO	Requires the kit approval and cylinder approval to exist
Insurance	Bi-fuel kits	IMT-25 (CNG/LPG kit endorsement)	Declared kit value; endorsement	Owner; insurer	No hydrogen endorsement exists; needs insurer product
OEM warranty	Retrofitted vehicles	Maruti owner's manual reportedly excludes retrofitted CNG	Programme warranty instead	Kit integrator	Powertrain warranty void after conversion (inference)
Prototype road trials	Converted prototypes	CMVR Rules 33 to 42 (trade certificate; Rule 41); Rule 126 test-agency evaluation; MNRE transport pilot scheme (buses, trucks and 4-wheelers)	Trade certificate; test-agency programme; pilot sanction	Kit maker; ARAI/ICAT; MNRE and MoRTH steering committee	Precedents: ICAT Mirai pilot (2022), Tata truck trials (2025)
Workshop safety	Installers and service	NFPA 2 (2023) repair garages: ventilation at or above 1 cfm per sq ft, high-level exhaust, interlock; PESO composite-cylinder training	Ventilation design, hydrogen detection, permit-to-work	Installer network	No Indian workshop code specific to hydrogen found

HCNG blend step	Existing CNG vehicles	GSR 585(E), 25 Sep 2020 (Rule 115): HCNG per IS 17314:2019; vehicles per AIS 024 / AIS 028 Rev 1	OEM confirmation that the CNG type approval covers HCNG; PESO-licensed HCNG dispensing	OEM; fuel supplier; PESO	Legal as a fuel; dispensing limited to demonstrations
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A5.2 Hazard analysis (prevention, detection and response, validation, residual risk)

Hazard	Prevention	Detection and response	Validation	Residual risk
Hydrogen leak into boot, cabin, garage or basement	Sealed boot bulkhead; boot vented to outside; no ignition sources in the cylinder bay; tank valve closed with ignition off; hydrogen-rated fittings	Two hydrogen sensors (alarm 1 percent, shutdown 2 percent); on-tank valve closes within 1 s; ECU switches to petrol; cabin warning; vent fan	Leak and concentration tests per GTR 13 / AIS-195 (below 4 percent in cabin, below 8 percent at vents); enclosed-space test with the vehicle parked	Residual: undetected slow permeation in an unventilated basement; mitigated by parking rules for the fleet and sensor self-test
Material degradation, embrittlement, permeation	316L stainless lines, hydrogen-compatible elastomers, composite liner; no reuse of CNG steel parts	Leak check at each service; 3-yearly cylinder requalification (GCR 2025); 15-year life	Supplier qualification to ISO 19881 / UN R134; cycling and permeation tests; field inspection log	Residual: long-term liner ageing in hot climates; tracked through pilot sampling
Crash damage	Cylinder behind the rear seat inside the crush-free zone; steel frame designed for 20 g longitudinal and 8 g lateral; protected valve orientation	Inertia switch closes the tank valve; post-crash leak limit	Sled test of the mounting; full-vehicle rear crash under AIS-195 procedures; Euro NCAP precedent (Mirai and Nexo five stars) shows hydrogen systems can be packaged safely	Residual: severe side or under-ride impacts beyond test conditions
Overpressure and fire	TPRD vents away from occupants, wheel wells and electrics; burst ratio 2.25 times working pressure; heat shielding from exhaust	TPRD opens on heat, giving a controlled release; hydrogen detection in the bay	Bonfire, burst, drop and pressure-cycling tests (GTR 13); vent-line routing review; workshop nitrogen purge for venting	Residual: localised fire impinging on the cylinder without activating the TPRD; mitigated by TPRD placement study
Detection or control failure	Redundant sensors; ECU watchdog; manual shut-off valve; limp-home on petrol	Fault codes; forced petrol mode; service lock-out	Hardware-in-the-loop and fault-injection tests; 500 h engine durability; software audit	Residual: common-mode sensor drift; addressed by periodic calibration
Workshop and refuelling incidents	NFPA 2 type ventilation, hydrogen detection, permit-to-work, PESO-trained staff; dispenser interlocks; SAE J2601 fill protocol; assembly torque verification (Kjørbo lesson); written procedures for trailer transfers (Santa Clara lesson)	Gas detection interlocks; emergency stop; incident reporting	PESO Form-H licence; commissioning tests; drills	Residual: human error during maintenance; mitigated by two-person rule on high-pressure work
Hydrogen combustion emissions	Lean burn (lambda 1.6 to 2.2) with retarded spark; lean NOx trap provision; load limiting with automatic petrol switch	Lambda sensor monitoring; OBD	Emission tests on the BS-VI cycle at the test agency; quarterly checks in the pilot	Residual: transient NOx above the limit in real driving; hence aftertreatment budget
Electrical risk	No high-voltage system in the H2-ICE retrofit (12 V only); sensor and valve wiring fused and routed away from hydrogen lines	Standard fault detection	Standard automotive electrical checks	Not applicable beyond conventional vehicle risk (a fuel-cell conversion would add high-voltage hazards)

Hydrogen properties used: flammability 4.0 to 75.0 percent in air; detonation 18.3 to 59 percent; minimum ignition energy 0.017 mJ; autoignition 585 C; diffusivity about 3.8 times natural gas; nearly invisible flame (H2Tools, US DOE). Prototype work and high-pressure testing are confined to qualified facilities (test cells at HORIBA Chakan, ARAI, IOCL R&D; PESO-approved test stations). This annexure is a development and validation plan, not conversion instructions.

A6. Market sizing method and bottom-up funnel

Step	Vehicles (base)	Share applied	Basis
CNG vehicles in India, all categories	8,195,000		PNGRB citing Vahan (Mar 2025), class A
Passenger cars among them	4,507,250	55 percent	Estimate from cumulative factory-CNG car sales (Maruti about 2.6 million FY20-FY26 plus other OEMs) and conversions; Round 1 said 65 percent (unverified)
Technically eligible: factory-fitted, boot-cylinder layout, under 6 years old at 2028	2,253,625	50 percent	Factory-fitted, boot cylinder, under 6 years old at 2028 (FY23-FY26 sales)
In commercial fleets (taxi, aggregator, corporate, institutional)	450,725	20 percent	Delhi alone registers 60,000 to 85,000 taxis; no national split published; assumption
In cities with a hydrogen hub station by 2030	180,290	40 percent	MNRE corridors: Delhi-NCR, Ahmedabad-Vadodara-Surat, Pune-Mumbai, Kochi and others
Plausible early adopters (return-to-base fleets willing to convert)	9,014	5 percent	Hypothesis to be validated in Phase 0
Supply-constrained capacity: stations x cars per station	1,394		8 hub stations x 174 cars per station (500 kg/day at 70 percent utilisation, 2.0 kg per car per day)
Realistic 2030 converted fleet (minimum of demand and supply)	1,394		Minimum of demand and supply
Round 1 claim for 2030	100,000		For comparison

Downside and upside funnels (workbook scenario selector) give roughly 700 to 2,600 cars by 2030; in every case supply (stations) rather than demand binds. Private owners are excluded from the first segment because they lack depot fuelling and demand aggregation; they could follow only after hub stations exist.

A7. Global benchmark evidence: progress and setbacks

Programme	Technology and class	Status (date)	Scale	Fuel-supply model	Commercial limitation	What transfers to an Indian retrofit
Toyota Mirai	FCEV sedan	Production since 2014; MY2026 refresh with fuel credits	US sales 499 (2024), 210 (2025)	Public retail (California, Japan)	Station closures and USD 32 to 36/kg fuel	Transfers: nothing on cost; warns against retail-network dependence
Hyundai Nexø (2nd gen)	FCEV SUV	Korea sales from June 2025	Korea about 36,000 FC cars in circulation	Public retail with subsidy	Dependence on Korean subsidies	Transfers: policy dependence
Honda CR-V e:FCEV	Plug-in FCEV	On lease in California since July 2024	About 300 per year	California retail	GM-Honda fuel-cell plant ends 2026	Transfers: OEM appetite is falling
Toyota GR Corolla H2	Liquid-hydrogen ICE race car	Racing development 2021 to 2025	1 car	Trackside liquid hydrogen	No production plan	Transfers: combustion durability learning only; not a cost proof
Toyota hydrogen HiAce (Australia)	3.4 L twin-turbo V6 H2-ICE van	Pilot with CPB Contractors (2024); hybrid variant announced 2025	Few prototypes	Captive depot (Toyota Altona)	Range below 200 to 250 km; SCR needed for Euro VI	Transfers: depot model, NOx aftertreatment, range reality
HySE (Kawasaki, Suzuki, Honda, Yamaha, Toyota)	Small H2-ICE research	Dakar 2024 and 2025 buggies	Research	Rally logistics	Pre-commercial	Transfers: small-engine hydrogen combustion know-how
Suzuki hydrogen Burgman; Suzuki-AVL hydrogen Swift	H2-ICE scooter concept (70 MPa); 1.4 L DI hydrogen Swift prototype (100 kW, 220 Nm)	JMS 2023 and 2025; Vienna Motor Symposium May 2026	Concepts	Not stated	'Not launching anytime soon' (Burgman); Swift is a prototype	Transfers: Suzuki-relevant small-car hydrogen combustion exists; DI, not retrofit
Cummins B6.7H (India)	6.7 L H2-ICE	First production engine Mar 2025, Jamshedpur, for Tata trucks	Limited	Truck stations (350 bar)	Fuel availability and TCO	Transfers: Indian supply of hydrogen injection and controls
JCB hydrogen engine	4.8 L off-highway H2-ICE	EU type approval (Stage V) May 2025; above 130 evaluation engines	Evaluation fleet	On-site bowzers	Not road-vehicle homologation	Transfers: regulatory precedent for hydrogen combustion approval
KEYOU (Germany)	H2-ICE conversion of 18 t trucks and 12 m buses	First truck delivered Dec 2024; Daimler Truck partnership for 2027	Tens of vehicles	H2 Mobility truck stations	Conversion cost; fuel price	Transfers: the only commercial H2-ICE retrofit model, heavy-duty, depot-based
ULEMCo (UK)	Hydrogen dual-fuel conversions of vans and municipal vehicles	Commercial conversions (Aberdeen fleet)	Tens of vehicles	Council depot hydrogen	No passenger-car conversions; prices not public	Transfers: municipal fleet model
Stellantis hydrogen vans	FCEV LCVs	Programme ended 16 Jul 2025	Series production cancelled	Public retail	Infrastructure, capital, incentives	Transfers: light-duty hydrogen retail model failed even for an OEM
Shell California; H2 Mobility Germany; OMV Austria	Retail hydrogen stations	Shell closed 7 car stations (Feb 2024); H2 Mobility closed 36 car stations (2025); OMV closed all (2025)	Network contraction	Retail	Low utilisation by cars	Transfers: utilisation, not technology, kills stations
India: MNRE pilots; Tata truck trials; NTPC Leh; IOCL HCNG buses; Mirai-ICAT pilot	Buses, trucks, HCNG, one FCEV car	12 projects, 70 vehicles, 16 stations sanctioned (₹410.77 crore, ₹41.58 crore released); Tata 16 trucks (2025), 40 more (2026); NTPC Leh 5 buses at 80 kg/day; 50 HCNG buses (2020); Mirai pilot 2022	Tens of heavy vehicles	Captive stations	No cars; only three captive dispensing points	Transfers: corridor stations that a car pilot can share; scheme names 4-wheelers

A8. Pilot design, instrumentation and first 90 days

Element	Design
Fleet size and why	30 converted cars: large enough to spread across three fleets and two duty patterns (staff shuttle and pool car), to accumulate about 45 vehicle-years of operation in 18 months, and to load one dispensing point at about 60 kg/day; small enough to insure and support from two workshops.
Duration and duty cycle	18 months of operation (April 2029 to September 2030), 150 to 210 km per operating day, 330 days per year, two depot fills per day, hydrogen mode by default with automatic petrol fallback.
Comparison vehicles	10 unmodified Dzire Tour S CNG cars and 10 compact BEVs from the same fleets on the same duty, instrumented identically, to give matched cost, availability and emissions references.
Data collected	Fuel and energy use per trip (dispenser logs, CAN mass-flow, meter readings); range and fills; availability and downtime by cause; maintenance events and cost; refuelling time and queue; emission tests quarterly (NOx, HC, PM); hydrogen sensor logs; luggage and passenger complaints; driver ratings; station utilisation and delivery reliability.
Instrumentation and logs	CAN logger with GPS on all 50 cars; hydrogen concentration logging in boot and engine bay; dispenser data (mass, pressure, temperature, fill time); workshop job cards; incident and near-miss register with a safety officer sign-off.
Maintenance and safety reporting	Monthly safety review; every hydrogen alarm investigated within 24 hours; 3-yearly cylinder requalification scheduled; PESO and test-agency informed of reportable events; independent evaluator receives raw data quarterly.
Customer and operator feedback	Quarterly structured interviews with fleet managers and drivers on power, boot space, refuelling, confidence; willingness-to-pay survey at month 12.
Independent validation	An academic or think-tank evaluator (proposed IIT Delhi or CEEW) publishes consumption, availability, emissions and cost results; test-agency emission and safety tests are the formal evidence.
What a pilot cannot prove	30 cars over 18 months cannot establish a statistical safety case; safety evidence comes from the type tests (storage, leak, crash, bonfire) and the standards route, not from incident-free operation.

A8.1 Success measures for gate 4 (September 2030)

- Consumption at or below 2.2 kg/100 km in fleet use; range per fill at or above 55 km; hydrogen share of km at or above 60 percent.
- Availability at or above 95 percent; conversion-related downtime below 2 percent of operating days.
- Zero reportable hydrogen incidents; all alarms investigated; no unplanned venting.
- Tailpipe NOx at or below 60 mg/km on the certification cycle and on-road spot checks within 1.5 times the limit.
- Measured service and maintenance cost within ₹1.2/km; refuelling time under 6 minutes per fill; luggage and passenger acceptance from fleet managers.
- Delivered hydrogen price at or below ₹250/kg with a documented path to ₹190/kg; station utilisation trend above 50 percent.

A8.2 First 90 days (from programme start, October 2026)

Window	Actions
Days 1 to 15	Appoint SRDI engineering lead and programme manager; agree governance with NIT Warangal team; freeze the evidence register; write to AISC secretariat, MoRTH, ARAI, ICAT and PESO requesting a technical meeting on the retrofit route.
Days 15 to 30	Shortlist fleets (PSU staff transport, airport-contract operators, corporate fleets in Faridabad and Gurugram); start 30 structured interviews; fit telematics to 20 donor cars for duty-cycle data.
Days 30 to 45	Request cylinder samples and quotations (Luxfer, Quantum, NPROXX, Time Technoplast); injector and regulator quotations (Bosch, PHINIA, OMB Saleri, Poppe + Potthoff); test-cell availability (HORIBA Chakan, ARAI, IOCL R&D).
Days 45 to 60	Boot mock-up of a Dzire Tour S with a 400 mm cylinder dummy; luggage and seating assessment; CAD packaging and mounting concept; preliminary hazard analysis workshop.
Days 60 to 75	Meet IOCL R&D Faridabad and the MNRE corridor awardees on shared dispensing; draft a proposal under the MNRE transport pilot scheme (4-wheeler category) with ARAI; check Gujarat station subsidy for the second hub.
Days 75 to 90	Gate 1 dossier: regulator view, partner letters, demand evidence, packaging result, updated cost model; proceed, redesign or stop decision by the steering group.

A9. Phase budgets, responsibilities and funding

Phase / line item	Quantity	Unit cost (₹)	Total (₹)
Phase 0: Feasibility and validation (Oct 2026 - Mar 2027)			
Regulator and test-agency consultations, route definition (AISC, MoRTH, ARAI/ICAT, PESO)	1.0	2,500,000	2,500,000
Fleet and fuel-partner validation: 30 operator interviews, duty-cycle telematics on 20 cars	1.0	2,500,000	2,500,000
Donor inspection, packaging study, CAD and mule vehicle (boot mock-up)	1.0	4,000,000	4,000,000
Phase subtotal			9,000,000
Phase 1: Engineering and controlled validation (Apr 2027 - Mar 2028)			
Hydrogen engine test-cell time and instrumentation (calibration, NOx, abnormal combustion, 500 h)	1.0	30,000,000	30,000,000
Prototype conversions (3 vehicles) at pilot kit cost	3.0	774,375	2,323,125
Prototype donor vehicles (used Tour S CNG)	3.0	500,000	1,500,000
Storage system qualification samples and tests (burst, cycling, bonfire, drop, permeation)	1.0	15,000,000	15,000,000
Vehicle-level safety tests (leak, sled/crash structure, ventilation) at test agency	1.0	12,000,000	12,000,000
Engineering team (8 FTE x 12 months)	96.0	250,000	24,000,000
Phase subtotal			84,823,125
Phase 2: Certification and pilot build (Apr 2028 - Mar 2029)			
Kit type approval and emissions testing at ARAI/ICAT (analogous to CNG kit approval)	1.0	10,000,000	10,000,000
Durability demonstration (2 vehicles x 80,000 km accelerated)	1.0	8,000,000	8,000,000
Pilot conversions (30 vehicles) at pilot kit cost	30.0	774,375	23,231,250
Instrumentation and telematics (30 cars + 20 comparison cars)	50.0	60,000	3,000,000
Hydrogen dispensing point at partner site (compressor, cascade, one dispenser, safety, PESO)	1.0	100,000,000	100,000,000
Workshop upgrade (ventilation, detection, training) at 2 service points	2.0	4,000,000	8,000,000
Phase subtotal			152,231,250
Phase 3: Pilot operation and evaluation (Apr 2029 - Sep 2030)			
Hydrogen fuel for 30 cars, 18 months (kg at delivered pilot price)	29,821	1,433	42,729,079
Operations, safety officer, data analysis and independent evaluation (18 months)	18.0	1,500,000	27,000,000
Quarterly emissions and consumption tests at test agency	6.0	800,000	4,800,000
Insurance, incident reserve and driver training	1.0	6,000,000	6,000,000
Phase subtotal			80,529,079
Subtotal all phases			326,583,454
Contingency (share of subtotal)	0.1		48,987,518
Total programme budget to September 2030 (Rs)			375,570,972
Total in Rs crore			38
Suzuki R&D Center India / Maruti Suzuki (engineering, prototypes, test cells)	0.4		150,228,389
Public support sought (MNRE NGHM R&D or pilot scheme, state hydrogen policy)	0.3		131,449,840
Fuel and station partner (OMC or gas utility): dispensing point	0.1		56,335,646
Fleet partner (vehicles, drivers, operations)	0.1		37,557,097
Item	2028	2029	2030
Converted cars in operation (cumulative)	3.0	30	1,394
Fleet-hub stations serving cars	1.0	1	8
Hydrogen demand from cars (t/yr)	2.0	20	924

Programme total to September 2030 including 15 percent contingency: ₹37.6 crore (phases ₹0.9, ₹8.5, ₹15.2, ₹8.1 crore). Proposed funding split: Suzuki R&D Center India and Maruti Suzuki 40 percent; public support 35 percent (MNRE transport pilot scheme, which names buses, trucks and 4-wheelers, or the R&D scheme with up to 80 percent support for industry at TRL 4 or higher; Gujarat's 30 percent station subsidy for the second hub); oil marketing company partner 15 percent; fleet partner 10 percent. For calibration, MNRE's heavy-vehicle pilots average ₹5.6 to 5.9 crore of support per vehicle including stations. All partners are proposed, not confirmed.

Responsibility	Proposed holder	Rationale
Engineering lead, calibration, prototypes	Suzuki R&D Center India with Maruti Suzuki engineering	CNG engine know-how; Suzuki hydrogen combustion prototypes
Kit integration, warranty, installer training	Kit integrator (aftermarket gaseous-fuel specialist) under SRDI oversight	Existing CNG kit approval experience
Type approval and testing	ARAI or ICAT (Rule 126 agencies); PESO for cylinders and station	Statutory
Dispensing point ownership and operation	Oil marketing company (proposed IOCL, Faridabad)	Existing hydrogen dispensing experience; corridor awardee
Hydrogen supply	SIGHT-awarded green hydrogen producer by tube trailer	Certified carbon intensity
Vehicles and operations	Fleet partner (PSU or corporate staff transport, airport contract)	Return-to-base duty
Independent evaluation	IIT Delhi or CEEW (proposed)	Credibility of results
Development risk	Programme (consortium)	Gated spend
Utilisation risk	Station partner with take-or-pay from the fleet	Aligns incentives
Residual-value risk	Fleet, backed by programme buy-back	Removes owner downside in the pilot

A10. Judge Q&A, validation log and change log

A10.1 Judge Q&A (20 questions)

Question	Answer
What is distinctive about the proposal?	It converts the installed factory-CNG base rather than building new vehicles, keeps petrol as an automatic backup so no car is stranded by missing stations, and, unlike most hydrogen pitches, states the price at which it works (₹188/kg delivered for CNG fuel parity) and the gates at which it stops.
Why retrofit this vehicle?	The Dzire Tour S CNG is the fleet workhorse with a 4-cylinder 1.2 L engine (power margin after the hydrogen derate), a boot bay that already carries a 55 L cylinder, hardened valve seats, a 37 L petrol tank for backup, and the largest donor pool (Maruti sells about 70 percent of India's CNG cars).
Why would the customer choose it over staying on CNG or buying a BEV?	On money alone they would not: continuing on CNG costs ₹5.2/km, a compact BEV ₹5.6/km including purchase, and the retrofit ₹13.7/km at 2028 hydrogen prices. The customer for the pilot is a fleet with a decarbonisation mandate or a public programme paying for abatement and data; the commercial customer appears only if delivered hydrogen falls below about ₹190/kg.
Does the hydrogen storage system actually fit?	A 60 L composite cylinder (OD 400 mm, 767 mm long, 36 kg) fits the transverse bay used by the 55 L CNG cylinder (OD 316 mm, about 900 mm), but it is 84 mm larger in diameter, so the boot loses a further 40 to 60 L. No under-floor option exists at 163 mm ground clearance. Phase 0 includes a physical mock-up because this is a concept, not validated design work.
What is sacrificed in passenger capacity, luggage or payload?	Seating is unchanged (five). Luggage falls from roughly 200 L with CNG to about 150 L. Kerb mass falls by about 13 kg, so payload is unchanged or slightly better. Peak power in hydrogen mode is about 42 PS, 48 percent below petrol; the car switches to petrol automatically for high load.
Which components cannot be reused?	Every CNG fuel-system part: the steel cylinder (ISO 11439 steels are limited to 2 percent hydrogen), regulator, rail, injectors, hoses and lines. Hydrogen parts need UN R134 or ISO 19881 class qualification. The engine, gearbox, petrol system, catalyst and body are retained.
What is the credible pilot conversion cost?	₹7.7 lakh per car including GST, dominated by the imported cylinder and valve at small quantities; ₹3.2 lakh installed at 10,000 kits per year (₹2.3 lakh ex-works). Round 1's ₹1.8 to 2.2 lakh is roughly the ex-works volume figure without taxes and margin.
Where will hydrogen come from and at what delivered price?	Certified green hydrogen by tube trailer from a SIGHT-awarded producer to a shared dispensing point in Faridabad. Our build-up gives about ₹1,433/kg at the pilot point (150 kg/day, 50 percent used) and about ₹584/kg at a 2030 hub (500 kg/day, 70 percent used); CEEW's central 2030 estimate is ₹360/kg. India's record refinery-gate tender is ₹279/kg before compression and dispensing.
What if fuel cost stays higher than expected?	The pilot still delivers its purpose (approval route, real-world data, supplier readiness) because fuel is a programme cost. The scale decision at gate 4 simply fails, and the programme stops and publishes.
What if the station is unavailable?	Every car runs on petrol automatically (555 km on a tank); the fleet loses the hydrogen benefit, not the service. The station partner carries utilisation risk through a take-or-pay contract, and a 1.5-day cascade buffer covers delivery slips.
What is the exact approval uncertainty?	There is no standard or rule for a hydrogen retrofit kit. AIS-195 covers new hydrogen ICE vehicles; AIS-157 (fuel cell) explicitly excludes retrofits; Rule 126 and Rule 115 provide the retrofit machinery only for CNG and LPG. We need either an AIS-195 extension or a Rule 115 hydrogen-kit provision; the trial itself can run under a trade certificate with test-agency evaluation, as the Mirai and Tata trials did.
How is NOx addressed?	Lean burn above lambda 2 gives near-zero engine-out NOx at steady state; transients do not stay lean, so the kit carries a lean NOx trap provision and lambda control, and the gate-2 test requires 60 mg/km or better on the BS-VI cycle. This is a target to demonstrate.
What is the environmental benefit under matched assumptions?	With certified green hydrogen (2 kg CO ₂ e/kg) the converted car emits about 89 g CO ₂ e/km well-to-wheel including the kit, against 137 on CNG and 134 for a new BEV on the 2030 grid including its manufacturing. With grid-electrolysis hydrogen it would be four times worse than CNG, so the fuel's certification is a condition, not a detail. Hydrogen combustion also emits NOx.
Who funds development and underutilised infrastructure?	Proposed split of ₹37.6 crore: Suzuki R&D Center India and Maruti 40 percent (engineering, prototypes, test cells), public support 35 percent (the MNRE transport pilot scheme names 4-wheelers; ARAI runs its calls), an oil marketing company 15 percent (dispensing point), the fleet 10 percent. Station utilisation risk sits with the station partner under take-or-pay. None of this is committed.
What result would make you stop?	No written regulator route by March 2027; backfire or NOx not controlled within 500 test-cell hours; delivered hydrogen still above ₹250/kg in 2030 with no credible path to ₹190/kg; or fleets refusing the power and boot loss in Phase 0 interviews.
What can realistically be achieved by 2030?	An approved retrofit route, 30 cars with 18 months of independent data, a shared hydrogen dispensing point in Faridabad, supplier quotations for car-size cylinders, and a decision. Conditional expansion to 1,000 to 1,500 cars at up to 8 hub stations is possible only after gate 4; Round 1's 1,00,000 cars is not.
Why not a fuel-cell conversion for better efficiency?	It halves hydrogen use per km but replaces the entire powertrain (stack, battery, motor, inverter, cooling, tank, controls) at ₹15 to 25 lakh per car in a pilot, more than a new BEV. It is not a retrofit in cost terms and has no retrofit approval route either.
Why keep the HCNG step if it displaces only 6 percent of energy?	Because it is legal today (GSR 585(E), IS 17314), needs no vehicle change, and lowers CO and HC; it is an air-quality measure that lets CGD companies start blending. We no longer present it as decarbonisation.
How do you know fleets want this?	We do not yet. Customer demand is a stated hypothesis; Phase 0 funds 30 structured interviews and telematics on 20 donor cars, and gate 1 requires evidence from at least three fleets before any engineering spend.
Is Suzuki endorsing this?	No. Suzuki R&D Center India is proposed as engineering lead because of its CNG engine and calibration know-how and Suzuki's own hydrogen combustion prototypes (Burgman, Swift with AVL). No endorsement or partnership is claimed.

A10.2 Validation log

Item	What was done
Attachments	Both PDFs rendered and inspected page by page (brief: 3 pages; submission: 4 slides at 1440 x 810 pt). Title slide preserved as the native 1376 x 768 raster with the source's 3.14 pt white bands; pixel comparison of renders confirms visual identity.
Research	Four workstreams (engineering; safety and regulation; market, infrastructure and benchmarks; economics and emissions) plus a gap-closure pass. Research cut-off 14 September 2026. 309 register entries.
Access limitation	The build environment's network policy blocked direct access to most primary hosts (ARAI, MoRTH, PIB, PESO, MNRE, DOE, NIST, IEA, OEM sites). Findings rely on indexed excerpts of those pages; PIB releases of 14 Feb 2024 and 3 Mar 2025 and the Scribd copy of AIS-157 were read in full through the search index. Entries marked 'excerpt' or 'summary' should be re-verified from the originals before external use.
Model	Workbook built with formulas only, recalculated with LibreOffice Calc; zero error cells; decomposition checks (a + b km + c km P) reproduce the TCO differences to zero; scenario selector tested for 1, 2, 3.
Consistency	Slides, annexure and pitch read numbers from the same output file exported from the recalculated workbook; budgets reconcile to the funding ask; rollout quantities reconcile to station capacity.
Design	Every slide rendered and inspected; overlaps and clipping fixed; fonts limited to Arial for cross-platform fidelity; PPTX and PDF each below 2 MB.
Not done	No customer interviews, letters of interest, test results, supplier quotations or partnerships exist; all are labelled as hypotheses or proposals. The submission portal's annexure handling could not be verified (site inaccessible).

A10.3 Change log: Round 1 to Round 2

Element	Round 1	Round 2
Core idea	Convert the installed CNG car base to hydrogen, fleets first, aggregate demand at hubs	Retained
Fallback fuel	CNG retained with automatic H2/CNG switching (second cylinder)	Petrol retained (already in every factory CNG car); CNG hardware removed; two cylinders do not fit
Donor	CNG cars in general (WagonR image)	Dzire Tour S CNG primary; WagonR, Ertiga, Eeco assessed as alternatives
Retrofit ladder	HCNG (2026-27), bi-fuel H2 plus CNG (2027-28), dedicated H2-ICE (2029-30)	HCNG as an air-quality step (about 6 percent of energy), H2-ICE retrofit pilot (2027-30), conditional 700 bar or hub-scale option after 2030
Hydrogen price basis	Production-cost trajectory to ₹160-200/kg treated as pump price	Delivered price build-up: ₹1,433/kg pilot, ₹584/kg 2030 hub (our conservative), ₹360/kg CEEW central
Economics	₹1.5 lakh annual saving versus petrol, 18-month payback	Fair three-way TCO (continue CNG, retrofit, sell and buy BEV): retrofit never pays back within 5 years; parity conditions stated
Kit cost	₹1.8 to 2.2 lakh	₹7.7 lakh pilot; ₹3.2 lakh installed at volume (₹2.3 lakh ex-works)
Range and power	400 plus km combined	61 km per hydrogen fill plus 555 km petrol; 42 PS in hydrogen mode
Regulation	AIS-157 plus PESO plus ISO/SAE	AIS-195 (new H2-ICE vehicles), Rule 126/115 retrofit template, PESO GCR 2025, Form-H; no retrofit route exists; gate 1 added
Roadmap	500 cars and 100 stations (2026-27), 25,000 kits (2027-28), 1,00,000 plus cars (2029-30)	3 prototypes (2027-28), 30 pilot cars (2028-30), conditional 1,000 to 1,500 cars after 2030; four gates
Impact	1,00,000 cars, 0.9 Mt CO ₂ , 15,000 t/yr hydrogen, ₹2,000 crore industry, 10,000 jobs	About 1,400 cars, 3,600 t CO ₂ e/yr, 924 t/yr hydrogen by 2030; jobs not claimed
Market figures	7.5 million CNG vehicles, 65 percent cars, 4,000 retrofit centres	8.2 million (Mar 2025); car share estimated 55 percent; retrofit-centre count unverified
Benchmarks	Toyota GR Corolla and Hilux H2-ICE	HiAce (Australia) is the hydrogen combustion van; Hilux was a fuel-cell prototype; Suzuki-AVL Swift added; global setbacks added
Carbon credits	USD 10 to 15/t as revenue	Immaterial; treated as upside only
Claims removed	Job creation, 'zero range anxiety' as absolute, 'CNG cylinder retained', implied component reuse, 'BS-VI compliance pathway' as achieved	Replaced by evidence-labelled statements

The Round 1 deck lists Raunak Jain first but does not name a team leader. The submission filename stem Raunak_TeamWookies_NITWarangal assumes Raunak Jain is the leader; confirm before uploading.